

Introduction to Real Time Quantum Field Theory: From S-matrices to the physics of relativistic thermal media

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Contents

1	Thermal Wick theorem and the Schwinger–Keldysh contour	2
1.1	Gaudin’s proof and its contour-ordered form	2
1.2	Full Schwinger–Keldysh contour and far-past thermal preparation	5
2	Free Schwinger–Keldysh propagators	9
2.1	Bosonic and fermionic quantum harmonic oscillators	9
2.2	Thermal free scalar field	15
2.3	Thermal free Dirac field	17
2.4	Free $U(1)$ gauge field	21
3	Schwinger–Keldysh diagrammatic rules	22
3.1	The usual $+/-$ basis and momentum-space rules	23
3.2	Why the r/a basis is better, and its rules	25
3.3	Examples of physical tree-level thermal effects	30
4	The zero-temperature limit: S-matrices, rates and cross-sections	32
4.1	The zero-temperature closed time contour and forward-branch Feynman diagrams	32
4.2	In-out states, the S-matrix, and LSZ	35
4.3	From amplitudes to rates and cross sections	39
4.4	S-matrix diagrammatics and tree-level examples	43
4.5	Spin-summed amplitudes and cross sections	48
5	Quasiparticles of a relativistic thermal medium and kinetic theory	52
5.1	Quasiparticles from the retarded propagator	52
5.2	One-loop self-energies in Yukawa theory from Schwinger–Keldysh rules	53
5.3	Explicit thermal masses, widths, and why resummation is necessary	56
5.4	From Schwinger–Dyson equations to the Boltzmann equation	66
6	Hard thermal loop effective theory of the QED plasma	68
6.1	Self-energy, Debye screening, plasma oscillations and Landau damping	68
6.2	Hard thermal loop effective action	73
7	Electrical conductivity of a relativistic plasma from Schwinger–Keldysh theory	76
7.1	The Kubo formula for conductivity from the Schwinger–Keldysh generating functional	77
7.2	The one-loop current bubble and the origin of the pinch problem	79
7.3	The leading-log transport rate in a weakly coupled QED plasma	83
7.4	Optical conductivity and the Drude peak	86

1 Thermal Wick theorem and the Schwinger–Keldysh contour

The purpose of this section is to put the finite-temperature Wick theorem in the form needed for real-time perturbation theory. The theorem is not a property of an arbitrary thermal state. It is a property of a Gaussian, or quasi-free, thermal state. In perturbation theory this is exactly the structure we use: the interacting theory is expanded around the thermal state of a quadratic Hamiltonian, while the interactions are inserted on a real-time contour.

There are two logically separate points. First, Gaudin’s proof gives Wick factorization for an arbitrary mixture of bosonic and fermionic creation and annihilation operators. Since free fields are linear combinations of these operators, the result immediately gives Wick’s theorem for free scalar and Dirac fields. Second, for Schwinger–Keldysh theory one needs the contour-ordered version. This is not a separate theorem; it follows by applying the same proof after the full product has first been contour ordered.

1.1 Gaudin’s proof and its contour-ordered form

Consider a free Hamiltonian

$$H_0 = \sum_{\alpha \in B} \omega_\alpha b_\alpha^\dagger b_\alpha + \sum_{r \in F} \omega_r c_r^\dagger c_r, \quad \omega_\alpha, \omega_r > 0, \quad (1)$$

with

$$[b_\alpha, b_\gamma^\dagger] = \delta_{\alpha\gamma}, \quad \{c_r, c_s^\dagger\} = \delta_{rs}, \quad (2)$$

and all independent graded commutators vanishing. The free thermal average is

$$\langle X \rangle_{\beta,0} := \frac{1}{Z_0} \text{Tr} \left(e^{-\beta H_0} X \right), \quad Z_0 = \text{Tr} e^{-\beta H_0}. \quad (3)$$

Each elementary operator has a Grassmann parity

$$\varepsilon(A) = \begin{cases} 0, & A \text{ bosonic,} \\ 1, & A \text{ fermionic,} \end{cases} \quad (4)$$

and the graded commutator is

$$[A, B]_g = AB - (-1)^{\varepsilon(A)\varepsilon(B)} BA. \quad (5)$$

Thus two bosonic operators give an ordinary commutator, two fermionic operators give an anticommutator, and a bosonic operator commutes with an independent fermionic operator.

Let A be one of the elementary creation or annihilation operators and suppose

$$[H_0, A] = -s_A \omega_A A, \quad s_A = +1 \text{ for annihilation,} \quad s_A = -1 \text{ for creation.} \quad (6)$$

Then

$$e^{-\beta H_0} A = e^{\beta s_A \omega_A} A e^{-\beta H_0}, \quad A e^{-\beta H_0} = e^{-\beta s_A \omega_A} e^{-\beta H_0} A. \quad (7)$$

This elementary identity is the mode version of the KMS relation.

Definition 1.1 (Thermal contraction). For elementary creation and annihilation operators we define

$$C_\beta(A_i, A_j) := \langle A_i A_j \rangle_{\beta,0}. \quad (8)$$

If the graded commutator of A_i and A_j is non-zero, then

$$C_\beta(A_i, A_j) = \frac{[A_i, A_j]_g}{1 - \zeta_i e^{-\beta s_i \omega_i}}, \quad \zeta_i := (-1)^{\varepsilon(A_i)}. \quad (9)$$

Here $s_i = s_{A_i}$ and $\omega_i = \omega_{A_i}$. If the operators belong to independent modes, the contraction vanishes.

For example, for one bosonic oscillator,

$$\langle bb^\dagger \rangle_{\beta,0} = 1 + n_B(\omega), \quad \langle b^\dagger b \rangle_{\beta,0} = n_B(\omega), \quad n_B(\omega) = \frac{1}{e^{\beta\omega} - 1}. \quad (10)$$

For one fermionic oscillator,

$$\langle cc^\dagger \rangle_{\beta,0} = 1 - n_F(\omega), \quad \langle c^\dagger c \rangle_{\beta,0} = n_F(\omega), \quad n_F(\omega) = \frac{1}{e^{\beta\omega} + 1}. \quad (11)$$

Thus the same algebraic formula contains both Bose enhancement and Pauli blocking.

Theorem 1.2 (Gaudin–Wick theorem at finite temperature). *Let $A_1, \dots, A_n$ be an arbitrary mixture of bosonic and fermionic elementary creation and annihilation operators of the free theory. Then*

$$\langle A_1 A_2 \cdots A_n \rangle_{\beta,0} = \sum_{\text{pairings } P} \sigma(P) \prod_{(ij) \in P} \langle A_i A_j \rangle_{\beta,0}. \quad (12)$$

If n is odd, or if the total fermion parity is odd, both sides vanish. The sign $\sigma(P)$ is the sign obtained by permuting the fermionic operators until the members of each contracted pair stand next to each other, while preserving the order inside each contraction. Bosonic permutations do not generate signs.

Proof. The proof is by induction. The case $n = 2$ is the definition of the contraction. Assume the theorem holds for all strings of length smaller than n and set

$$G_n = \langle A_1 A_2 \cdots A_n \rangle_{\beta,0}. \quad (13)$$

Move A_1 through the string $A_2 \cdots A_n$ by using the graded commutator. The exact operator identity is

$$\begin{aligned} A_1 A_2 \cdots A_n &= \sum_{j=2}^n (-1)^{\varepsilon_1(\varepsilon_2 + \cdots + \varepsilon_{j-1})} [A_1, A_j]_g A_2 \cdots \widehat{A_j} \cdots A_n \\ &\quad + (-1)^{\varepsilon_1(\varepsilon_2 + \cdots + \varepsilon_n)} A_2 \cdots A_n A_1. \end{aligned} \quad (14)$$

The hat means that the operator is omitted. The graded commutator of two elementary operators is a c-number, so it can be pulled out of the thermal trace.

Taking the free thermal average, the last term is reduced by cyclicity of the trace and by (7):

$$\begin{aligned} \langle A_2 \cdots A_n A_1 \rangle_{\beta,0} &= \frac{1}{Z_0} \text{Tr} \left(e^{-\beta H_0} A_2 \cdots A_n A_1 \right) \\ &= \frac{1}{Z_0} \text{Tr} \left(A_1 e^{-\beta H_0} A_2 \cdots A_n \right) \\ &= e^{-\beta s_1 \omega_1} \langle A_1 A_2 \cdots A_n \rangle_{\beta,0}. \end{aligned} \quad (15)$$

If the total fermion parity of the product is odd, the thermal trace vanishes because $e^{-\beta H_0}$ is even under fermion parity. Hence, for the non-vanishing case,

$$(-1)^{\varepsilon_1(\varepsilon_2 + \cdots + \varepsilon_n)} = (-1)^{\varepsilon_1} = \zeta_1. \quad (16)$$

Equation (14) therefore gives the recursion relation

$$\begin{aligned} G_n &= \sum_{j=2}^n (-1)^{\varepsilon_1(\varepsilon_2 + \cdots + \varepsilon_{j-1})} \frac{[A_1, A_j]_g}{1 - \zeta_1 e^{-\beta s_1 \omega_1}} \langle A_2 \cdots \widehat{A_j} \cdots A_n \rangle_{\beta,0} \\ &= \sum_{j=2}^n (-1)^{\varepsilon_1(\varepsilon_2 + \cdots + \varepsilon_{j-1})} C_\beta(A_1, A_j) \langle A_2 \cdots \widehat{A_j} \cdots A_n \rangle_{\beta,0}. \end{aligned} \quad (17)$$

By the induction hypothesis, each lower-point thermal average is a sum over pairings of the remaining $n - 2$ operators. Adding the fixed pair $(1j)$ gives all pairings of the original n operators exactly once, with the correct graded sign. This proves (12). $\square$

The theorem immediately applies to free relativistic fields. For a real scalar field,

$$\phi(t, \mathbf{x}) = \int \frac{d^3\mathbf{k}}{(2\pi)^3} \frac{1}{\sqrt{2\omega_{\mathbf{k}}}} \left(a_{\mathbf{k}} e^{-i\omega_{\mathbf{k}}t + i\mathbf{k}\cdot\mathbf{x}} + a_{\mathbf{k}}^\dagger e^{i\omega_{\mathbf{k}}t - i\mathbf{k}\cdot\mathbf{x}} \right), \quad (18)$$

and every free thermal scalar correlator factorizes as

$$\langle \phi(x_1) \cdots \phi(x_n) \rangle_{\beta,0} = \sum_{\text{pairings } P} \prod_{(ij) \in P} \langle \phi(x_i) \phi(x_j) \rangle_{\beta,0}. \quad (19)$$

For a free Dirac field, the mode expansion is linear in the fermionic operators $b_s(\mathbf{p})$, $d_s(\mathbf{p})$, $b_s^\dagger(\mathbf{p})$ and $d_s^\dagger(\mathbf{p})$. Hence thermal correlators vanish unless the total fermion parity is even. In particular,

$$\langle \psi_{\alpha_1}(x_1) \cdots \psi_{\alpha_m}(x_m) \bar{\psi}_{\beta_1}(y_1) \cdots \bar{\psi}_{\beta_m}(y_m) \rangle_{\beta,0} = \det \left[\langle \psi_{\alpha_i}(x_i) \bar{\psi}_{\beta_j}(y_j) \rangle_{\beta,0} \right]_{i,j=1}^m, \quad (20)$$

up to the obvious generalization when the external fermionic fields are not written in the displayed order. The determinant is just the compact way of writing the signed sum over fermionic pairings.

We now pass to contour ordering. Let $\mathcal{C}$ be an oriented contour, with contour parameter λ increasing along the direction of the contour. We write

$$z_1 \succ_{\mathcal{C}} z_2 \iff z_1 \text{ is later than } z_2 \text{ on } \mathcal{C}. \quad (21)$$

The contour-ordering operation $T_{\mathcal{C}}$ places later contour points to the left. Whenever two fermionic fields are interchanged, one includes the corresponding minus sign. Thus, for fields $\Phi_i(z_i)$ of definite Grassmann parity,

$$T_{\mathcal{C}} \Phi_1(z_1) \cdots \Phi_n(z_n) = \eta(\pi) \Phi_{\pi(1)}(z_{\pi(1)}) \cdots \Phi_{\pi(n)}(z_{\pi(n)}), \quad (22)$$

where

$$z_{\pi(1)} \succ_{\mathcal{C}} z_{\pi(2)} \succ_{\mathcal{C}} \cdots \succ_{\mathcal{C}} z_{\pi(n)}, \quad (23)$$

and $\eta(\pi)$ is the fermionic sign of the permutation.

Theorem 1.3 (Contour Wick theorem). *For free fields in the Gaussian thermal state of H_0 ,*

$$\langle T_{\mathcal{C}} \Phi_1(z_1) \Phi_2(z_2) \cdots \Phi_n(z_n) \rangle_{\beta,0} = \sum_{\text{pairings } P} \sigma_{\mathcal{C}}(P) \prod_{(ij) \in P} \langle T_{\mathcal{C}} \Phi_i(z_i) \Phi_j(z_j) \rangle_{\beta,0}. \quad (24)$$

The sign $\sigma_{\mathcal{C}}(P)$ is the total fermionic sign obtained after contour ordering and then pairing.

Proof. This is the point which must be stated explicitly. The proof has two steps.

First choose the permutation π in (22) so that all arguments are already contour ordered as in (23). After this step there is no contour-ordering operation left to perform; we have an ordinary ordered string of free fields, multiplied by the sign $\eta(\pi)$.

Second, apply Gaudin's theorem to this ordinary ordered string. A term in Gaudin's expansion contracts two entries in positions $r < s$ of the already ordered product. The two-point factor produced by the proof is therefore

$$\langle \Phi_{\pi(r)}(z_{\pi(r)}) \Phi_{\pi(s)}(z_{\pi(s)}) \rangle_{\beta,0}. \quad (25)$$

But $z_{\pi(r)} \succ_{\mathcal{C}} z_{\pi(s)}$ by construction. Hence this pair is already in contour order. Therefore

$$\langle \Phi_{\pi(r)}(z_{\pi(r)}) \Phi_{\pi(s)}(z_{\pi(s)}) \rangle_{\beta,0} = \langle T_{\mathcal{C}} \Phi_{\pi(r)}(z_{\pi(r)}) \Phi_{\pi(s)}(z_{\pi(s)}) \rangle_{\beta,0}. \quad (26)$$

When the result is rewritten in terms of the original labels i, j , the sign from the first contour-ordering permutation and the signs from Gaudin's pairing combine into $\sigma_{\mathcal{C}}(P)$. This proves (24). $\square$

For a free real scalar field, (24) gives

$$\langle T_{\mathcal{C}} \phi(z_1) \cdots \phi(z_n) \rangle_{\beta,0} = \sum_{\text{pairings } P} \prod_{(ij) \in P} \langle T_{\mathcal{C}} \phi(z_i) \phi(z_j) \rangle_{\beta,0}. \quad (27)$$

There are no extra signs. For a free Dirac field,

$$\begin{aligned} & \langle T_{\mathcal{C}} \psi_{\alpha_1}(z_1) \cdots \psi_{\alpha_m}(z_m) \bar{\psi}_{\beta_1}(w_1) \cdots \bar{\psi}_{\beta_m}(w_m) \rangle_{\beta,0} \\ &= \sum_P \sigma_{\mathcal{C}}(P) \prod_{(ij) \in P} \langle T_{\mathcal{C}} \psi_{\alpha_i}(z_i) \bar{\psi}_{\beta_j}(w_j) \rangle_{\beta,0}. \end{aligned} \quad (28)$$

Equivalently, after one fixes an ordering of the external fermionic fields, this is the determinant of the matrix of contour-ordered Dirac two-point functions.

The same proof also gives the Euclidean thermal version. Introduce Euclidean time

$$\tau = it, \quad X(\tau) = e^{\tau H_0} X e^{-\tau H_0}, \quad 0 \leq \tau \leq \beta. \quad (29)$$

The Euclidean thermal contour C_E is the oriented segment

$$C_E: \quad z(\tau) = -i\tau, \quad 0 \leq \tau \leq \beta. \quad (30)$$

The Euclidean ordering operation T_{τ} , equivalently T_{C_E} , places larger Euclidean time τ to the left and includes a minus sign for each interchange of fermionic fields. The KMS relation says that bosonic operators are periodic and fermionic operators are antiperiodic around this thermal contour. Since T_{τ} first converts the product into an ordinary ordered string, the same two-step proof gives

$$\langle T_{\tau} X_1(\tau_1) X_2(\tau_2) \cdots X_n(\tau_n) \rangle_{\beta,0} = \sum_{\text{pairings } P} \sigma_{\tau}(P) \prod_{(ij) \in P} \langle T_{\tau} X_i(\tau_i) X_j(\tau_j) \rangle_{\beta,0}. \quad (31)$$

No additional argument is needed: after the initial Euclidean ordering permutation, every contracted pair is already Euclidean-time ordered.

The proof is equally valid on a mixed contour. The mixed thermal contour is obtained by joining real-time and Euclidean segments, for example

$$C_{\text{mix}} = C_+ \cup C_- \cup C_{\beta}, \quad (32)$$

where the real-time branches are followed by the vertical thermal branch. A point on C_{β} is later on the contour than any point on C_- , and a point on C_- is later than any point on C_+ . The same definition of $T_{C_{\text{mix}}}$ applies. Since the proof of (24) used only the initial contour-ordering permutation and Gaudin's algebraic recursion, it applies unchanged to this mixed contour.

1.2 Full Schwinger–Keldysh contour and far-past thermal preparation

Thermal real-time expectation values are in-in quantities. One evolves a state forward in time, inserts the desired operators, and then evolves back again in order to take a trace. At finite temperature there is one additional ingredient: the trace is weighted by the thermal density matrix. The most economical way to keep these two facts together is to use the full finite-temperature Schwinger–Keldysh contour

$$C = C_+ \cup C_- \cup C_{\beta}. \quad (33)$$

We take a finite return time t_f first and send it to $+\infty$ only after the formal manipulations. The three pieces are

$$C_+ : \quad z = t, \quad t_0 \leq t \leq t_f, \quad (34)$$

$$C_- : \quad z = t, \quad t_f \geq t \geq t_0, \quad (35)$$

$$C_{\beta} : \quad z = t_0 - i\tau, \quad 0 \leq \tau \leq \beta. \quad (36)$$

Thus

$$\int_{C_+} dz A(z) = \int_{t_0}^{t_f} dt A_+(t), \quad (37)$$

$$\int_{C_-} dz A(z) = \int_{t_f}^{t_0} dt A_-(t) = - \int_{t_0}^{t_f} dt A_-(t), \quad (38)$$

$$\int_{C_\beta} dz A(z) = -i \int_0^\beta d\tau A(t_0 - i\tau). \quad (39)$$

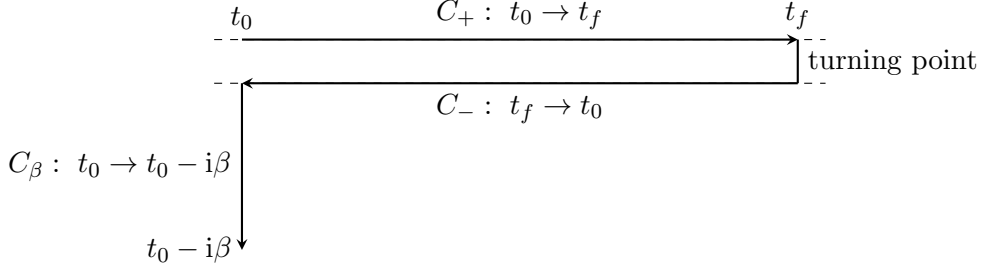


Figure 1: The full finite-temperature Schwinger–Keldysh contour. The branch C_+ gives the forward real-time evolution, C_- gives the inverse real-time evolution, and C_β represents the thermal trace. For the real-time perturbation theory used below the initial point is taken to the far past, $t_0 \rightarrow -\infty$, and real-time interactions and sources are switched on adiabatically.

Let

$$H = H_0 + H_I. \quad (40)$$

We shall distinguish carefully between two normalizations. The interacting thermal ensemble is normalized by

$$Z = \text{Tr } e^{-\beta H}, \quad (41)$$

whereas the free thermal Wick theorem uses

$$Z_0 = \text{Tr } e^{-\beta H_0}, \quad \langle X \rangle_{\beta,0} = \frac{1}{Z_0} \text{Tr } (e^{-\beta H_0} X). \quad (42)$$

This distinction is the source of many apparent normalization puzzles in the closed-time-path formalism.

Let $T_C \mathcal{O}$ denote a contour-ordered string of Heisenberg-picture local operators. The exact interacting thermal expectation value is

$$\langle T_C \mathcal{O} \rangle_\beta = \frac{1}{Z} \text{Tr } (e^{-\beta H} T_C \mathcal{O}). \quad (43)$$

In the interaction picture with respect to H_0 ,

$$e^{-\beta H} = e^{-\beta H_0} T_\tau \exp \left[- \int_0^\beta d\tau H_I(t_0 - i\tau) \right]. \quad (44)$$

Equivalently, using (39), the exponential on the right is

$$T_C \exp \left[-i \int_{C_\beta} dz H_I(z) \right]. \quad (45)$$

Including also the real-time evolution on C_+ and C_- gives the standard full-contour perturbative representation

$$\langle T_C \mathcal{O} \rangle_\beta = \frac{\langle T_C \mathcal{O} \exp [-i \int_C dz H_I(z)] \rangle_{\beta,0}}{\langle T_C \exp [-i \int_C dz H_I(z)] \rangle_{\beta,0}}. \quad (46)$$

This equation is still a representation of the interacting thermal state. At this stage the internal vertices are integrated over the whole contour

$$\mathcal{C} = C_+ \cup C_- \cup C_\beta. \quad (47)$$

The vertical branch has not disappeared; it is precisely the perturbative representation of the thermal Boltzmann factor.

It is useful to see the normalization first without dividing by Z_0 . Define the unnormalized numerator

$$\tilde{N}[\mathcal{O}] = \text{Tr} \left[e^{-\beta H_0} T_{\mathcal{C}} \left\{ \mathcal{O}_I \exp \left[-i \int_{\mathcal{C}} dz H_I(z) \right] \right\} \right]. \quad (48)$$

Then

$$\tilde{N}[\mathcal{O}] = Z \langle T_{\mathcal{C}} \mathcal{O} \rangle_\beta. \quad (49)$$

For $\mathcal{O} = 1$ one obtains the interacting partition function. Indeed, the real-time pieces of the contour give forward evolution followed by inverse evolution,

$$U_I(t_0, t_f) U_I(t_f, t_0) = 1, \quad (50)$$

so the vacuum evolution on $C_+ \cup C_-$ cancels. The nontrivial normalization is therefore carried only by the vertical branch:

$$\tilde{N}[1] = \text{Tr} \left[e^{-\beta H_0} T_{C_\beta} \exp \left[-i \int_{C_\beta} dz H_I(z) \right] \right] = Z. \quad (51)$$

Equivalently, in the normalized free notation of (42),

$$\begin{aligned} \left\langle T_{\mathcal{C}} \exp \left[-i \int_{\mathcal{C}} dz H_I(z) \right] \right\rangle_{\beta,0} &= \frac{1}{Z_0} \text{Tr} \left[e^{-\beta H_0} T_{C_\beta} \exp \left[-i \int_{C_\beta} dz H_I(z) \right] \right] \\ &= \frac{Z}{Z_0} = \exp(V_{\text{vac}}[C_\beta]). \end{aligned} \quad (52)$$

The notation $V_{\text{vac}}[C_\beta]$ emphasizes that, in the denominator, the real-time vacuum bubbles cancel between C_+ and C_- . The remaining vacuum factor is the imaginary-branch perturbative correction to the partition function.

Now consider $\tilde{N}[\mathcal{O}]$ with operator insertions on the real-time branches. Its perturbative expansion contains two kinds of factors. First, there are vacuum subdiagrams which do not attach to $\mathcal{O}_I$; as above, their nontrivial part is the vertical thermal factor Z . Second, there are diagrams connected to the real-time operator string $\mathcal{O}_I$. Before the far-past limit, such connected diagrams may still contain vertices on C_β : a vertical vertex can be attached to a real-time insertion by an ordinary free thermal contraction. For a free mode, a mixed line contains factors of the form

$$e^{-i\omega(t-t_0)} e^{-\omega\tau}, \quad (53)$$

up to the usual thermal occupation factors. Thus the full contour genuinely describes the interacting thermal density matrix.

The real-time Schwinger–Keldysh perturbation theory used in practice is obtained after far-past preparation. The interaction and the sources on the real-time branches are switched on adiabatically, and the initial point is sent to the remote past:

$$H_I(t) \longrightarrow H_I^\epsilon(t) = e^{-\epsilon|t|} H_I(t), \quad \epsilon > 0, \quad (54)$$

with $t_0 \rightarrow -\infty$ taken before the regulator is removed. The point is not that the operator H_I on C_β is set to zero. Rather, any connected diagram in which a vertical-branch vertex remains

attached to a finite-time real-time insertion contains at least one propagator stretching from $t_0 - i\tau$ to a finite real time. With the adiabatic prescription, and equivalently by the usual cluster property of the initial state, such mixed connected contributions vanish in the far-past limit. Hence the vertical branch can no longer remain connected to the real-time operator insertion. It factorizes.

The factorization has to be written with the normalizations visible. In unnormalized notation it gives

$$\tilde{N}[\mathcal{O}] \longrightarrow Z \frac{1}{Z_0} \text{Tr} \left[e^{-\beta H_0} T_{\mathcal{C}_R} \left\{ \mathcal{O}_I \exp \left[-i \int_{\mathcal{C}_R} dz H_I(z) \right] \right\} \right], \quad \mathcal{C}_R = C_+ \cup C_-. \quad (55)$$

The first factor, Z , is the interacting thermal normalization produced by the imaginary branch. The second factor is a real-time expectation value in the normalized free thermal ensemble. The factor $1/Z_0$ is therefore not an additional vacuum-bubble denominator. It is simply the normalization of the free density matrix with respect to which the real-time Wick theorem is applied.

Dividing (55) by (49), the factor Z cancels and one obtains the real-time SK formula

$$\langle T_{\mathcal{C}_R} \mathcal{O} \rangle_\beta = \frac{1}{Z_0} \text{Tr} \left[e^{-\beta H_0} T_{\mathcal{C}_R} \left\{ \mathcal{O}_I \exp \left[-i \int_{\mathcal{C}_R} dz H_I(z) \right] \right\} \right]. \quad (56)$$

Equivalently,

$$\langle T_{\mathcal{C}_R} \mathcal{O} \rangle_\beta = \left\langle T_{\mathcal{C}_R} \left\{ \mathcal{O}_I \exp \left[-i \int_{\mathcal{C}_R} dz H_I(z) \right] \right\} \right\rangle_{\beta,0}. \quad (57)$$

This is the precise meaning of real-time perturbation theory with a free thermal initial state: the vertical branch has not been thrown away. It has supplied the interacting normalization Z , which cancels against the exact thermal normalization, and it has left behind the free density matrix

$$\rho_0 = \frac{e^{-\beta H_0}}{Z_0} \quad (58)$$

for the real-time Wick contractions. All interaction vertices which remain connected to the real-time observable are integrated only over $\mathcal{C}_R$.

The same statement can be expressed diagrammatically. After far-past preparation, the expansion of (57) is built from real-time vertices on C_+ and C_- only. Vacuum subdiagrams on the real-time closed contour sum to unity because forward and backward evolution cancel. Consequently one may compute (57) by retaining only diagrams connected to at least one operator in $\mathcal{O}$:

$$\langle T_{\mathcal{C}_R} \mathcal{O} \rangle_\beta = \left[\begin{array}{c} \text{sum of diagrams connected to } \mathcal{O}, \\ \text{with all internal vertices integrated only over } \mathcal{C}_R = C_+ \cup C_- \end{array} \right]. \quad (59)$$

This final restriction to $\mathcal{C}_R$ is a consequence of far-past preparation and factorization. It is not a property of the full contour before that limit is taken.

It is useful to state the corresponding generating functional. For a scalar source on the real-time branches,

$$\mathcal{Z}[J_+, J_-] = \left\langle T_{\mathcal{C}_R} \exp \left[i \int_{\mathcal{C}_R} d^4x J(x) \phi(x) \right] \exp \left[-i \int_{\mathcal{C}_R} dz H_I(z) \right] \right\rangle_{\beta,0}. \quad (60)$$

The compact contour source integral means

$$\int_{\mathcal{C}_R} d^4x J \phi = \int_{t_0}^{t_f} d^4x J_+(x) \phi_+(x) - \int_{t_0}^{t_f} d^4x J_-(x) \phi_-(x). \quad (61)$$

The general SK unitarity identity is

$$\mathcal{Z}[J, J] = 1, \quad \mathcal{W}[J, J] := -i \log \mathcal{Z}[J, J] = 0. \quad (62)$$

When $J_+ = J_- = J$, the source-deformed evolution on C_+ is followed by the inverse source-deformed evolution on C_- . The operator product inside the trace is therefore the identity, and the normalized free trace of the identity is one. Thus the vanishing of the connected generating functional on the physical line $J_+ = J_-$ is a consequence of unitarity and of the normalization by Z_0 , not an additional diagrammatic rule.

Correlation functions are obtained by differentiating (60) with respect to J_+ and J_- . The four real-time two-point functions are the restrictions of one contour-ordered object. For the free theory,

$$G^{ab}(x, y) = -i \langle T_{C_R} \phi_a(x) \phi_b(y) \rangle_{\beta, 0}, \quad a, b = +, -. \quad (63)$$

Thus

$$G^{++} = G_F^\beta, \quad G^{--} = G_{\bar{F}}^\beta, \quad G^{-+} = G^>, \quad G^{+-} = G^<. \quad (64)$$

The retarded function, spectral function, self-energies, damping rates, conductivities and susceptibilities are all obtained from these real-time correlators. This is why the contour formulation is the natural continuation of Gaudin's finite-temperature Wick theorem into finite-temperature dynamics.

2 Free Schwinger–Keldysh propagators

The Wick theorem of Section 1 reduces perturbative real-time finite-temperature field theory to the computation of free two-point functions on the contour. In this section we compute these two-point functions directly from the operator algebra. No path integral argument is used. The logic is always the same:

- (i) solve the free Heisenberg equations for the creation and annihilation operators;
- (ii) evaluate the thermal expectation values using the oscillator density matrix;
- (iii) assemble the four Schwinger–Keldysh components by contour ordering;
- (iv) Fourier transform and identify the pole prescriptions.

Throughout this section the real-time contour has two real branches, C_+ and C_- . A point on C_+ is denoted t_+ , and a point on C_- is denoted t_- . The contour runs from the initial time to the final time along C_+ and then returns along C_- . Therefore every point on C_- is later on the contour than every point on C_+ . For a bosonic operator X , we define

$$G_X^{ab}(t, t') = \langle T_{C_R} X(t_a) X(t'_b) \rangle_{\beta, 0}, \quad a, b = +, -. \quad (65)$$

For a fermionic annihilation operator f , we define

$$g^{ab}(t, t') = \langle T_{C_R} f(t_a) f^\dagger(t'_b) \rangle_{\beta, 0}. \quad (66)$$

The minus signs in the fermionic case are not put in by hand. They are produced by contour ordering.

2.1 Bosonic and fermionic quantum harmonic oscillators

This subsection is deliberately detailed because every free relativistic field is a continuum of oscillators. The scalar field is a continuum of bosonic oscillators, the Dirac field is a continuum of fermionic oscillators, and the free gauge field in a fixed gauge is again a collection of bosonic oscillators with tensor structure. We therefore begin with one bosonic oscillator and one fermionic oscillator.

Bosonic oscillator in real time

Let

$$H_0 = \omega \left(a^\dagger a + \frac{1}{2} \right), \quad [a, a^\dagger] = 1, \quad q(t) = \frac{1}{\sqrt{2\omega}} \left(a e^{-i\omega t} + a^\dagger e^{i\omega t} \right). \quad (67)$$

The thermal occupation number is

$$n_B(\omega) = \frac{1}{e^{\beta\omega} - 1}, \quad \langle a^\dagger a \rangle_{\beta,0} = n_B(\omega), \quad \langle a a^\dagger \rangle_{\beta,0} = 1 + n_B(\omega). \quad (68)$$

By time-translation invariance the propagators depend only on $t - t'$. We set $t' = 0$ and write t for $t - t'$. Direct use of (67) and (68) gives

$$G^>(t) := \langle q(t)q(0) \rangle_{\beta,0} = \frac{1}{2\omega} \left[(1 + n_B) e^{-i\omega t} + n_B e^{i\omega t} \right], \quad (69)$$

$$G^<(t) := \langle q(0)q(t) \rangle_{\beta,0} = \frac{1}{2\omega} \left[n_B e^{-i\omega t} + (1 + n_B) e^{i\omega t} \right], \quad (70)$$

where $n_B = n_B(\omega)$. Contour ordering gives the four real-time components

$$G^{++}(t) = \theta(t)G^>(t) + \theta(-t)G^<(t), \quad (71)$$

$$G^{--}(t) = \theta(-t)G^>(t) + \theta(t)G^<(t), \quad (72)$$

$$G^{-+}(t) = G^>(t), \quad G^{+-}(t) = G^<(t). \quad (73)$$

The $++$ component is the time-ordered correlator, the $--$ component is the anti-time-ordered correlator, and the off-diagonal components are the two Wightman functions.

The Fourier convention is

$$G(t) = \int_{-\infty}^{\infty} \frac{dk_0}{2\pi} e^{-ik_0 t} G(k_0), \quad G(k_0) = \int_{-\infty}^{\infty} dt e^{ik_0 t} G(t). \quad (74)$$

The Wightman functions contain no step functions, so their Fourier transforms are immediate:

$$G^{-+}(k_0) = \frac{\pi}{\omega} \left[(1 + n_B) \delta(k_0 - \omega) + n_B \delta(k_0 + \omega) \right], \quad (75)$$

$$G^{+-}(k_0) = \frac{\pi}{\omega} \left[n_B \delta(k_0 - \omega) + (1 + n_B) \delta(k_0 + \omega) \right]. \quad (76)$$

The diagonal functions contain the pole prescriptions. We use

$$\int_0^{\infty} dt e^{ixt} = \frac{i}{x + i0^+}, \quad \int_{-\infty}^0 dt e^{ixt} = -\frac{i}{x - i0^+}. \quad (77)$$

Let us spell out the substitution, because it is the origin of the Feynman and anti-Feynman prescriptions. From (71),

$$\begin{aligned} G^{++}(k_0) &= \frac{1 + n_B}{2\omega} \frac{i}{k_0 - \omega + i0^+} + \frac{n_B}{2\omega} \frac{i}{k_0 + \omega + i0^+} - \frac{n_B}{2\omega} \frac{i}{k_0 - \omega - i0^+} - \frac{1 + n_B}{2\omega} \frac{i}{k_0 + \omega - i0^+} \\ &= \frac{1}{2\omega} \left(\frac{i}{k_0 - \omega + i0^+} - \frac{i}{k_0 + \omega - i0^+} \right) + \frac{n_B}{2\omega} \left[\frac{i}{k_0 - \omega + i0^+} - \frac{i}{k_0 - \omega - i0^+} \right. \\ &\quad \left. + \frac{i}{k_0 + \omega + i0^+} - \frac{i}{k_0 + \omega - i0^+} \right]. \end{aligned}$$

Using

$$\frac{1}{x + i0^+} - \frac{1}{x - i0^+} = -2\pi i \delta(x)$$

and

$$\frac{1}{2\omega} \left(\frac{1}{k_0 - \omega + i0^+} - \frac{1}{k_0 + \omega - i0^+} \right) = \frac{1}{k_0^2 - \omega^2 + i0^+},$$

we obtain

$$G^{++}(k_0) = \frac{i}{k_0^2 - \omega^2 + i0^+} + 2\pi n_B(\omega)\delta(k_0^2 - \omega^2). \quad (78)$$

Similarly, using (77) in (72),

$$\begin{aligned} G^{--}(k_0) &= -\frac{1+n_B}{2\omega} \frac{i}{k_0 - \omega - i0^+} - \frac{n_B}{2\omega} \frac{i}{k_0 + \omega - i0^+} \\ &\quad + \frac{n_B}{2\omega} \frac{i}{k_0 - \omega + i0^+} + \frac{1+n_B}{2\omega} \frac{i}{k_0 + \omega + i0^+} \\ &= -\frac{i}{k_0^2 - \omega^2 - i0^+} + 2\pi n_B(\omega)\delta(k_0^2 - \omega^2). \end{aligned}$$

Thus

$$G^{--}(k_0) = -\frac{i}{k_0^2 - \omega^2 - i0^+} + 2\pi n_B(\omega)\delta(k_0^2 - \omega^2). \quad (79)$$

Equations (78) and (79) show the useful separation into a Feynman-like vacuum pole part and an on-shell thermal correction. The zero-temperature limit follows immediately because

$$\lim_{\beta \rightarrow \infty} n_B(\omega) = 0. \quad (80)$$

Therefore

$$G_0^{++}(k_0) = \frac{i}{k_0^2 - \omega^2 + i0^+}, \quad G_0^{--}(k_0) = -\frac{i}{k_0^2 - \omega^2 - i0^+}, \quad (81)$$

$$G_0^{-+}(k_0) = 2\pi\theta(k_0)\delta(k_0^2 - \omega^2), \quad G_0^{+-}(k_0) = 2\pi\theta(-k_0)\delta(k_0^2 - \omega^2). \quad (82)$$

The $i\varepsilon$ prescription in the Feynman part means

$$\frac{i}{k_0^2 - \omega^2 + i0^+} = \frac{i}{(k_0 - \omega + i0^+)(k_0 + \omega - i0^+)}, \quad (83)$$

so the positive-energy pole is just below the real axis and the negative-energy pole is just above it. For $t > 0$ the contour is closed in the lower half-plane and picks the positive-energy pole; for $t < 0$ it is closed in the upper half-plane and picks the negative-energy pole. This gives the contour integral

$$G_0^{++}(t) = \frac{1}{2\omega} e^{-i\omega|t|} = \int_{-\infty}^{\infty} \frac{dk_0}{2\pi} e^{-ik_0 t} \frac{i}{k_0^2 - \omega^2 + i0^+}. \quad (84)$$

For the anti-time-ordered part,

$$-\frac{i}{k_0^2 - \omega^2 - i0^+} = -\frac{i}{(k_0 - \omega - i0^+)(k_0 + \omega + i0^+)}, \quad (85)$$

so the pole locations are reflected.

The Wick-rotation trick is this same contour statement in imaginary time. Since no Feynman pole lies on the imaginary k_0 axis, for $t = -i\tau$ with $\tau > 0$ one may rotate the contour to $k_0 = ik_E$. This gives

$$G_0^{++}(-i\tau) = \int_{-\infty}^{\infty} \frac{dk_E}{2\pi} \frac{e^{ik_E \tau}}{k_E^2 + \omega^2}, \quad \tau > 0. \quad (86)$$

At finite temperature imaginary time is compact, and the integral is replaced by a Fourier series over Matsubara frequencies.

It is also useful to introduce the retarded, advanced, and spectral functions. With the present convention, in which no extra factor of $-i$ is included in the definition of the two-point function, define

$$\rho_B(t) = G^>(t) - G^<(t) = \langle [q(t), q(0)] \rangle_{\beta,0}, \quad G_R(t) = \theta(t)\rho_B(t), \quad G_A(t) = -\theta(-t)\rho_B(t). \quad (87)$$

The thermal factors cancel in the commutator:

$$\rho_B(t) = \frac{1}{2\omega} (e^{-i\omega t} - e^{i\omega t}). \quad (88)$$

Therefore the spectral function is temperature independent:

$$\rho_B(k_0) = 2\pi \operatorname{sgn}(k_0) \delta(k_0^2 - \omega^2) = \frac{\pi}{\omega} [\delta(k_0 - \omega) - \delta(k_0 + \omega)]. \quad (89)$$

The retarded and advanced functions are also temperature independent:

$$G_R(k_0) = \frac{i}{(k_0 + i0^+)^2 - \omega^2}, \quad G_A(k_0) = \frac{i}{(k_0 - i0^+)^2 - \omega^2}. \quad (90)$$

Temperature changes the Wightman functions, but it does not change the free commutator. The occupation numbers only tell us how the two spectral weights are populated.

Fermionic oscillator in real time

Now take a single fermionic oscillator

$$H_0 = \omega f^\dagger f, \quad \{f, f^\dagger\} = 1, \quad f(t) = f e^{-i\omega t}. \quad (91)$$

The thermal occupation number is

$$n_F(\omega) = \frac{1}{e^{\beta\omega} + 1}, \quad \langle f^\dagger f \rangle_{\beta,0} = n_F(\omega), \quad \langle f f^\dagger \rangle_{\beta,0} = 1 - n_F(\omega). \quad (92)$$

The greater and lesser functions are defined with the sign produced by fermionic contour ordering:

$$g^>(t) := \langle f(t) f^\dagger(0) \rangle_{\beta,0} = (1 - n_F) e^{-i\omega t}, \quad (93)$$

$$g^<(t) := -\langle f^\dagger(0) f(t) \rangle_{\beta,0} = -n_F e^{-i\omega t}. \quad (94)$$

Then

$$g^{++}(t) = \theta(t) g^>(t) + \theta(-t) g^<(t), \quad (95)$$

$$g^{--}(t) = \theta(-t) g^>(t) + \theta(t) g^<(t), \quad (96)$$

$$g^{-+}(t) = g^>(t), \quad g^{+-}(t) = g^<(t). \quad (97)$$

Fourier transforming gives

$$g^{-+}(k_0) = 2\pi(1 - n_F) \delta(k_0 - \omega), \quad (98)$$

$$g^{+-}(k_0) = -2\pi n_F \delta(k_0 - \omega), \quad (99)$$

$$g^{++}(k_0) = \frac{i}{k_0 - \omega + i0^+} - 2\pi n_F(\omega) \delta(k_0 - \omega), \quad (100)$$

$$g^{--}(k_0) = -\frac{i}{k_0 - \omega - i0^+} - 2\pi n_F(\omega) \delta(k_0 - \omega). \quad (101)$$

The diagonal components again contain a Feynman-like pole part and an on-shell thermal correction. The correction has the Pauli-blocking sign. Since

$$\lim_{\beta \rightarrow \infty} n_F(\omega) = 0 \quad (102)$$

for a positive-energy oscillator, the zero-temperature limit is

$$g_0^{++}(k_0) = \frac{i}{k_0 - \omega + i0^+}, \quad g_0^{--}(k_0) = -\frac{i}{k_0 - \omega - i0^+}, \quad (103)$$

$$g_0^{-+}(k_0) = 2\pi \delta(k_0 - \omega), \quad g_0^{+-}(k_0) = 0. \quad (104)$$

The corresponding contour integrals are

$$g_0^{++}(t) = \theta(t)e^{-i\omega t} = \int_{-\infty}^{\infty} \frac{dk_0}{2\pi} e^{-ik_0 t} \frac{i}{k_0 - \omega + i0^+}, \quad (105)$$

$$g_0^{--}(t) = \theta(-t)e^{-i\omega t} = \int_{-\infty}^{\infty} \frac{dk_0}{2\pi} e^{-ik_0 t} \left(-\frac{i}{k_0 - \omega - i0^+} \right). \quad (106)$$

Thus the Feynman pole is below the real axis and contributes when the contour is closed downward for $t > 0$. The anti-Feynman pole is above the real axis and contributes when the contour is closed upward for $t < 0$. There is no negative-frequency pole because this elementary oscillator only annihilates one species. Negative-frequency poles appear in relativistic Dirac theory because the field contains both particle annihilation and antiparticle creation operators.

The fermionic spectral function is the thermal expectation value of the anticommutator, with the sign convention inherited from $g^<$:

$$\rho_F(t) = g^>(t) - g^<(t) = \left\langle \{f(t), f^\dagger(0)\} \right\rangle_{\beta,0} = e^{-i\omega t}. \quad (107)$$

Again the occupation number cancels. Hence

$$\rho_F(k_0) = 2\pi\delta(k_0 - \omega), \quad g_R(k_0) = \frac{i}{k_0 - \omega + i0^+}, \quad g_A(k_0) = \frac{i}{k_0 - \omega - i0^+}. \quad (108)$$

The free fermionic spectral function, retarded propagator, and advanced propagator are temperature independent. Temperature only changes the statistical weights in $g^>$ and $g^<$.

Euclidean oscillator propagators

The Euclidean operator is obtained by imaginary-time evolution with the Hamiltonian:

$$X(\tau) = e^{\tau H_0} X e^{-\tau H_0}, \quad 0 \leq \tau \leq \beta. \quad (109)$$

For the bosonic oscillator,

$$q(\tau) = \frac{1}{\sqrt{2\omega}} \left(a e^{-\omega\tau} + a^\dagger e^{\omega\tau} \right), \quad (110)$$

and the Euclidean time-ordered propagator is

$$G_E(\tau) = \langle T_\tau q(\tau) q(0) \rangle_{\beta,0}. \quad (111)$$

For $0 < \tau < \beta$, no reordering is needed, so

$$\begin{aligned} G_E(\tau) &= \frac{1}{2\omega} \left\langle \left(a e^{-\omega\tau} + a^\dagger e^{\omega\tau} \right) (a + a^\dagger) \right\rangle_{\beta,0} \\ &= \frac{1}{2\omega} \left(\left\langle a a^\dagger \right\rangle_{\beta,0} e^{-\omega\tau} + \left\langle a^\dagger a \right\rangle_{\beta,0} e^{\omega\tau} \right). \end{aligned}$$

Using (68), this gives

$$G_E(\tau) = \frac{1}{2\omega} \left[(1 + n_B) e^{-\omega\tau} + n_B e^{\omega\tau} \right], \quad 0 < \tau < \beta. \quad (112)$$

The bosonic KMS relation follows directly from cyclicity of the trace:

$$\begin{aligned} G_E(\tau + \beta) &= Z_0^{-1} \text{Tr} \left(e^{-\beta H_0} q(\tau + \beta) q(0) \right) \\ &= Z_0^{-1} \text{Tr} \left(q(0) e^{-\beta H_0} q(\tau) \right) = Z_0^{-1} \text{Tr} \left(e^{-\beta H_0} q(\tau) q(0) \right). \end{aligned}$$

Thus

$$G_E(\tau + \beta) = G_E(\tau). \quad (113)$$

Equivalently, in the explicit expression (112), periodicity uses $n_B e^{\beta\omega} = 1 + n_B$. The Fourier series is therefore over bosonic Matsubara frequencies

$$\omega_n^B = \frac{2\pi n}{\beta}, \quad n \in \mathbb{Z}. \quad (114)$$

The Fourier coefficient is derived by direct integration:

$$\begin{aligned} G_E(i\omega_n^B) &= \int_0^\beta d\tau e^{i\omega_n^B \tau} G_E(\tau) \\ &= \frac{1}{2\omega} \left[(1 + n_B) \frac{e^{(i\omega_n^B - \omega)\beta} - 1}{i\omega_n^B - \omega} + n_B \frac{e^{(i\omega_n^B + \omega)\beta} - 1}{i\omega_n^B + \omega} \right] \\ &= \frac{1}{2\omega} \left[(1 + n_B) \frac{e^{-\beta\omega} - 1}{i\omega_n^B - \omega} + n_B \frac{e^{\beta\omega} - 1}{i\omega_n^B + \omega} \right]. \end{aligned}$$

Since $(1 + n_B)(e^{-\beta\omega} - 1) = -1$ and $n_B(e^{\beta\omega} - 1) = 1$, this becomes

$$G_E(i\omega_n^B) = \frac{1}{2\omega} \left[-\frac{1}{i\omega_n^B - \omega} + \frac{1}{i\omega_n^B + \omega} \right] = \frac{1}{(\omega_n^B)^2 + \omega^2}.$$

Therefore

$$G_E(\tau) = \frac{1}{\beta} \sum_{n \in \mathbb{Z}} e^{-i\omega_n^B \tau} G_E(i\omega_n^B), \quad G_E(i\omega_n^B) = \frac{1}{(\omega_n^B)^2 + \omega^2}. \quad (115)$$

For the fermionic oscillator,

$$f(\tau) = f e^{-\omega\tau}, \quad f^\dagger(\tau) = f^\dagger e^{\omega\tau}. \quad (116)$$

We define the raw Euclidean operator correlator by

$$g_E(\tau) = \left\langle T_\tau f(\tau) f^\dagger(0) \right\rangle_{\beta,0}. \quad (117)$$

For $0 < \tau < \beta$,

$$g_E(\tau) = \left\langle f(\tau) f^\dagger(0) \right\rangle_{\beta,0} = (1 - n_F) e^{-\omega\tau}. \quad (118)$$

This is the fermionic analogue of (112). For $-\beta < \tau < 0$, Euclidean time ordering moves $f^\dagger(0)$ past $f(\tau)$ and produces a minus sign:

$$g_E(\tau) = - \left\langle f^\dagger(0) f(\tau) \right\rangle_{\beta,0} = -n_F e^{-\omega\tau}. \quad (119)$$

The fermionic KMS relation also follows from cyclicity, but now cyclically moving one fermionic operator through the trace gives the graded sign:

$$g_E(\tau + \beta) = -g_E(\tau). \quad (120)$$

Equivalently, using (118), antiperiodicity is just $(1 - n_F) e^{-\beta\omega} = n_F$ together with the minus sign from fermionic ordering. The Fourier series uses fermionic Matsubara frequencies

$$\omega_n^F = \frac{(2n + 1)\pi}{\beta}, \quad n \in \mathbb{Z}. \quad (121)$$

The coefficient is again obtained by integration over $0 < \tau < \beta$:

$$\begin{aligned} g_E(i\omega_n^F) &= \int_0^\beta d\tau e^{i\omega_n^F \tau} (1 - n_F) e^{-\omega \tau} \\ &= (1 - n_F) \frac{e^{(i\omega_n^F - \omega)\beta} - 1}{i\omega_n^F - \omega} = (1 - n_F) \frac{-e^{-\beta\omega} - 1}{i\omega_n^F - \omega} = -\frac{1}{i\omega_n^F - \omega}. \end{aligned}$$

Thus

$$g_E(\tau) = \frac{1}{\beta} \sum_{n \in \mathbb{Z}} e^{-i\omega_n^F \tau} g_E(i\omega_n^F), \quad g_E(i\omega_n^F) = -\frac{1}{i\omega_n^F - \omega}. \quad (122)$$

Many books instead define the Euclidean fermion Green function with an extra minus sign,

$$\mathfrak{g}_E(\tau) := -g_E(\tau), \quad \mathfrak{g}_E(i\omega_n^F) = \frac{1}{i\omega_n^F - \omega}. \quad (123)$$

Both conventions contain the same operator algebra. One only has to keep the convention fixed when analytically continuing to real time.

Finally let us make precise the analytic continuation between the Euclidean and Feynman correlators. For $0 < \tau < \beta$,

$$G_E(\tau) = G^>(-i\tau) = G^{++}(t = -i\tau), \quad (124)$$

because the point at $t = -i\tau$ is reached by continuing the $t > 0$ time-ordered branch. Substituting $t = -i\tau$ into (69) gives exactly (112). The interval $-\beta < \tau < 0$ is obtained from the other ordering and is related to the first by KMS periodicity. For the fermionic oscillator the same argument gives

$$g_E(\tau) = g^>(-i\tau) = g^{++}(t = -i\tau), \quad 0 < \tau < \beta, \quad (125)$$

and the negative- τ branch follows from antiperiodicity.

In frequency space the cleanest statement is obtained from the pole part. For the bosonic oscillator,

$$G_0^{++}(k_0) = -i G_E(k_E) \Big|_{k_E^2 = -(k_0^2 + i0^+)} = \frac{i}{k_0^2 - \omega^2 + i0^+}. \quad (126)$$

This is exactly the Wick-rotation relation: the Euclidean denominator $k_E^2 + \omega^2$ becomes the Feynman denominator $-(k_0^2 - \omega^2 + i0^+)$. For the fermionic oscillator, with the conventional Euclidean propagator $\mathfrak{g}_E$, one has

$$g_0^{++}(k_0) = i \mathfrak{g}_E(i\omega_n^F) \Big|_{i\omega_n^F \rightarrow k_0 + i0^+} = \frac{i}{k_0 - \omega + i0^+}. \quad (127)$$

At finite temperature the time-domain analytic continuation (124) and (125) remains true. In frequency space the Matsubara correlator gives the analytic function away from the real axis, while the real-time $++$ correlator is the corresponding Feynman boundary value together with the on-shell thermal term fixed by the KMS relation. Thus the Euclidean correlator, the spectral function, and KMS data together reconstruct the finite-temperature Feynman correlator.

2.2 Thermal free scalar field

Consider a real scalar field with

$$H_0 = \int \frac{d^3 \mathbf{p}}{(2\pi)^3} E_{\mathbf{p}} a_{\mathbf{p}}^\dagger a_{\mathbf{p}}, \quad E_{\mathbf{p}} = \sqrt{\mathbf{p}^2 + m^2}, \quad (128)$$

where the zero-point energy has been omitted. The mode expansion is

$$\phi(x) = \int \frac{d^3 \mathbf{p}}{(2\pi)^3} \frac{1}{\sqrt{2E_{\mathbf{p}}}} \left(a_{\mathbf{p}} e^{-iE_{\mathbf{p}}t + i\mathbf{p} \cdot \mathbf{x}} + a_{\mathbf{p}}^\dagger e^{iE_{\mathbf{p}}t - i\mathbf{p} \cdot \mathbf{x}} \right), \quad (129)$$

with

$$[a_{\mathbf{p}}, a_{\mathbf{q}}^\dagger] = (2\pi)^3 \delta^3(\mathbf{p} - \mathbf{q}), \quad \left\langle a_{\mathbf{p}}^\dagger a_{\mathbf{q}} \right\rangle_{\beta,0} = (2\pi)^3 \delta^3(\mathbf{p} - \mathbf{q}) n_B(E_{\mathbf{p}}). \quad (130)$$

Each momentum mode is just a bosonic oscillator. Therefore

$$G^>(x) = \langle \phi(x) \phi(0) \rangle_{\beta,0} = \int \frac{d^3\mathbf{p}}{(2\pi)^3} \frac{1}{2E_{\mathbf{p}}} [(1 + n_B(E_{\mathbf{p}})) e^{-iE_{\mathbf{p}}t + i\mathbf{p}\cdot\mathbf{x}} + n_B(E_{\mathbf{p}}) e^{iE_{\mathbf{p}}t - i\mathbf{p}\cdot\mathbf{x}}], \quad (131)$$

$$G^<(x) = \langle \phi(0) \phi(x) \rangle_{\beta,0} = \int \frac{d^3\mathbf{p}}{(2\pi)^3} \frac{1}{2E_{\mathbf{p}}} [n_B(E_{\mathbf{p}}) e^{-iE_{\mathbf{p}}t + i\mathbf{p}\cdot\mathbf{x}} + (1 + n_B(E_{\mathbf{p}})) e^{iE_{\mathbf{p}}t - i\mathbf{p}\cdot\mathbf{x}}]. \quad (132)$$

The four Schwinger–Keldysh components are

$$G^{++} = \theta(t)G^> + \theta(-t)G^<, \quad G^{--} = \theta(-t)G^> + \theta(t)G^<, \quad G^{-+} = G^>, \quad G^{+-} = G^<. \quad (133)$$

Let us derive the first momentum-space formula explicitly. Substituting (131) and (132) into the first relation in (133) gives

$$G^{++}(t, \mathbf{x}) = \int \frac{d^3\mathbf{p}}{(2\pi)^3} \frac{1}{2E_{\mathbf{p}}} \left\{ \theta(t) [(1 + n_B(E_{\mathbf{p}})) e^{-iE_{\mathbf{p}}t + i\mathbf{p}\cdot\mathbf{x}} + n_B(E_{\mathbf{p}}) e^{iE_{\mathbf{p}}t - i\mathbf{p}\cdot\mathbf{x}}] \right. \\ \left. + \theta(-t) [n_B(E_{\mathbf{p}}) e^{-iE_{\mathbf{p}}t + i\mathbf{p}\cdot\mathbf{x}} + (1 + n_B(E_{\mathbf{p}})) e^{iE_{\mathbf{p}}t - i\mathbf{p}\cdot\mathbf{x}}] \right\}. \quad (134)$$

Fourier transforming the terms with step functions uses exactly the same contour integrals as for the oscillator,

$$\int_{-\infty}^{\infty} dt \theta(t) e^{iqt} = \frac{i}{q + i0^+}, \quad \int_{-\infty}^{\infty} dt \theta(-t) e^{iqt} = -\frac{i}{q - i0^+}. \quad (135)$$

The vacuum pieces therefore combine as

$$\frac{1}{2E_{\mathbf{k}}} \left(\frac{i}{k_0 - E_{\mathbf{k}} + i0^+} - \frac{i}{k_0 + E_{\mathbf{k}} - i0^+} \right) = \frac{i}{k_0^2 - E_{\mathbf{k}}^2 + i0^+}, \quad (136)$$

where the compact symbol $i0^+$ means that the positive-energy pole lies below the real k_0 axis and the negative-energy pole lies above it. The thermal pieces are not time ordered in the same way; they multiply the full oscillatory factors and give the on-shell distributional contribution

$$\frac{2\pi n_B(E_{\mathbf{k}})}{2E_{\mathbf{k}}} [\delta(k_0 - E_{\mathbf{k}}) + \delta(k_0 + E_{\mathbf{k}})] = 2\pi n_B(|k_0|) \delta(k^2 - m^2). \quad (137)$$

Combining (136) and (137) gives the first line below. The other three components are derived in the same way. Relative to the single oscillator, the only new input is the thermal covariance (130), which fixes the occupation number multiplying each positive- or negative-frequency mode. In momentum space,

$$G^{++}(k) = \frac{i}{k^2 - m^2 + i0^+} + 2\pi n_B(|k_0|) \delta(k^2 - m^2), \quad (138)$$

$$G^{--}(k) = -\frac{i}{k^2 - m^2 - i0^+} + 2\pi n_B(|k_0|) \delta(k^2 - m^2), \quad (139)$$

$$G^{-+}(k) = 2\pi [\theta(k_0) + n_B(|k_0|)] \delta(k^2 - m^2), \quad (140)$$

$$G^{+-}(k) = 2\pi [\theta(-k_0) + n_B(|k_0|)] \delta(k^2 - m^2). \quad (141)$$

Here $k^2 = k_0^2 - \mathbf{k}^2$. The $i0^+$ prescription in the first two lines is inherited from the time-ordering and anti-time-ordering step exactly as in the oscillator.

The scalar spectral function and retarded/advanced functions are the continuum versions of the oscillator formulas:

$$\begin{aligned}\rho_\phi(k) &= G^>(k) - G^<(k) = 2\pi \operatorname{sgn}(k_0) \delta(k^2 - m^2), \\ G_R(k) &= \frac{i}{(k_0 + i0^+)^2 - \mathbf{k}^2 - m^2}, \quad G_A(k) = \frac{i}{(k_0 - i0^+)^2 - \mathbf{k}^2 - m^2}.\end{aligned}\tag{142}$$

These objects are temperature independent in the free theory because they are fixed by the canonical commutator. Temperature only changes the Wightman weights.

The Euclidean scalar field is

$$\phi(\tau, \mathbf{x}) = e^{\tau H_0} \phi(0, \mathbf{x}) e^{-\tau H_0}.\tag{143}$$

For $0 < \tau < \beta$,

$$G_E(\tau, \mathbf{p}) = \frac{1}{2E_{\mathbf{p}}} \left[(1 + n_B(E_{\mathbf{p}})) e^{-E_{\mathbf{p}}\tau} + n_B(E_{\mathbf{p}}) e^{E_{\mathbf{p}}\tau} \right], \quad G_E(\tau + \beta, \mathbf{p}) = G_E(\tau, \mathbf{p}).\tag{144}$$

Therefore

$$G_E(\tau, \mathbf{p}) = \frac{1}{\beta} \sum_{n \in \mathbb{Z}} e^{-i\omega_n^B \tau} G_E(i\omega_n^B, \mathbf{p}), \quad G_E(i\omega_n^B, \mathbf{p}) = \frac{1}{(\omega_n^B)^2 + \mathbf{p}^2 + m^2}.\tag{145}$$

This is the field-theoretic version of the bosonic oscillator result, with ω replaced by $E_{\mathbf{p}}$.

2.3 Thermal free Dirac field

Let

$$H_0 = \sum_s \int \frac{d^3\mathbf{p}}{(2\pi)^3} E_{\mathbf{p}} \left(b_{\mathbf{p}s}^\dagger b_{\mathbf{p}s} + d_{\mathbf{p}s}^\dagger d_{\mathbf{p}s} \right), \quad E_{\mathbf{p}} = \sqrt{\mathbf{p}^2 + m^2},\tag{146}$$

with the usual anticommutation relations. The mode expansion is

$$\psi(x) = \sum_s \int \frac{d^3\mathbf{p}}{(2\pi)^3} \frac{1}{\sqrt{2E_{\mathbf{p}}}} \left(b_{\mathbf{p}s} u_s(p) e^{-ip \cdot x} + d_{\mathbf{p}s}^\dagger v_s(p) e^{ip \cdot x} \right), \quad p^0 = E_{\mathbf{p}}.\tag{147}$$

The thermal averages are

$$\left\langle b_{\mathbf{p}s}^\dagger b_{\mathbf{q}r} \right\rangle_{\beta,0} = (2\pi)^3 \delta^3(\mathbf{p} - \mathbf{q}) \delta_{sr} n_F(E_{\mathbf{p}}), \quad \left\langle d_{\mathbf{p}s}^\dagger d_{\mathbf{q}r} \right\rangle_{\beta,0} = (2\pi)^3 \delta^3(\mathbf{p} - \mathbf{q}) \delta_{sr} n_F(E_{\mathbf{p}}).\tag{148}$$

We use the standard relativistic normalization

$$\bar{u}_s(p) u_r(p) = 2m \delta_{sr}, \quad \bar{v}_s(p) v_r(p) = -2m \delta_{sr}, \quad u_s^\dagger(p) u_r(p) = v_s^\dagger(p) v_r(p) = 2E_{\mathbf{p}} \delta_{sr}.\tag{149}$$

The corresponding spin sums are

$$\sum_s u_s(p) \bar{u}_s(p) = \not{p} + m, \quad \sum_s v_s(p) \bar{v}_s(p) = \not{p} - m, \quad p^0 = E_{\mathbf{p}}.\tag{150}$$

Here is a short proof, assuming the Dirac equations

$$(\not{p} - m) u_s(p) = 0, \quad (\not{p} + m) v_s(p) = 0.\tag{151}$$

At rest, $p_* = (m, \mathbf{0})$, the equations become

$$(\gamma^0 - 1) u_s(p_*) = 0, \quad (\gamma^0 + 1) v_s(p_*) = 0.\tag{152}$$

Thus the two u spinors span the $+1$ eigenspace of γ^0 , while the two v spinors span the -1 eigenspace. The projectors onto these two subspaces are

$$P_+ = \frac{1 + \gamma^0}{2}, \quad P_- = \frac{1 - \gamma^0}{2}. \quad (153)$$

The normalization (149) fixes the completeness matrices on these subspaces. Since $\bar{u}_s = u_s^\dagger \gamma^0$ and $\gamma^0 u_s(p_*) = u_s(p_*)$, the matrix $\sum_s u_s(p_*) \bar{u}_s(p_*)$ acts as $2m$ times the identity on the positive-energy subspace and annihilates the negative-energy subspace. Hence

$$\sum_s u_s(p_*) \bar{u}_s(p_*) = 2mP_+ = m(1 + \gamma^0) = \not{p}_* + m. \quad (154)$$

Similarly, $\gamma^0 v_s(p_*) = -v_s(p_*)$, so $\bar{v}_s(p_*) = -v_s^\dagger(p_*)$ on the negative-energy subspace. Therefore the completeness matrix is $-2m$ times the identity on that subspace:

$$\sum_s v_s(p_*) \bar{v}_s(p_*) = -2mP_- = m(\gamma^0 - 1) = \not{p}_* - m. \quad (155)$$

For a general on-shell momentum $p^\mu = \Lambda^\mu{}_\nu p_*^\nu$, choose the corresponding spinor representation $S(\Lambda)$, so that

$$S(\Lambda) \gamma^\mu S(\Lambda)^{-1} = \Lambda^\mu{}_\nu \gamma^\nu. \quad (156)$$

The boosted spinors are linear combinations of $S(\Lambda)u_s(p_*)$ and $S(\Lambda)v_s(p_*)$ inside each spin sector. The spin rotation inside the degenerate two-dimensional spin space drops out of the sum over s . Thus

$$\sum_s u_s(p) \bar{u}_s(p) = S(\Lambda) (\not{p}_* + m) S(\Lambda)^{-1} = \not{p} + m, \quad (157)$$

and the same argument gives

$$\sum_s v_s(p) \bar{v}_s(p) = S(\Lambda) (\not{p}_* - m) S(\Lambda)^{-1} = \not{p} - m. \quad (158)$$

This proves (150). The massless formulas follow by continuity from the massive ones.

The real-time contour propagator is

$$S_{\alpha\beta}^{ab}(x-y) = \langle T_{\mathcal{C}_R} \psi_\alpha(x_a) \bar{\psi}_\beta(y_b) \rangle_{\beta,0}. \quad (159)$$

The two operator-ordered Dirac correlators are

$$S_{\alpha\beta}^>(x-y) = \langle \psi_\alpha(x) \bar{\psi}_\beta(y) \rangle_{\beta,0}, \quad \tilde{S}_{\alpha\beta}^<(x-y) = \langle \bar{\psi}_\beta(y) \psi_\alpha(x) \rangle_{\beta,0}. \quad (160)$$

The tilde is useful because the Schwinger–Keldysh component with the later point on the contour to the left carries the fermionic exchange sign:

$$S_{\alpha\beta}^{+-}(x-y) \equiv S_{\alpha\beta}^<(x-y) = -\tilde{S}_{\alpha\beta}^<(x-y). \quad (161)$$

Thus the Dirac spectral function is explicitly the thermal expectation value of the anticommutator,

$$\rho_{D,\alpha\beta}(x-y) = \langle \{ \psi_\alpha(x), \bar{\psi}_\beta(y) \} \rangle_{\beta,0} = S_{\alpha\beta}^>(x-y) + \tilde{S}_{\alpha\beta}^<(x-y) = S_{\alpha\beta}^>(x-y) - S_{\alpha\beta}^<(x-y). \quad (162)$$

This is the exact analogue of the fermionic oscillator result $\rho_F(t) = g^>(t) - g^<(t) = \langle \{ f(t), f^\dagger(0) \} \rangle_{\beta,0}$. Equivalently, if one uses the raw lesser correlator without the Schwinger–Keldysh minus sign, the fermionic spectral function is the sum of the two raw Wightman functions. For bosons the corresponding object is a commutator, $G^> - G^<$; for fermions it is an anticommutator.

Let us derive its momentum-space form. Setting $y = 0$ and inserting the mode expansion, all occupation numbers cancel between the two raw Wightman functions because $1 - n_F + n_F = 1$. Therefore

$$\begin{aligned}\rho_D(x) &= \sum_s \int \frac{d^3\mathbf{p}}{(2\pi)^3} \frac{1}{2E_{\mathbf{p}}} [u_s(p)\bar{u}_s(p)e^{-ip\cdot x} + v_s(p)\bar{v}_s(p)e^{ip\cdot x}] \\ &= \int \frac{d^3\mathbf{p}}{(2\pi)^3} \frac{1}{2E_{\mathbf{p}}} [(\not{p} + m)e^{-ip\cdot x} + (\not{p} - m)e^{ip\cdot x}], \quad p^0 = E_{\mathbf{p}},\end{aligned}\quad (163)$$

where the second line used the spin sums (150). Fourier transforming (163) gives the compact covariant expression

$$\rho_D(k) = 2\pi \operatorname{sgn}(k_0)(\not{k} + m)\delta(k^2 - m^2). \quad (164)$$

For $k_0 > 0$, this is the particle spin sum $\not{k} + m$. For $k_0 < 0$, the factor $\operatorname{sgn}(k_0)$ converts the negative-frequency part into the antiparticle spin sum $\not{k} - m$, with $p^0 = E_{\mathbf{p}}$. This is why the same covariant spectral function packages both particles and antiparticles.

With the convention

$$n_F(k_0) = \frac{1}{e^{\beta k_0} + 1}, \quad (165)$$

which obeys $n_F(-E) = 1 - n_F(E)$, the operator algebra gives

$$S^{-+}(k) = S^>(k) = [1 - n_F(k_0)]\rho_D(k), \quad S^{+-}(k) = S^<(k) = -n_F(k_0)\rho_D(k). \quad (166)$$

Equivalently, written with $n_F(|k_0|)$,

$$S^{-+}(k) = 2\pi(\not{k} + m)[\theta(k_0) - n_F(|k_0|)]\delta(k^2 - m^2), \quad (167)$$

$$S^{+-}(k) = -2\pi(\not{k} + m)[\theta(k_0)n_F(|k_0|) - \theta(-k_0)(1 - n_F(|k_0|))]\delta(k^2 - m^2). \quad (168)$$

These formulae also explain an important point about the relation between a relativistic Dirac field and the single fermionic oscillator studied in (104). For a single positive-energy fermionic annihilation operator $f(t) = fe^{-i\omega t}$, the zero-temperature result was

$$g_0^{-+}(k_0) = 2\pi\delta(k_0 - \omega), \quad g_0^{+-}(k_0) = 0. \quad (169)$$

This is correct for that simple oscillator. The vacuum contains no fermion which can be annihilated first, and therefore the lesser function vanishes. However, a relativistic Dirac field is not merely a positive-energy annihilation operator. Its mode expansion contains both a particle-annihilation part and an antiparticle-creation part,

$$\psi(x) = \sum_s \int \frac{d^3\mathbf{p}}{(2\pi)^3} \frac{1}{\sqrt{2E_{\mathbf{p}}}} [a_s(\mathbf{p})u_s(p)e^{-ip\cdot x} + b_s^\dagger(\mathbf{p})v_s(p)e^{ip\cdot x}]. \quad (170)$$

The first term behaves like the oscillator of (104): $\bar{\psi}(y)$ contains $a_s^\dagger$, so the vacuum matrix element $\langle a_s(\mathbf{p})a_{s'}^\dagger(\mathbf{p}') \rangle_{\beta,0}$ produces the positive-frequency contribution to S^{-+} . The second term has no analogue in the single oscillator: $\psi(x)$ contains $b_s^\dagger$, while $\bar{\psi}(y)$ contains b_s . Consequently the vacuum matrix element $\langle b_s(\mathbf{p})b_{s'}^\dagger(\mathbf{p}') \rangle_{\beta,0}$ produces a nonzero negative-frequency contribution to the lesser function. Thus, taking $T \rightarrow 0$ in (167)–(168) gives

$$S_0^{-+}(k) = 2\pi(\not{k} + m)\theta(k_0)\delta(k^2 - m^2), \quad (171)$$

$$S_0^{+-}(k) = 2\pi(\not{k} + m)\theta(-k_0)\delta(k^2 - m^2). \quad (172)$$

The difference from (104) is therefore not a contradiction. Equation (104) refers to one positive-energy annihilation operator only. The Dirac field contains a second oscillator, written as an

antiparticle creation operator, and this second oscillator supplies the negative-energy support of S_0^{+-} . Equivalently, the covariant spectral function (164) packages both particle propagation at $k_0 > 0$ and antiparticle propagation at $k_0 < 0$. This is why the off-diagonal Dirac propagators have both positive- and negative-frequency vacuum pieces, whereas the single oscillator in (104) has only the positive-frequency piece.

The diagonal components are obtained by the same time-ordering step as for the fermionic oscillator:

$$S^{++}(k) = (\not{k} + m) \left[\frac{i}{k^2 - m^2 + i0^+} - 2\pi n_F(|k_0|)\delta(k^2 - m^2) \right], \quad (173)$$

$$S^{--}(k) = (\not{k} + m) \left[-\frac{i}{k^2 - m^2 - i0^+} - 2\pi n_F(|k_0|)\delta(k^2 - m^2) \right]. \quad (174)$$

The Dirac retarded and advanced propagators are obtained from the same spectral function:

$$S_R(k) = \frac{i(\not{k} + m)}{(k_0 + i0^+)^2 - \mathbf{k}^2 - m^2}, \quad S_A(k) = \frac{i(\not{k} + m)}{(k_0 - i0^+)^2 - \mathbf{k}^2 - m^2}. \quad (175)$$

The spectral function (164), and hence S_R and S_A , are temperature independent in the free theory; the Fermi factors only redistribute the particle and antiparticle Wightman weights.

At zero temperature, the thermal occupation number vanishes for positive energy modes. Hence

$$S_0^{++}(k) = \frac{i(\not{k} + m)}{k^2 - m^2 + i0^+}, \quad S_0^{--}(k) = -\frac{i(\not{k} + m)}{k^2 - m^2 - i0^+}, \quad (176)$$

while the off-diagonal components reduce to the positive- and negative-energy on-shell parts of the vacuum Dirac two-point function.

For Euclidean time, the Dirac field is evolved as

$$\psi(\tau, \mathbf{x}) = e^{\tau H_0} \psi(0, \mathbf{x}) e^{-\tau H_0}. \quad (177)$$

The fermionic KMS relation gives antiperiodicity,

$$S_E(\tau + \beta, \mathbf{p}) = -S_E(\tau, \mathbf{p}). \quad (178)$$

Therefore the Fourier frequencies are fermionic,

$$\omega_n^F = \frac{(2n + 1)\pi}{\beta}. \quad (179)$$

Using the same spin sums as above, or equivalently replacing each particle and antiparticle oscillator by its Euclidean correlator, one obtains the antiperiodic Fourier series. As in the single fermionic oscillator, there are two common sign conventions. If

$$\tilde{S}_E(\tau, \mathbf{p}) := \langle T_\tau \psi(\tau, \mathbf{p}) \bar{\psi}(0, -\mathbf{p}) \rangle_{\beta, 0} \quad (180)$$

is the raw operator-ordered correlator, then the conventional Matsubara Dirac propagator is

$$S_E(\tau, \mathbf{p}) := -\tilde{S}_E(\tau, \mathbf{p}). \quad (181)$$

With this convention,

$$S_E(\tau, \mathbf{p}) = \frac{1}{\beta} \sum_{n \in \mathbb{Z}} e^{-i\omega_n^F \tau} S_E(i\omega_n^F, \mathbf{p}), \quad (182)$$

where, in Euclidean gamma-matrix notation $\{\gamma_E^\mu, \gamma_E^\nu\} = 2\delta^{\mu\nu}$,

$$S_E(i\omega_n^F, \mathbf{p}) = \frac{-i\omega_n^F \gamma_E^0 - i\mathbf{p}_i \gamma_E^i + m}{(\omega_n^F)^2 + \mathbf{p}^2 + m^2}. \quad (183)$$

This is not a path-integral result. It is the Fourier series of the antiperiodic operator two-point function; the only extra input is the conventional overall minus sign used for Euclidean fermion propagators.

2.4 Free $U(1)$ gauge field

For a free Abelian gauge field one must first specify the gauge. We use covariant Feynman gauge,

$$\mathcal{L}_0 = -\frac{1}{4}F_{\mu\nu}F^{\mu\nu} - \frac{1}{2}(\partial_\mu A^\mu)^2, \quad (184)$$

for which the free equation is simply

$$\square A_\mu = 0. \quad (185)$$

In an Abelian theory the Faddeev–Popov ghosts are free and decouple from gauge-invariant observables. In canonical Gupta–Bleuler quantization the four components of A_μ are quantized with an indefinite metric; after imposing the physical-state condition, gauge-invariant observables are independent of the unphysical polarizations. For perturbative thermal calculations in Feynman gauge the free contour propagator is the massless scalar contour propagator multiplied by $-g_{\mu\nu}$.

The mode expansion may be written schematically as

$$A_\mu(x) = \sum_\lambda \int \frac{d^3\mathbf{p}}{(2\pi)^3} \frac{1}{\sqrt{2|\mathbf{p}|}} \left[\epsilon_\mu^{(\lambda)}(\mathbf{p}) a_{\lambda\mathbf{p}} e^{-i|\mathbf{p}|t + i\mathbf{p}\cdot\mathbf{x}} + \epsilon_\mu^{(\lambda)*}(\mathbf{p}) a_{\lambda\mathbf{p}}^\dagger e^{i|\mathbf{p}|t - i\mathbf{p}\cdot\mathbf{x}} \right], \quad (186)$$

with the covariant polarization sum producing $-g_{\mu\nu}$. Since each momentum-polarization mode is a bosonic oscillator, the SK propagator is

$$D_{\mu\nu}^{ab}(k) = -g_{\mu\nu} G_{m=0}^{ab}(k), \quad (187)$$

where $G_{m=0}^{ab}$ is the scalar propagator (138)–(141) with $m = 0$. Explicitly,

$$D_{\mu\nu}^{++}(k) = -g_{\mu\nu} \left[\frac{i}{k^2 + i0^+} + 2\pi n_B(|k_0|) \delta(k^2) \right], \quad (188)$$

$$D_{\mu\nu}^{--}(k) = -g_{\mu\nu} \left[-\frac{i}{k^2 - i0^+} + 2\pi n_B(|k_0|) \delta(k^2) \right], \quad (189)$$

$$D_{\mu\nu}^{-+}(k) = -g_{\mu\nu} 2\pi [\theta(k_0) + n_B(|k_0|)] \delta(k^2), \quad (190)$$

$$D_{\mu\nu}^{+-}(k) = -g_{\mu\nu} 2\pi [\theta(-k_0) + n_B(|k_0|)] \delta(k^2). \quad (191)$$

The $i0^+$ prescriptions are again exactly the scalar prescriptions because the free Maxwell equations in Feynman gauge reduce to four oscillator equations.

The Euclidean photon propagator uses bosonic Matsubara frequencies,

$$\omega_n^B = \frac{2\pi n}{\beta}. \quad (192)$$

In Euclidean Feynman gauge,

$$D_{\mu\nu}^E(i\omega_n^B, \mathbf{k}) = \frac{\delta_{\mu\nu}}{(\omega_n^B)^2 + \mathbf{k}^2}. \quad (193)$$

The symbol ξ labels the usual family of covariant gauges. In real time one replaces the Feynman-gauge fixing term by

$$\mathcal{L}_{\text{gf}}^{(\xi)} = -\frac{1}{2\xi}(\partial_\mu A^\mu)^2, \quad \xi = 1 \text{ Feynman gauge}, \quad \xi \rightarrow 0 \text{ Landau gauge}. \quad (194)$$

The free equation of motion is then

$$\left[g_{\mu\nu} \square - \left(1 - \frac{1}{\xi} \right) \partial_\mu \partial_\nu \right] A^\nu = 0. \quad (195)$$

After continuation to Euclidean frequency and momentum, the corresponding quadratic operator is

$$Q_{\mu\nu}(K) = K^2 \delta_{\mu\nu} - \left(1 - \frac{1}{\xi}\right) K_\mu K_\nu = K^2 P_{\mu\nu}^T + \frac{K^2}{\xi} P_{\mu\nu}^L, \quad (196)$$

where

$$P_{\mu\nu}^L = \frac{K_\mu K_\nu}{K^2}, \quad P_{\mu\nu}^T = \delta_{\mu\nu} - \frac{K_\mu K_\nu}{K^2}. \quad (197)$$

Since P^T and P^L are orthogonal projectors, the inverse is immediate:

$$Q_{\mu\nu}^{-1}(K) = \frac{P_{\mu\nu}^T}{K^2} + \xi \frac{P_{\mu\nu}^L}{K^2}. \quad (198)$$

Thus, in a general covariant ξ -gauge, the Euclidean free propagator is

$$D_{\mu\nu}^E(K) = \frac{1}{K^2} \left(\delta_{\mu\nu} - (1 - \xi) \frac{K_\mu K_\nu}{K^2} \right), \quad K = (\omega_n^B, \mathbf{k}), \quad K^2 = (\omega_n^B)^2 + \mathbf{k}^2. \quad (199)$$

For the real-time SK propagators above we have chosen $\xi = 1$. Other covariant gauges modify only the tensor structure and not the thermal occupation factors, which are still bosonic because the gauge field oscillators have integer spin-statistics and periodic Euclidean boundary conditions.

3 Schwinger–Keldysh diagrammatic rules

We now turn the operator formula of Section 1 into practical diagrammatic rules. The derivation is still operator based. One expands the interaction-picture evolution operator on the closed real-time contour and then evaluates the resulting free thermal expectation values by the finite-temperature Wick theorem. No path integral argument is needed.

Let

$$H = H_0 + H_I, \quad (200)$$

where all fields below are interaction-picture fields evolved with H_0 . For an operator $\mathcal{O}$ whose insertions lie on the two real-time branches, the normalized real-time formula is

$$\langle T_{\mathcal{C}_R} \mathcal{O} \rangle_\beta = \frac{\left\langle T_{\mathcal{C}_R} \mathcal{O} \exp \left[-i \int_{\mathcal{C}_R} dt H_I(t) \right] \right\rangle_{\beta,0}}{\left\langle T_{\mathcal{C}_R} \exp \left[-i \int_{\mathcal{C}_R} dt H_I(t) \right] \right\rangle_{\beta,0}}. \quad (201)$$

The real-time contour is $\mathcal{C}_R = C_+ \cup C_-$. Hence

$$\int_{\mathcal{C}_R} dt H_I(t) = \int_{t_0}^{t_f} dt H_I[\Phi_+(t)] - \int_{t_0}^{t_f} dt H_I[\Phi_-(t)]. \quad (202)$$

Equivalently,

$$\int_{\mathcal{C}_R} dt H_I(t) = \sum_{\sigma=\pm} \eta_\sigma \int_{t_0}^{t_f} dt H_I[\Phi_\sigma(t)], \quad \eta_+ = +1, \quad \eta_- = -1. \quad (203)$$

For interactions without time derivatives, $H_I = - \int d^3\mathbf{x} \mathcal{L}_I$. Therefore

$$-i \int_{\mathcal{C}_R} dt H_I(t) = i \sum_{\sigma=\pm} \eta_\sigma \int_{t_0}^{t_f} d^4x \mathcal{L}_I[\Phi_\sigma(x)]. \quad (204)$$

This equation is the origin of the doubled vertices. A vertex on the $+$ branch has the ordinary in-out vertex factor, while the same vertex on the $-$ branch has the opposite sign.

Expanding the numerator of (201) gives

$$\begin{aligned} \left\langle T_{\mathcal{C}_R} \mathcal{O} \exp \left[-i \int_{\mathcal{C}_R} dt H_I(t) \right] \right\rangle_{\beta,0} &= \sum_{n=0}^{\infty} \frac{(-i)^n}{n!} \int_{\mathcal{C}_R} dt_1 \cdots dt_n \\ &\times \langle T_{\mathcal{C}_R} \mathcal{O} H_I(t_1) \cdots H_I(t_n) \rangle_{\beta,0}. \end{aligned} \quad (205)$$

The expectation value on the right is a free thermal contour-ordered expectation value. Wick's theorem factorizes it into free Schwinger–Keldysh propagators. The denominator cancels vacuum diagrams, so the answer is the sum of diagrams connected to the external operator insertions.

3.1 The usual $+/-$ basis and momentum-space rules

The $+/-$ basis is the most direct translation of the closed time contour. Every external insertion and every internal vertex carries a branch label $\sigma = \pm$. A scalar, Dirac, or gauge line joining branches σ and σ' contributes the corresponding free thermal propagator $G_0^{\sigma\sigma'}$, $S_0^{\sigma\sigma'}$, or $D_{0,\mu\nu}^{\sigma\sigma'}$. Each internal vertex is integrated over spacetime and summed over its branch label. The branch vertex is the ordinary zero-temperature vertex multiplied by η_σ .

For actual calculations students should see how the momentum-space rule follows. We use the convention

$$X_\sigma(x) = \int \frac{d^4 p}{(2\pi)^4} e^{-ip \cdot x} X_\sigma(p). \quad (206)$$

Suppose a local interaction contains n fields,

$$\mathcal{L}_I(x) = -\frac{g}{S} X_1(x) X_2(x) \cdots X_n(x). \quad (207)$$

Here S is not yet a symmetry factor of a whole Feynman graph. It is the *local factorial already put into the interaction Lagrangian* in order not to overcount identical fields sitting at the same spacetime point. More explicitly, if the same field appears m_1 times, another identical field appears m_2 times, and so on, then

$$S = m_1! m_2! \cdots. \quad (208)$$

Thus $S = 4!$ for $\lambda\phi^4$, $S = 2!$ for an interaction such as $g\phi^2\chi$, and $S = 1$ for the Yukawa interaction $g\phi\bar{\psi}\psi$, since ϕ , ψ , and $\bar{\psi}$ are different fields. The purpose of this factor is simple: when Wick's theorem contracts the fields at the vertex with external or internal lines, permutations of identical fields give the same contraction. The factorial in the Lagrangian removes precisely this local overcounting.

On branch σ , the contribution to the perturbative exponential is

$$\begin{aligned} i\eta_\sigma \int d^4 x \mathcal{L}_I[X_\sigma] &= -i\eta_\sigma \frac{g}{S} \int d^4 x \prod_{j=1}^n \int \frac{d^4 p_j}{(2\pi)^4} e^{-ip_j \cdot x} X_{j,\sigma}(p_j) \\ &= -i\eta_\sigma \frac{g}{S} \int \prod_{j=1}^n \frac{d^4 p_j}{(2\pi)^4} (2\pi)^4 \delta^{(4)} \left(\sum_{j=1}^n p_j \right) \prod_{j=1}^n X_{j,\sigma}(p_j). \end{aligned} \quad (209)$$

The x -integral is the origin of four-momentum conservation at the vertex. The factor $1/S$ has a different role: it is a bookkeeping factor for Wick contractions. To see the cancellation, take $\lambda\phi^4$ as the simplest example. The interaction contains

$$\mathcal{L}_I = -\frac{\lambda}{4!} \phi^4. \quad (210)$$

If the four fields at this vertex are contracted with four labelled lines, there are $4!$ ways to decide which copy of ϕ in ϕ^4 is contracted with which line. These $4!$ contractions give the same

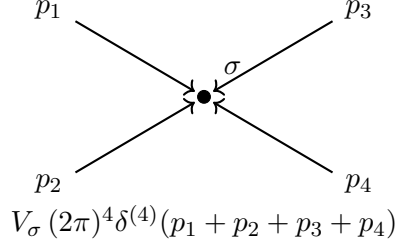


Figure 2: A local branch vertex in momentum space. The branch label gives the sign η_σ ; the spacetime integral gives four-momentum conservation.

momentum-space expression, and therefore cancel the $1/4!$ in the Lagrangian. The local vertex factor is then just

$$-i\eta_\sigma \lambda (2\pi)^4 \delta^{(4)} \left(\sum_{i=1}^4 p_i \right). \quad (211)$$

Similarly, for $-g\phi^2\chi/2!$, the two identical ϕ fields give two identical Wick contractions, cancelling the $1/2!$. For Yukawa theory there is no such factorial to cancel because the three fields in $\phi\bar{\psi}\psi$ are distinct.

This is why the factor S no longer appears in the final vertex rule. It has already done its job: it has cancelled the identical-field contractions localized at one vertex. After this local cancellation, the remaining momentum-space vertex is the ordinary zero-temperature vertex multiplied by the branch sign η_σ , together with

$$(2\pi)^4 \delta^{(4)} \left(\sum_{\text{all incoming}} p_i \right). \quad (212)$$

All momenta in the vertex formulae below are taken incoming. This local factorial should be distinguished from the symmetry factor of a complete Feynman graph. A graph may still have a residual automorphism symmetry, for example an exchange of two identical internal lines leaving the external legs fixed. Such residual graph symmetries are the “usual symmetry factors” mentioned in the general Feynman rules below. They are not the same thing as the factorial S in (207).

The complete $+/-$ momentum-space prescription is:

- (i) Assign a branch label $\sigma = \pm$ to every internal vertex and every external insertion.
- (ii) Assign a momentum to every line. For a scalar line use $G_0^{\sigma\sigma'}(k)$; for a Dirac line use $S_0^{\sigma\sigma'}(k)$ with the fermion arrow fixing the order of spinor indices; for a gauge line use $D_{0,\mu\nu}^{\sigma\sigma'}(k)$.
- (iii) At every vertex insert the branch vertex factor and the factor $(2\pi)^4 \delta^{(4)}(\sum p_i)$.
- (iv) Integrate over every independent loop momentum with $\int d^4\ell/(2\pi)^4$.
- (v) Sum over all internal branch labels. Include the usual residual graph symmetry factors, the usual fermion signs, and a minus sign for every closed fermion loop. The graph symmetry factor means an automorphism factor of the completed diagram, not the local factorial S already absorbed into the vertex rule.

For $\lambda\phi^4$ theory,

$$\mathcal{L} = \frac{1}{2}\partial_\mu\phi\partial^\mu\phi - \frac{1}{2}m^2\phi^2 - \frac{\lambda}{4!}\phi^4. \quad (213)$$

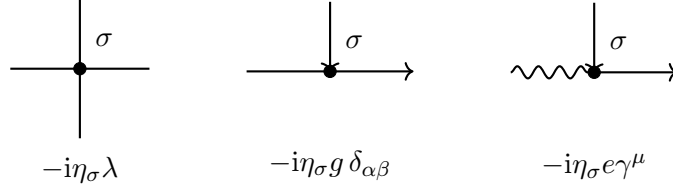


Figure 3: The $+/-$ vertices for $\lambda\phi^4$, Yukawa theory, and QED. In momentum space each vertex also carries the delta function enforcing the sum of incoming four-momenta to vanish.

The four-scalar vertex on branch σ is

$$V_{\sigma}^{\phi^4}(p_1, p_2, p_3, p_4) = -i\eta_{\sigma}\lambda(2\pi)^4\delta^{(4)}(p_1 + p_2 + p_3 + p_4). \quad (214)$$

The $+$ vertex is $-i\lambda$ times the delta function and the $-$ vertex is $+i\lambda$ times the delta function.

For Yukawa theory,

$$\mathcal{L} = \frac{1}{2}\partial_{\mu}\phi\partial^{\mu}\phi - \frac{1}{2}m_{\phi}^2\phi^2 + \bar{\psi}(i\gamma^{\mu}\partial_{\mu} - m_{\psi})\psi - g\phi\bar{\psi}\psi. \quad (215)$$

With all momenta incoming, the vertex is a matrix in spinor space,

$$(V_{\sigma}^{\text{Yukawa}})_{\alpha\beta}(p_{\phi}, p_{\psi}, p_{\bar{\psi}}) = -i\eta_{\sigma}g\delta_{\alpha\beta}(2\pi)^4\delta^{(4)}(p_{\phi} + p_{\psi} + p_{\bar{\psi}}). \quad (216)$$

Here α and β are the spinor indices on the two fermion legs, ordered along the fermion arrow. In this scalar Yukawa theory the spinor matrix is the identity, $\delta_{\alpha\beta}$. Later, in an amplitude, this identity matrix is contracted with the adjacent fermion propagators or external spinors.

For QED in a covariant gauge,

$$\mathcal{L} = -\frac{1}{4}F_{\mu\nu}F^{\mu\nu} - \frac{1}{2\xi}(\partial_{\mu}A^{\mu})^2 + \bar{\psi}(i\gamma^{\mu}\partial_{\mu} - m)\psi - e\bar{\psi}\gamma^{\mu}A_{\mu}\psi. \quad (217)$$

The photon propagator depends on ξ , but the branch structure does not. The branch QED vertex is

$$V_{\sigma,\mu}^{\text{QED}}(p_A, p_{\psi}, p_{\bar{\psi}}) = -i\eta_{\sigma}e\gamma_{\mu}(2\pi)^4\delta^{(4)}(p_A + p_{\psi} + p_{\bar{\psi}}). \quad (218)$$

Since QED is abelian, the Faddeev–Popov ghosts decouple (ignore this statement if you do not understand it).

3.2 Why the r/a basis is better, and its rules

The $+/-$ basis is the literal basis of the closed time contour. It is the right basis for deriving the rules, because every insertion remembers whether it lies on the forward or the backward branch. It is not, however, the most transparent basis for physics. A retarded response function is not equal to one of the four $+/-$ functions by itself. It is a difference of two of them. The thermal occupation number is also not tied to a retarded pole prescription; it appears in the symmetric, or statistical, two-point function. The r/a basis is just the linear change of basis which separates these two pieces of information.

Nothing in this subsection uses a path integral or an action principle. We use only the free Schwinger–Keldysh propagators derived in Section 2 and then rotate the two branch labels. The calculation is therefore ordinary linear algebra.

The four contour functions and the physical combinations

For a bosonic field we write

$$G_F \equiv G^{++}, \quad G_{\bar{F}} \equiv G^{--}, \quad G_{>} \equiv G^{-+}, \quad G_{<} \equiv G^{+-}. \quad (219)$$

Here G_F is the thermal Feynman function, $G_{\bar{F}}$ is the thermal anti-Feynman function, and $G_>, G_<$ are the Wightman functions. From Section 2,

$$\begin{aligned} G_F(x, y) &= \theta(x^0 - y^0)G_>(x, y) + \theta(y^0 - x^0)G_<(x, y), \\ G_{\bar{F}}(x, y) &= \theta(y^0 - x^0)G_>(x, y) + \theta(x^0 - y^0)G_<(x, y). \end{aligned} \quad (220)$$

These two equations are the only input needed. Subtracting $G_<$ from the first line gives

$$\begin{aligned} G_F(x, y) - G_<(x, y) &= \theta(x^0 - y^0)G_>(x, y) + \theta(y^0 - x^0)G_<(x, y) - G_<(x, y) \\ &= \theta(x^0 - y^0) [G_>(x, y) - G_<(x, y)]. \end{aligned} \quad (221)$$

This is the retarded function,

$$G_R(x, y) = \theta(x^0 - y^0) [G_>(x, y) - G_<(x, y)] = G_F(x, y) - G_<(x, y). \quad (222)$$

Equally, subtracting $G_>$ from G_F gives

$$\begin{aligned} G_F(x, y) - G_>(x, y) &= \theta(x^0 - y^0)G_>(x, y) + \theta(y^0 - x^0)G_<(x, y) - G_>(x, y) \\ &= -\theta(y^0 - x^0) [G_>(x, y) - G_<(x, y)]. \end{aligned} \quad (223)$$

This is the advanced function,

$$G_A(x, y) = -\theta(y^0 - x^0) [G_>(x, y) - G_<(x, y)] = G_F(x, y) - G_>(x, y). \quad (224)$$

Finally, adding the two equations in (220) gives

$$G_F(x, y) + G_{\bar{F}}(x, y) = G_>(x, y) + G_<(x, y). \quad (225)$$

Equivalently,

$$G_{\bar{F}}(x, y) = G_>(x, y) + G_<(x, y) - G_F(x, y). \quad (226)$$

This identity is the two-point version of closed-time-contour unitarity. It says that only three of the four $+/-$ components are independent. The third useful combination is the Keldysh, or statistical, function

$$G_K(x, y) = G_>(x, y) + G_<(x, y). \quad (227)$$

Thus the three physical combinations are

$$\boxed{G_R = G_F - G_<, \quad G_A = G_F - G_>, \quad G_K = G_> + G_<}. \quad (228)$$

The inverse relations are also useful. Solving (228) and (225) gives

$$\begin{aligned} G^{++} &= \frac{1}{2}G_K + \frac{1}{2}(G_R + G_A), & G^{--} &= \frac{1}{2}G_K - \frac{1}{2}(G_R + G_A), \\ G^{-+} &= \frac{1}{2}G_K + \frac{1}{2}(G_R - G_A), & G^{+-} &= \frac{1}{2}G_K - \frac{1}{2}(G_R - G_A). \end{aligned} \quad (229)$$

These formulae are often the quickest way of checking a real-time calculation. The Feynman propagator contains both response and fluctuations; the retarded propagator contains only the commutator, or spectral, part.

The Keldysh rotation as a basis conversion

Now let us see how the same combinations arise as a change of basis. It is pedagogically clean to use the normalized average/difference variables

$$\hat{X}_r = \frac{1}{\sqrt{2}}(X_+ - X_-), \quad \hat{X}_a = \frac{1}{\sqrt{2}}(X_+ + X_-). \quad (230)$$

The hat reminds us that this is the normalized version of the average/difference basis. Later, when writing vertices, it is convenient to remove the square roots and use $X_a = (X_+ + X_-)/2$ and $X_r = X_+ - X_-$. The two conventions differ only by constant rescalings of external r/a labels. The physical objects G_R, G_A, G_K are the same.

Define the rotation matrix

$$R = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix}, \quad \begin{pmatrix} \hat{X}_r \\ \hat{X}_a \end{pmatrix} = R \begin{pmatrix} X_+ \\ X_- \end{pmatrix}. \quad (231)$$

This matrix is orthogonal:

$$RR^T = R^T R = 1, \quad R^{-1} = R^T. \quad (232)$$

Therefore the basis conversion of the two-point table is a similarity transformation. With the ordering of Wightman functions used in (219), write the $+/-$ table as

$$\mathbf{G}_{+/-} = \begin{pmatrix} G_F & G_> \\ G_< & G_{\bar{F}} \end{pmatrix}. \quad (233)$$

This is only a compact way of arranging the four known functions. If one uses the transposed arrangement, the retarded and advanced entries below are transposed. No physical statement changes; one has only changed the convention for drawing the direction of a line.

The rotated table is

$$\mathbf{G}_{r/a} = R \mathbf{G}_{+/-} R^T. \quad (234)$$

Let us multiply it out explicitly. First,

$$\begin{aligned} \mathbf{G}_{r/a} &= \frac{1}{2} \begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} G_F & G_> \\ G_< & G_{\bar{F}} \end{pmatrix} \begin{pmatrix} 1 & 1 \\ -1 & 1 \end{pmatrix} \\ &= \frac{1}{2} \begin{pmatrix} G_F - G_< & G_> - G_{\bar{F}} \\ G_F + G_< & G_> + G_{\bar{F}} \end{pmatrix} \begin{pmatrix} 1 & 1 \\ -1 & 1 \end{pmatrix} \\ &= \frac{1}{2} \begin{pmatrix} G_F - G_> - G_< + G_{\bar{F}} & G_F + G_> - G_< - G_{\bar{F}} \\ G_F - G_> + G_< - G_{\bar{F}} & G_F + G_> + G_< + G_{\bar{F}} \end{pmatrix}. \end{aligned} \quad (235)$$

Now use the identities already proved. The upper-left entry vanishes because $G_F + G_{\bar{F}} = G_> + G_<$. The upper-right entry is

$$\begin{aligned} \frac{1}{2} (G_F + G_> - G_< - G_{\bar{F}}) &= \frac{1}{2} (G_F + G_> - G_< - (G_> + G_< - G_F)) \\ &= G_F - G_< = G_R. \end{aligned} \quad (236)$$

Similarly,

$$\begin{aligned} \frac{1}{2} (G_F - G_> + G_< - G_{\bar{F}}) &= \frac{1}{2} (G_F - G_> + G_< - (G_> + G_< - G_F)) \\ &= G_F - G_> = G_A, \end{aligned} \quad (237)$$

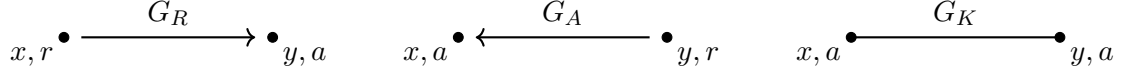


Figure 4: The three free lines in the r/a basis. Retarded and advanced lines carry causal propagation. The Keldysh line carries the thermal occupation factor.

and the lower-right entry is

$$\begin{aligned} \frac{1}{2} (G_F + G_> + G_< + G_{\bar{F}}) &= \frac{1}{2} ((G_F + G_{\bar{F}}) + (G_> + G_<)) \\ &= G_> + G_< = G_K. \end{aligned} \quad (238)$$

Therefore the free propagator table in the r/a basis is

$$G_{r/a} = \begin{array}{c|cc} & r & a \\ \hline r & 0 & G_R \\ a & G_A & G_K \end{array} \quad (239)$$

This is the result that one should remember. It has been obtained only by rotating the four free $+/-$ propagators. The zero in the rr entry is not a new dynamical assumption; it is just the unitarity identity (225) written in a better basis.

For a free scalar field in equilibrium,

$$G_K(k) = [1 + 2n_B(|k_0|)] \rho_\phi(k), \quad \rho_\phi(k) = G_R(k) - G_A(k) = 2\pi \operatorname{sgn}(k_0) \delta(k^2 - m^2). \quad (240)$$

For a Dirac field one has the analogous formula

$$S_K(k) = [1 - 2n_F(k_0)] \rho_D(k), \quad \rho_D(k) = S_R(k) - S_A(k), \quad (241)$$

with

$$\rho_D(k) = 2\pi \operatorname{sgn}(k_0) (\not{k} + m) \delta(k^2 - m^2). \quad (242)$$

For the photon in Feynman gauge,

$$D_{K,\mu\nu}(k) = -g_{\mu\nu} [1 + 2n_B(|k_0|)] 2\pi \operatorname{sgn}(k_0) \delta(k^2). \quad (243)$$

In a general covariant ξ -gauge, the usual longitudinal tensor structure is added to the propagator, but the r/a branch structure is unchanged. Gauge-invariant observables are independent of ξ when all required diagrams are included.

How the interaction vertices are rotated

The propagators came from rotating the free two-point table. The vertices come from rotating the branch labels of the interaction insertions in the operator expansion. This is again not a path-integral statement. It is the algebraic statement that an interaction insertion on C_+ comes with the opposite sign from the same insertion on C_- , as derived in (204).

For vertices it is convenient to use the unnormalised average/difference fields

$$X_a = \frac{1}{2}(X_+ + X_-), \quad X_r = X_+ - X_-, \quad X_+ = X_a + \frac{1}{2}X_r, \quad X_- = X_a - \frac{1}{2}X_r. \quad (244)$$

The source term transforms dually:

$$J_+X_+ - J_-X_- = J_aX_a + J_rX_r, \quad J_a = J_+ - J_-, \quad J_r = \frac{1}{2}(J_+ + J_-). \quad (245)$$

A physical source has $J_+ = J_-$. Therefore it has $J_a = 0$ and only J_r is nonzero. Closed-time-contour unitarity gives

$$\mathcal{Z}[J_r, J_a = 0] = 1. \quad (246)$$

This is the diagrammatic origin of the absence of a bare all-average vertex.

For a local interaction, the branch contribution is the difference between the $+$ and $-$ insertions. Thus one rewrites

$$\mathcal{L}_I[X_+] - \mathcal{L}_I[X_-] = \mathcal{L}_I[X_a + \tfrac{1}{2}X_r] - \mathcal{L}_I[X_a - \tfrac{1}{2}X_r]. \quad (247)$$

The right-hand side is odd in the difference field X_r . Hence every bare vertex contains an odd number of r legs. In particular,

$$\Gamma_{aaa\dots a}^{(0)} = 0. \quad (248)$$

If all fields are average fields, then $X_+ = X_-$ and the two branch insertions cancel exactly.

Momentum conservation in the r/a basis is inherited from the same locality integral as in (209). The only new step is the algebraic expansion in X_a and X_r . For $\lambda\phi^4$ theory,

$$\begin{aligned} \mathcal{L}_I(\phi_+) - \mathcal{L}_I(\phi_-) &= -\frac{\lambda}{4!} [(\phi_a + \tfrac{1}{2}\phi_r)^4 - (\phi_a - \tfrac{1}{2}\phi_r)^4] \\ &= -\frac{\lambda}{6}\phi_r\phi_a^3 - \frac{\lambda}{24}\phi_r^3\phi_a. \end{aligned} \quad (249)$$

Thus the nonzero momentum-space vertices are

$$V_{raaa}^{\phi^4}(p_1, p_2, p_3, p_4) = -i\lambda(2\pi)^4\delta^{(4)}\left(\sum_i p_i\right), \quad (250)$$

$$V_{rrra}^{\phi^4}(p_1, p_2, p_3, p_4) = -i\frac{\lambda}{4}(2\pi)^4\delta^{(4)}\left(\sum_i p_i\right), \quad (251)$$

plus permutations of the external labels. The all-average $aaaa$ vertex is absent.

For Yukawa theory, using the same transformation for ψ and $\bar{\psi}$,

$$\mathcal{L}_I(+) - \mathcal{L}_I(-) = -g\left(\phi_r\bar{\psi}_a\psi_a + \phi_a\bar{\psi}_r\psi_a + \phi_a\bar{\psi}_a\psi_r + \frac{1}{4}\phi_r\bar{\psi}_r\psi_r\right). \quad (252)$$

Therefore the one- r Yukawa vertices carry the spinor-space matrix

$$-ig\delta_{\alpha\beta}(2\pi)^4\delta^{(4)}(p_\phi + p_\psi + p_{\bar{\psi}}), \quad (253)$$

and the three- r vertex carries the same spinor matrix divided by 4.

For QED,

$$\mathcal{L}_I(+) - \mathcal{L}_I(-) = -e\left(A_{r\mu}\bar{\psi}_a\gamma^\mu\psi_a + A_{a\mu}\bar{\psi}_r\gamma^\mu\psi_a + A_{a\mu}\bar{\psi}_a\gamma^\mu\psi_r + \frac{1}{4}A_{r\mu}\bar{\psi}_r\gamma^\mu\psi_r\right). \quad (254)$$

The one- r QED vertices carry

$$-ie\gamma^\mu(2\pi)^4\delta^{(4)}(p_A + p_\psi + p_{\bar{\psi}}), \quad (255)$$

and the three- r vertex carries the same factor divided by 4.

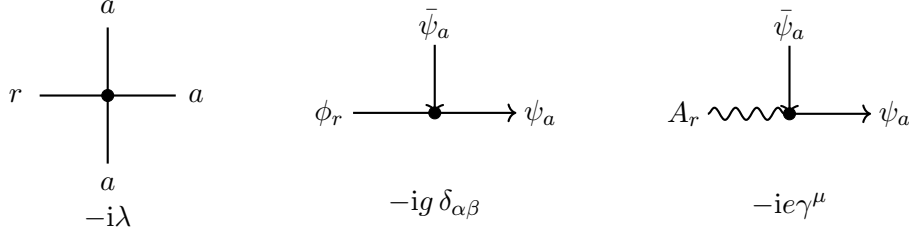


Figure 5: Representative one- r vertices in the r/a basis. Each vertex also carries four-momentum conservation. Vertices with the r leg placed on a different field are obtained from the same algebraic branch difference.

3.3 Examples of physical tree-level thermal effects

We now use one simple model throughout. Let Φ be a heavy scalar and ϕ a light scalar,

$$\mathcal{L} = \frac{1}{2}(\partial\Phi)^2 - \frac{1}{2}M^2\Phi^2 + \frac{1}{2}(\partial\phi)^2 - \frac{1}{2}m^2\phi^2 - \frac{g}{2}\Phi\phi^2, \quad M > 2m. \quad (256)$$

The r/a interaction is

$$\begin{aligned} \mathcal{L}_I(+) - \mathcal{L}_I(-) &= -\frac{g}{2} [(\Phi_a + \frac{1}{2}\Phi_r)(\phi_a + \frac{1}{2}\phi_r)^2 - (\Phi_a - \frac{1}{2}\Phi_r)(\phi_a - \frac{1}{2}\phi_r)^2] \\ &= -\frac{g}{2} \left(\Phi_r\phi_a^2 + 2\Phi_a\phi_a\phi_r + \frac{1}{4}\Phi_r\phi_r^2 \right). \end{aligned} \quad (257)$$

The all-average vertex $\Phi_a\phi_a\phi_a$ is absent. In momentum space, for instance,

$$V_{\Phi_r\phi_a\phi_a}(P, K_1, K_2) = -ig(2\pi)^4\delta^{(4)}(P + K_1 + K_2). \quad (258)$$

Observable 1: nonlinear response

Couple a classical source j to the light field. The second-order response of the heavy field defines a nonlinear response function,

$$\langle\Phi(1)\rangle_j = \langle\Phi(1)\rangle_0 + \int d^4x_2 G_R^{\Phi\phi}(1, 2)j(2) + \frac{1}{2} \int d^4x_2 d^4x_3 G_R^{\Phi\phi\phi}(1; 2, 3)j(2)j(3) + \dots \quad (259)$$

It is better to call this a nonlinear response function rather than a nonlinear susceptibility unless one has already taken the static, homogeneous limit. The causal kernel is the nested-commutator correlator

$$\begin{aligned} G_R^{\Phi\phi\phi}(1; 2, 3) &= (-i)^2\theta(t_1 - t_2)\theta(t_2 - t_3) \langle[[\Phi(1), \phi(2)], \phi(3)]] \\ &\quad + (-i)^2\theta(t_1 - t_3)\theta(t_3 - t_2) \langle[[\Phi(1), \phi(3)], \phi(2)]] \rangle. \end{aligned} \quad (260)$$

This follows by differentiating the real-time evolution operator twice with respect to the physical source. Each difference between a $+$ and a $-$ insertion converts the corresponding source derivative into a commutator, and contour ordering supplies the two possible causal orderings.

In the r/a basis this is the raa correlator in the convention of these notes. At tree level,

$$G_{raa}^{\Phi\phi\phi}(1, 2, 3) = -ig \int d^4z G_R^{\Phi}(1 - z)G_A^{\phi}(z - 2)G_A^{\phi}(z - 3). \quad (261)$$

In momentum space,

$$G_{raa}^{\Phi\phi\phi}(P, K_1, K_2) = -ig(2\pi)^4\delta^{(4)}(P + K_1 + K_2)G_R^{\Phi}(P)G_A^{\phi}(K_1)G_A^{\phi}(K_2). \quad (262)$$

Only G_R and G_A appear. Therefore the tree-level causal nonlinear response is temperature independent in a free theory: the spectral commutator fixes G_R and G_A , while the Bose occupation number appears only in G_K .

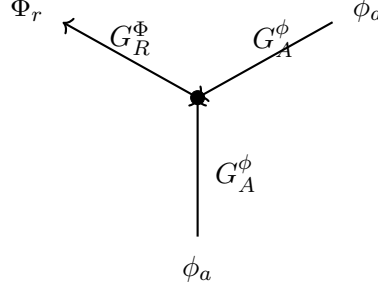


Figure 6: The tree-level nonlinear response in the $\Phi\phi^2$ model. The raa correlator contains only retarded and advanced propagators, so no thermal occupation factor appears at tree level.

Observable 2: thermal gain, absorption, and non-Gaussian noise

A second physical observable is the thermally modified conversion between one heavy excitation and two light excitations. This is the microscopic process behind the imaginary part of the heavy retarded self-energy, but its thermal factors are already visible in three-point building blocks. The relevant mixed response-fluctuation correlator is of rra type. At tree level one obtains sums of products with exactly one Keldysh line, for example

$$G_{rra}^{\Phi\phi\phi}(1,2,3) = -ig \int d^4z \left[G_R^\Phi(1-z) G_R^\phi(2-z) G_K^\phi(z-3) + G_R^\Phi(1-z) G_K^\phi(2-z) G_A^\phi(z-3) \right], \quad (263)$$

with the additional term obtained by exchanging the two identical light legs when required by the external assignment. The concrete point is that the rra correlator is not just “some” thermal vertex; it is a causal response with one fluctuating external leg. It contains precisely the products $G_R G_R G_K$ and $G_R G_K G_A$ at tree level.

The thermal interpretation comes from

$$G_K(k) = G^>(k) + G^<(k). \quad (264)$$

For a positive-energy light mode,

$$G^>(k) \propto 1 + n_B(E_{\mathbf{k}}), \quad G^<(k) \propto n_B(E_{\mathbf{k}}). \quad (265)$$

The factor $1 + n_B$ is emission into a bosonic final state: the 1 is spontaneous emission and the n_B is Bose stimulation into an already occupied state. The factor n_B is absorption from the bath: the process is possible only if a bath particle is present. Thus the processes

$$\Phi \rightarrow \phi\phi, \quad \phi\phi \rightarrow \Phi \quad (266)$$

carry the Bose factors

$$\Phi \rightarrow \phi\phi : (1 + n_1)(1 + n_2), \quad \phi\phi \rightarrow \Phi : n_1 n_2. \quad (267)$$

The net absorptive part of the retarded heavy self-energy is proportional to

$$(1 + n_1)(1 + n_2) - n_1 n_2 = 1 + n_1 + n_2. \quad (268)$$

This is the finite-temperature enhancement of the heavy excitation’s decay width.

The all-average three-point correlator is not a response function. It is a fluctuation observable: the third cumulant, or skewness, of thermal fluctuations. The absence of a bare aaa vertex does

not imply that the aaa correlator vanishes. It means that it must be assembled from allowed vertices and statistical lines. At tree level the $\Phi\phi^2$ model gives

$$\begin{aligned}
G_{aaa}^{\Phi\phi\phi}(1, 2, 3) = & -ig \int d^4z \left[G_K^\Phi(1-z)G_A^\phi(z-2)G_A^\phi(z-3) \right. \\
& + G_A^\Phi(z-1)G_K^\phi(2-z)G_A^\phi(z-3) \\
& \left. + G_A^\Phi(z-1)G_A^\phi(z-2)G_K^\phi(3-z) \right] \\
& + \text{retarded mirror terms fixed by the external ordering.} \tag{269}
\end{aligned}$$

Every displayed term contains a G_K line and therefore a thermal occupation factor. This is why G_{aaa} is naturally interpreted as non-Gaussian noise rather than causal response.

4 The zero-temperature limit: S-matrices, rates and cross-sections

The Schwinger–Keldysh formalism was introduced as an in-in formalism: it computes expectation values in a specified initial state by evolving the system forward and then backward in time. At finite temperature the initial state is thermal, and both branches of the closed contour are essential. At zero temperature the initial density matrix becomes the vacuum projector. This limit is important for two related, but conceptually distinct, reasons.

First, it gives the vacuum version of the contour Wick theorem. If all operator insertions are placed on the forward branch, contour ordering reduces to ordinary time ordering, and the propagator is the usual Feynman propagator. This is the precise sense in which the zero-temperature Schwinger–Keldysh rules contain the ordinary time-ordered vacuum diagrams.

Second, scattering theory asks a different question. It does not merely ask for an expectation value in one state. It compares states prepared in the remote past with states measured in the remote future. That comparison requires in states, out states, and the S-matrix. The purpose of this section is therefore to keep two facts separate. Section 4.1 proves the decoupling of the backward branch in the zero-temperature closed-time-contour theory. Section 4.2 then explains how the same forward-branch diagrams acquire their standard in-out interpretation as S-matrix elements after introducing asymptotic in and out states.

4.1 The zero-temperature closed time contour and forward-branch Feynman diagrams

Let us first ask only what follows from the results of Section 1. The free thermal density matrix is

$$\rho_\beta = \frac{e^{-\beta H_0}}{Z_0}. \tag{270}$$

If the free Hamiltonian has a unique vacuum, then

$$\rho_\beta \longrightarrow |0\rangle\langle 0| \quad (\beta \rightarrow \infty). \tag{271}$$

Thus Gaudin’s theorem becomes the vacuum contour Wick theorem,

$$\langle 0|T_C \phi_{a_1}(x_1) \cdots \phi_{a_n}(x_n)|0\rangle = \sum_{\text{pairings } P} \prod_{(ij) \in P} G_0^{a_i a_j}(x_i - x_j), \quad a_i = \pm. \tag{272}$$

This is the direct zero-temperature consequence of the contour formalism. It is still a closed-time-contour statement; the indices $+$ and $-$ have not disappeared.

Now specialize to the case in which all insertions lie on the forward branch. Along the forward branch, contour ordering is ordinary time ordering. Therefore

$$T_C \phi_+(x_1) \cdots \phi_+(x_n) = T \phi(x_1) \cdots \phi(x_n), \quad (273)$$

and the contour Wick theorem gives

$$\langle 0 | T \phi(x_1) \cdots \phi(x_n) | 0 \rangle = \sum_{\text{pairings } P} \prod_{(ij) \in P} \Delta_F(x_i - x_j), \quad \Delta_F(x - y) = \langle 0 | T \phi(x) \phi(y) | 0 \rangle. \quad (274)$$

For Dirac fields the same statement holds with the usual fermionic signs, or equivalently with determinants of the Feynman propagator

$$S_F(x - y) = \langle 0 | T \psi(x) \bar{\psi}(y) | 0 \rangle. \quad (275)$$

This is the first lesson: the ordinary vacuum Wick theorem is the all-forward-branch part of the zero-temperature contour Wick theorem.

The four free zero-temperature contour propagators are

$$G_0^{++}(x - y) = \Delta_F(x - y), \quad G_0^{--}(x - y) = \Delta_{\bar{F}}(x - y), \quad (276)$$

$$G_0^{-+}(x - y) = G_0^>(x - y), \quad G_0^{+-}(x - y) = G_0^<(x - y), \quad (277)$$

where $\Delta_{\bar{F}}$ is the anti-time-ordered propagator. In momentum space the two Wightman functions have one-sided energy support. For a scalar field,

$$G_0^{-+}(k) = 2\pi\theta(k_0)\delta(k^2 - m^2), \quad (278)$$

$$G_0^{+-}(k) = 2\pi\theta(-k_0)\delta(k^2 - m^2). \quad (279)$$

Dirac and gauge fields have the same support property, with spinor or tensor numerators multiplying these distributions. This support property is the simple reason why the backward branch decouples from forward-branch correlators at zero temperature.

Theorem 4.1 (Backward-branch decoupling at zero temperature). *Consider a connected zero-temperature Schwinger–Keldysh diagram whose external operator insertions all lie on the + branch. Then every connected diagram containing at least one internal --branch vertex vanishes. The surviving connected diagrams contain only +-vertices and only ++ internal lines.*

Proof. Take a connected diagram with all external insertions on C_+ , and suppose that it contains at least one internal --branch vertex. Collect a connected set of --vertices into a maximal blob B , where vertices inside the blob are connected to one another by -- lines. If the diagram contains several such blobs, the following argument applies to each one separately.

There are no external insertions on C_- . Therefore every line which leaves B must end in the + part of the diagram. Choose the momentum q_j on each boundary line to flow out of the -- blob and into the + region. Such a boundary line is a Wightman line of type $-+$. By (278), it is supported only when

$$q_j^0 > 0 \quad \text{for every boundary line of } B. \quad (280)$$

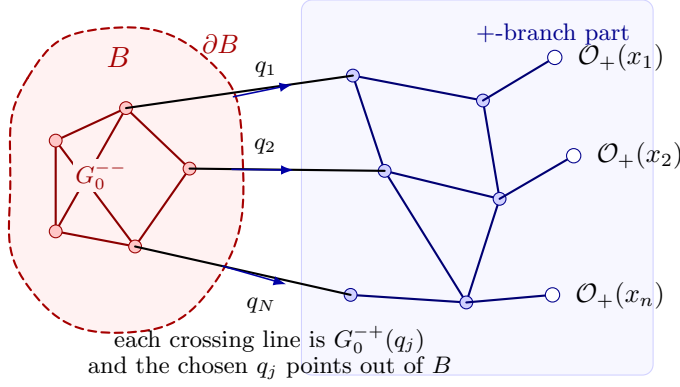
On the other hand, no external momentum is injected into the interior of the -- blob. Momentum conservation at each vertex inside the blob implies conservation of the total momentum crossing its boundary,

$$\sum_{j \in \partial B} q_j = 0. \quad (281)$$

Taking the energy component gives

$$\sum_{j \in \partial B} q_j^0 = 0. \quad (282)$$

(a) A mixed Schwinger–Keldysh graph



(b) Why the graph vanishes

Vacuum Wightman support $G_0^{-+}(q_j) = 2\pi\theta(q_j^0)\delta(q_j^2 - m^2)$ $\implies q_j^0 > 0$ for every j
Momentum conservation in B $\sum q_j^\mu = 0 \implies \sum q_j^0 = 0$
$q_j^0 > 0 \forall j$ gives $\sum_j q_j^0 > 0$, but conservation gives $\sum_j q_j^0 = 0$.
no common support $\implies \mathcal{D}_B = 0$

Figure 7: Graphical proof of backward-branch decoupling at zero temperature. (a) In a connected diagram whose external insertions all lie on C_+ , isolate a maximal component B made of $-$ vertices connected by $--$ propagators. Every line crossing ∂B is then a vacuum Wightman propagator $G_0^{-+}(q_j)$. The thin blue arrows merely define all boundary momenta to point out of B . (b) Vacuum Wightman support requires $q_j^0 > 0$ on every crossing line, while momentum conservation inside B gives $\sum_j q_j^0 = 0$. These conditions have no common support, so every diagram containing a $-$ blob vanishes.

Equations (280) and (282) cannot both hold on the support of the integrand, because every term in the sum is positive. Hence the diagram vanishes. For massless fields, possible zero-energy boundary points are infrared boundary cases; they are treated by the usual wave-packet smearing or infrared regulator before the regulator is removed.

Thus no connected diagram with a $-$ -branch vertex survives. The remaining connected diagrams use only $++$ lines, namely Feynman propagators, and only $+$ -branch vertices. This proves the statement. $\square$

The conclusion is deliberately modest but important. For all external operators on the forward branch,

$$\langle 0|T_C \mathcal{O}_+(x_1) \cdots \mathcal{O}_+(x_n)|0\rangle_{\text{conn}} = \langle 0|T \mathcal{O}(x_1) \cdots \mathcal{O}(x_n)|0\rangle_{\text{conn}}, \quad (283)$$

with the right-hand side computed by diagrams made only of Feynman propagators and ordinary $+$ -branch vertices. These local diagrammatic rules are exactly the rules that appear in the usual in-out formalism. Their interpretation as scattering amplitudes, however, requires the separate in/out construction reviewed in the next subsection.

In momentum space the forward-branch rules are the familiar ones.

Scalar $\lambda\phi^4$ theory. For

$$\mathcal{L} = \frac{1}{2}\partial_\mu\phi\partial^\mu\phi - \frac{1}{2}m^2\phi^2 - \frac{\lambda}{4!}\phi^4, \quad (284)$$

use the scalar Feynman propagator

$$\Delta_F(k) = \frac{i}{k^2 - m^2 + i0^+}, \quad (285)$$

and the local four-point vertex

$$-i\lambda(2\pi)^4\delta^{(4)}(p_1 + p_2 + p_3 + p_4), \quad (286)$$

where all momenta are taken incoming. Integrate each independent loop momentum with $\int d^4\ell/(2\pi)^4$, and include the usual symmetry factor.

Yukawa theory. For

$$\mathcal{L} = \frac{1}{2}\partial_\mu\phi\partial^\mu\phi - \frac{1}{2}m_\phi^2\phi^2 + \bar{\psi}(i\gamma^\mu\partial_\mu - m_\psi)\psi - g\phi\bar{\psi}\psi, \quad (287)$$

use

$$\Delta_F(k) = \frac{i}{k^2 - m_\phi^2 + i0^+}, \quad S_F(k) = \frac{i(\not{k} + m_\psi)}{k^2 - m_\psi^2 + i0^+}, \quad (288)$$

and the Yukawa vertex

$$-ig\delta_{\alpha\beta}(2\pi)^4\delta^{(4)}(p_\phi + p_\psi + p_{\bar{\psi}}). \quad (289)$$

The indices α, β are the spinor indices on the two fermion legs. Since this is a scalar Yukawa coupling, the vertex contains the identity matrix in spinor space. Follow the fermion arrows to order the spinor matrices, and include a minus sign for every closed fermion loop.

QED. For covariant-gauge QED,

$$\mathcal{L} = -\frac{1}{4}F_{\mu\nu}F^{\mu\nu} - \frac{1}{2\xi}(\partial_\mu A^\mu)^2 + \bar{\psi}(i\gamma^\mu\partial_\mu - m)\psi - e\bar{\psi}\gamma^\mu A_\mu\psi, \quad (290)$$

use

$$S_F(k) = \frac{i(\not{k} + m)}{k^2 - m^2 + i0^+}, \quad (291)$$

and the photon propagator

$$D_{\mu\nu}^F(k) = \frac{-i}{k^2 + i0^+} \left[g_{\mu\nu} - (1 - \xi) \frac{k_\mu k_\nu}{k^2 + i0^+} \right], \quad (292)$$

which reduces to $-ig_{\mu\nu}/(k^2 + i0^+)$ in Feynman gauge $\xi = 1$. The QED vertex is

$$-ie\gamma^\mu(2\pi)^4\delta^{(4)}(p_A + p_\psi + p_{\bar{\psi}}), \quad (293)$$

with the Lorentz index μ contracted with the photon line. Again, loop momenta are integrated with $\int d^4\ell/(2\pi)^4$, closed fermion loops carry a minus sign, and the usual symmetry factors are included.

4.2 In-out states, the S-matrix, and LSZ

The previous subsection was still an in-in statement. It concerned vacuum expectation values on a closed time contour, and it showed that, at zero temperature, connected diagrams with only forward-branch external insertions are built from the same Feynman propagators and local vertices that appear in ordinary vacuum perturbation theory. Scattering theory now asks a different question. We no longer want only an expectation value in a single state. We want the amplitude for a state prepared in the remote past to be observed in the detector as another state in the remote future.

This is the reason for introducing in and out states. The interaction need not be literally switched off in a laboratory. Rather, localized wave packets which are very far apart interact negligibly because the theory is local and the wave packets have separated spatial support. In the remote past and future such states move as free relativistic particles. The in/out construction turns the forward-branch Feynman diagrams of the previous subsection into the standard objects used for scattering amplitudes.

This is why the split

$$H = H_0 + H_I \quad (294)$$

needs a physical interpretation. The Hamiltonian H_0 is not the Hamiltonian of bare fields. It is the free Hamiltonian of the stable physical particles, with their physical masses and spin or

polarization labels. The interaction H_I is the residual interaction between these asymptotic particles. If the mass in H_0 were not the physical mass, a one-particle wave packet would not approach the correct free motion at large positive or negative times.

Let $|\alpha\rangle$ be a normalizable free wave-packet state in the Fock space of H_0 . Plane waves will be used later as shorthand, but the following norm limits are meaningful only for wave packets. The incoming scattering state $|\alpha, \text{in}\rangle$ is defined by

$$\lim_{t \rightarrow -\infty} \|e^{-iHt}|\alpha, \text{in}\rangle - e^{-iH_0t}|\alpha\rangle\| = 0. \quad (295)$$

Similarly, the outgoing scattering state $|\beta, \text{out}\rangle$ is defined by

$$\lim_{t \rightarrow +\infty} \|e^{-iHt}|\beta, \text{out}\rangle - e^{-iH_0t}|\beta\rangle\| = 0. \quad (296)$$

These equations say that if the exact interacting state is evolved far enough into the past or future, it becomes indistinguishable in norm from a freely moving wave-packet state.

Multiplying the first condition by e^{iHt} gives the incoming Moller operator

$$\Omega_- = \text{s-lim}_{t \rightarrow -\infty} e^{iHt} e^{-iH_0t}, \quad |\alpha, \text{in}\rangle = \Omega_- |\alpha\rangle. \quad (297)$$

The outgoing Moller operator is

$$\Omega_+ = \text{s-lim}_{t \rightarrow +\infty} e^{iHt} e^{-iH_0t}, \quad |\beta, \text{out}\rangle = \Omega_+ |\beta\rangle. \quad (298)$$

Here “s-lim” means strong limit: the limit is taken after acting on normalizable wave packets. The S-matrix is then

$$S_{\beta\alpha} := \langle\beta, \text{out}|\alpha, \text{in}\rangle = \langle\beta|\Omega_+^\dagger \Omega_-|\alpha\rangle. \quad (299)$$

For finite times the interaction-picture evolution operator is

$$U_I(t_2, t_1) := e^{iH_0t_2} e^{-iH(t_2-t_1)} e^{-iH_0t_1}. \quad (300)$$

It satisfies

$$i \frac{\partial}{\partial t_2} U_I(t_2, t_1) = H_I^I(t_2) U_I(t_2, t_1), \quad (301)$$

$$H_I^I(t) = e^{iH_0t} H_I e^{-iH_0t}, \quad U_I(t_1, t_1) = 1. \quad (302)$$

Therefore, perturbatively,

$$U_I(t_2, t_1) = T \exp \left[-i \int_{t_1}^{t_2} dt H_I^I(t) \right]. \quad (303)$$

Combining this with the Moller operators gives

$$S = \Omega_+^\dagger \Omega_- = \text{s-lim}_{\substack{t_+ \rightarrow +\infty \\ t_- \rightarrow -\infty}} U_I(t_+, t_-). \quad (304)$$

Thus the familiar expression

$$S = T \exp \left[-i \int_{-\infty}^{+\infty} dt H_I^I(t) \right] \quad (305)$$

is a compact perturbative notation for the strong limit on scattering wave packets.

For a scalar particle, define free wave-packet creation and annihilation operators by

$$a_0^\dagger(f) = \int \frac{d^3\mathbf{p}}{(2\pi)^3 \sqrt{2E_{\mathbf{p}}}} f(\mathbf{p}) a_0^\dagger(\mathbf{p}), \quad (306)$$

$$a_0(f) = \int \frac{d^3\mathbf{p}}{(2\pi)^3 \sqrt{2E_{\mathbf{p}}}} f^*(\mathbf{p}) a_0(\mathbf{p}). \quad (307)$$

Then the corresponding in and out operators are

$$a_{\text{in}}^\dagger(f) = \Omega_- a_0^\dagger(f) \Omega_-^\dagger, \quad a_{\text{in}}(f) = \Omega_- a_0(f) \Omega_-^\dagger, \quad (308)$$

$$a_{\text{out}}^\dagger(f) = \Omega_+ a_0^\dagger(f) \Omega_+^\dagger, \quad a_{\text{out}}(f) = \Omega_+ a_0(f) \Omega_+^\dagger. \quad (309)$$

Thus an n -particle incoming wave packet is

$$|f_1, \dots, f_n; \text{in}\rangle = a_{\text{in}}^\dagger(f_1) \cdots a_{\text{in}}^\dagger(f_n) |\Omega\rangle, \quad (310)$$

and similarly for out states. This formula includes both a and $a^\dagger$: creation operators build the external particles, while annihilation operators strip them off in the LSZ reduction below.

Theorem 4.2 (Vacuum phase and vacuum bubbles). *Assume that H_0 has a unique free vacuum $|0\rangle$, that H has a unique physical vacuum $|\Omega\rangle$, that the vacuum energies have been shifted to zero, and that the Moller operators exist on the vacuum. Then*

$$\Omega_- |0\rangle = e^{i\alpha_-} |\Omega\rangle, \quad \Omega_+ |0\rangle = e^{i\alpha_+} |\Omega\rangle. \quad (311)$$

Consequently

$$\langle 0|S|0\rangle = e^{i(\alpha_- - \alpha_+)}. \quad (312)$$

Perturbatively, $\log\langle 0|S|0\rangle$ is the sum of connected vacuum bubbles.

Proof. The Moller operators intertwine the two time evolutions on scattering states:

$$e^{-iHt} \Omega_\pm = \Omega_\pm e^{-iH_0 t}. \quad (313)$$

Differentiating this identity gives $H\Omega_\pm = \Omega_\pm H_0$ on the domain where the Hamiltonians act. Applying it to the free vacuum gives

$$H\Omega_\pm |0\rangle = \Omega_\pm H_0 |0\rangle = 0. \quad (314)$$

Thus $\Omega_\pm |0\rangle$ is a zero-energy eigenvector of H . By uniqueness of the physical vacuum, it must be proportional to $|\Omega\rangle$. Since $\Omega_\pm$ preserve norms on the scattering subspace, the proportionality constants have unit modulus. This proves (311); Eq. (312) follows immediately from $S = \Omega_+^\dagger \Omega_-$.

Diagrammatically, the numerator of

$$\frac{\langle 0|T \phi_I(x_1) \cdots \phi_I(x_n) S|0\rangle}{\langle 0|S|0\rangle} \quad (315)$$

contains all Feynman diagrams, including components disconnected from the external insertions. The logarithm of the denominator is precisely the sum of connected vacuum bubbles. Dividing by $\langle 0|S|0\rangle$ therefore removes exactly those disconnected vacuum components. This is the in-out version of the vacuum-bubble cancellation previously seen on the Schwinger–Keldysh contour. $\square$

We now prove the tree-level scalar LSZ rule slowly. Let $f(x)$ be a positive-frequency solution of the free Klein–Gordon equation,

$$(\square + m^2)f(x) = 0, \quad f(x) = \int \frac{d^3\mathbf{p}}{(2\pi)^3 \sqrt{2E_{\mathbf{p}}}} f(\mathbf{p}) e^{-ip \cdot x}. \quad (316)$$

For any Heisenberg scalar field $\phi(x)$ define the time-dependent wave-packet annihilation operator

$$A_f(t) := i \int d^3\mathbf{x} f^*(x) \overleftrightarrow{\partial}_0 \phi(x), \quad u \overleftrightarrow{\partial}_0 v := u(\partial_0 v) - (\partial_0 u)v. \quad (317)$$

Using the free equation for f and integrating by parts in space gives

$$\frac{d}{dt} A_f(t) = i \int d^3\mathbf{x} f^*(x) (\square + m^2) \phi(x). \quad (318)$$

Hence

$$A_f(+\infty) - A_f(-\infty) = i \int d^4x f^*(x) (\square + m^2) \phi(x). \quad (319)$$

In the asymptotic regions, $A_f(-\infty) = a_{\text{in}}(f)$ and $A_f(+\infty) = a_{\text{out}}(f)$. Therefore

$$a_{\text{out}}(f) - a_{\text{in}}(f) = i \int d^4x f^*(x) (\square + m^2) \phi(x). \quad (320)$$

The adjoint identity strips an incoming creation operator.

Consider first one outgoing particle. For connected matrix elements,

$$\begin{aligned} \langle f, \text{out}; \beta, \text{out} | \alpha, \text{in} \rangle_c &= \langle \beta, \text{out} | a_{\text{out}}(f) | \alpha, \text{in} \rangle_c \\ &= i \int d^4x f^*(x) (\square_x + m^2) \langle \beta, \text{out} | \phi(x) | \alpha, \text{in} \rangle_c. \end{aligned} \quad (321)$$

The term with $a_{\text{in}}(f)$ has been dropped because it only removes a particle already present in the incoming state; for a connected scattering process with the specified external particle it gives a disconnected overlap. This is the first “stripping” step.

If there are two outgoing particles, apply the same identity again to the remaining matrix element. After stripping m outgoing particles and then n incoming particles one obtains, at tree level and with $Z = 1$,

$$\begin{aligned} &\langle g_1, \dots, g_m; \text{out} | f_1, \dots, f_n; \text{in} \rangle_c \\ &= \prod_{r=1}^m \left[i \int d^4x_r g_r^*(x_r) (\square_{x_r} + m^2) \right] \prod_{s=1}^n \left[i \int d^4y_s f_s(y_s) (\square_{y_s} + m^2) \right] \\ &\quad \times \langle \Omega | T \phi(x_1) \cdots \phi(x_m) \phi(y_1) \cdots \phi(y_n) | \Omega \rangle_c. \end{aligned} \quad (322)$$

This is the wave-packet LSZ formula. The plane-wave version is obtained by taking sharply peaked wave packets. The factors $(\square + m^2)$ amputate the external Feynman propagators because

$$(\square_x + m^2) \Delta_F(x - y) = -i \delta^{(4)}(x - y), \quad \Delta_F(k) = \frac{i}{k^2 - m^2 + i0^+}. \quad (323)$$

At tree level each external leg of a Green function is exactly one such propagator attached to a truncated diagram. LSZ removes those external propagators one by one.

Scalar example. For $\lambda\phi^4$ theory, the connected four-point Green function at order λ is

$$G_{4,c}(x_1, x_2, x_3, x_4) = -i\lambda \int d^4z \prod_{r=1}^4 \Delta_F(x_r - z). \quad (324)$$

Applying $i(\square_{x_r} + m^2)$ to each external point gives one delta function per leg. Therefore

$$\left[\prod_{r=1}^4 i(\square_{x_r} + m^2) \right] G_{4,c}(x_1, x_2, x_3, x_4) = -i\lambda \prod_{r=1}^4 \delta^{(4)}(x_r - z) \quad \text{after the } z \text{ integral.} \quad (325)$$

In momentum space this gives

$$S_{fi,c} = i(2\pi)^4 \delta^{(4)}(P_f - P_i) \mathcal{M}, \quad \mathcal{M} = -\lambda. \quad (326)$$

Thus the usual tree-level Feynman rule is recovered as the amputated, on-shell part of the vacuum time-ordered Green function.

For external fermions one replaces the Klein–Gordon inverse propagator by the Dirac inverse propagator. An outgoing fermion contributes an external spinor $\bar{u}_s(p)$, an incoming fermion contributes $u_s(p)$, an outgoing antifermion contributes $v_s(p)$, and an incoming antifermion contributes $\bar{v}_s(p)$. Equivalently, LSZ amputates external fermion propagators by multiplying by $i(\not{\partial} + m)$ or its adjoint form and then putting the spinors on shell. For a photon in a covariant gauge, LSZ amputates the external photon propagator and contracts the result with a physical polarization vector $\varepsilon_\mu^{(\lambda)}(p)$ satisfying $p \cdot \varepsilon^{(\lambda)}(p) = 0$. Only physical external polarizations are used in S-matrix elements; gauge-dependent longitudinal pieces cancel from gauge-invariant amplitudes.

4.3 From amplitudes to rates and cross sections

The invariant amplitude $\mathcal{M}$ is not itself what an experiment records. An experiment records counts: how many decays occur in a time interval, or how many scattering events are registered by a detector surrounding the collision region. The task of this subsection is to explain, in a pedestrian way, how the amplitude appearing in

$$S_{fi,c} = i(2\pi)^4 \delta^{(4)}(P_f - P_i) \mathcal{M}_{fi} \quad (327)$$

becomes a decay rate or a cross section. The essential point is simple: one squares the amplitude, sums over final states, and then divides by the appropriate normalization of the initial state. The only subtlety is that the square of the energy-momentum delta function must be interpreted correctly. For this reason we first work in a finite box and observe the system for a finite time.

Box normalization and the meaning of δ^2

Put the system in a cubic box of volume $V = L^3$ with periodic boundary conditions. The allowed momenta are discrete, and one-particle box states are normalized by

$$\langle \mathbf{p} | \mathbf{q} \rangle_{\text{box}} = \delta_{\mathbf{p}\mathbf{q}}, \quad \sum_{\mathbf{p}} \longrightarrow V \int \frac{d^3 \mathbf{p}}{(2\pi)^3}. \quad (328)$$

The continuum one-particle states used elsewhere in these notes have the Lorentz-covariant normalization

$$\langle \mathbf{p} | \mathbf{q} \rangle_{\text{cont}} = 2E_{\mathbf{p}} (2\pi)^3 \delta^{(3)}(\mathbf{p} - \mathbf{q}). \quad (329)$$

The two conventions are related by

$$|\mathbf{p}\rangle_{\text{cont}} = \sqrt{2E_{\mathbf{p}} V} |\mathbf{p}\rangle_{\text{box}}. \quad (330)$$

This conversion is the origin of the factors $(2EV)^{-1/2}$ carried by each external line in a finite-box transition amplitude.

Let the process have a incoming particles and n outgoing particles,

$$p_1 + \cdots + p_a \longrightarrow q_1 + \cdots + q_n. \quad (331)$$

For a large but finite observation time T , energy conservation is represented by a regulated delta function $\delta_T(E_f - E_i)$, while momentum conservation is represented by a Kronecker delta. The connected finite-box amplitude is

$$S_{fi,c}^{\text{box}} = i \mathcal{M}_{fi} (2\pi) \delta_T(E_f - E_i) V \delta_{\mathbf{P}_f, \mathbf{P}_i} \left[\prod_{\alpha=1}^a \frac{1}{\sqrt{2E_{\alpha} V}} \right] \left[\prod_{r=1}^n \frac{1}{\sqrt{2E_r V}} \right]. \quad (332)$$

Here $E_i = \sum_{\alpha=1}^a E_\alpha$, $E_f = \sum_{r=1}^n E_r$, and similarly for the total three-momentum. Squaring this expression produces two apparently singular objects. In the finite box and finite time they are ordinary large factors:

$$[2\pi\delta_T(E)]^2 \longrightarrow T 2\pi\delta(E), \quad [V\delta_{\mathbf{P},\mathbf{0}}]^2 = V [V\delta_{\mathbf{P},\mathbf{0}}]. \quad (333)$$

Thus the square of the four-dimensional delta function supplies the spacetime volume VT . This is not a new physical effect; it simply says that the transition can occur anywhere in the box and at any time during the observation interval.

After summing over final momenta using $\sum_{\mathbf{q}} \rightarrow V \int d^3\mathbf{q}/(2\pi)^3$, the probability per unit time into the final phase-space element is

$$dW_{i \rightarrow f}^{\text{box}} = V^{1-a} \left[\prod_{\alpha=1}^a \frac{1}{2E_\alpha} \right] |\mathcal{M}_{fi}|^2 d\Phi_n(P_i; q_1, \dots, q_n), \quad (334)$$

where

$$d\Phi_n(P_i; q_1, \dots, q_n) = (2\pi)^4 \delta^{(4)}\left(P_i - \sum_{r=1}^n q_r\right) \prod_{r=1}^n \frac{d^3\mathbf{q}_r}{(2\pi)^3 2E_r} \quad (335)$$

is the Lorentz-invariant n -body phase-space element. Equation (334) is the master transition formula. It is the Weinberg-style box derivation, written in the four-dimensionally normalized convention used in these notes. The final formulas for decay rates and cross sections do not depend on the temporary box. A real experiment uses wave packets rather than perfectly sharp momentum eigenstates; the spatial overlap volume and coherence time of those wave packets play the same role as V and T and drop out of inclusive rates.

It is also useful to state the same result in the language of a macroscopic experiment. For two incoming particles, $a = 2$, (334) gives an event rate for one pair in the box proportional to $1/V$. Since one particle of each kind in the box has densities $n_1 = n_2 = 1/V$, the event rate per unit volume is

$$\frac{dR}{V} = n_1 n_2 \frac{|\mathcal{M}|^2}{4E_1 E_2} d\Phi_n. \quad (336)$$

This is the transition probability per unit time per unit volume before it has been divided by the incident flux. The cross section introduced below is the same statement rewritten as $dR/V = n_1 n_2 v_{\text{rel}} d\sigma$.

If some identical particles occur in the final state, the phase space overcounts physically identical points. One must then include

$$d\Phi_n \longrightarrow \frac{1}{S_f} d\Phi_n, \quad S_f = \prod_A n_A!, \quad (337)$$

where n_A is the number of final particles of species A . This is the phase-space analogue of the familiar symmetry factor: it prevents us from counting the same physical final state several times.

Decay rates and the status of unstable particles

For one initial particle, $a = 1$, the factor V^{1-a} in (334) is absent. The differential transition probability per unit time is therefore

$$d\Gamma(P) = \frac{1}{2E_P} |\mathcal{M}|^2 d\Phi_n(P; q_1, \dots, q_n). \quad (338)$$

This is called the differential decay rate. In the rest frame of the parent, $P = (M, \mathbf{0})$, it becomes

$$d\Gamma_0 = \frac{1}{2M} |\mathcal{M}|^2 d\Phi_n. \quad (339)$$

Integrating over the final phase space gives the total width,

$$\Gamma = \int d\Gamma, \quad \tau = \Gamma^{-1}. \quad (340)$$

The phase-space measure and $|\mathcal{M}|^2$ are invariant. Hence the rate measured for a moving parent differs from the rest-frame rate only through the prefactor:

$$\Gamma_{\text{lab}} = \frac{M}{E_P} \Gamma_0 = \frac{\Gamma_0}{\gamma}, \quad \tau_{\text{lab}} = \gamma \tau_0. \quad (341)$$

This is the usual relativistic time dilation of lifetimes.

There is, however, an important conceptual point. A genuinely unstable particle is not an exact asymptotic eigenstate of the full Hamiltonian. Strictly speaking, therefore, its decay is not an ordinary S-matrix element between stable states at $t = -\infty$ and $t = +\infty$. The formula (338) is instead the leading long-time transition rate obtained from a finite-time calculation. For a finite observation time the energy-conserving delta function is really a peaked function of width $1/T$, for example a sinc-squared function. Only when T is large compared with microscopic time scales does this peak become narrow enough that one may replace it by

$$[2\pi\delta_T(E_f - E_i)]^2 \simeq T 2\pi\delta(E_f - E_i). \quad (342)$$

This is the same large-time limit that underlies Fermi's golden rule. At the same time, if one wants to interpret ΓT as a probability, one should remain in the regime in which the decay probability is small. Thus the simple derivation assumes the window

$$\Delta E^{-1} \ll T \ll \Gamma^{-1}, \quad (343)$$

where ΔE denotes a typical microscopic energy scale controlling the final-state energy resolution. For times comparable to or larger than Γ^{-1} , repeated self-energy insertions must be resummed and the survival probability becomes exponential, $P_{\text{surv}}(T) \simeq e^{-\Gamma T}$. The same number Γ appears in that resummed propagator: for a narrow resonance the pole moves from M^2 to approximately $M^2 - iM\Gamma$. This is why the on-shell amplitude formula is trustworthy: it is the leading term in the long-time, narrow-width limit, even though the unstable particle is not an exact external S-matrix state.

Example: $\Phi \rightarrow \phi\phi$. For

$$\mathcal{L}_I = -\frac{g}{2}\Phi\phi^2, \quad M_\Phi = M, \quad m_\phi = m, \quad (344)$$

the tree amplitude is $\mathcal{M} = -g$. The two final particles are identical, so (337) contributes a factor $1/2!$. In the rest frame,

$$\Gamma_0(\Phi \rightarrow \phi\phi) = \frac{g^2}{32\pi M} \sqrt{1 - \frac{4m^2}{M^2}}, \quad M > 2m. \quad (345)$$

Scattering, incoming flux, and cross sections

For two incoming particles, $a = 2$, the master formula gives a transition probability per unit time proportional to $1/V$. This is exactly what it should do: in a box containing one particle of type 1 and one particle of type 2, the density of either particle is $1/V$, so the collision probability per unit time decreases when the box is made larger.

A cross section is defined by dividing the event rate by the incident flux. In the box, the flux of particle 1 through the single particle of type 2 is

$$\text{box flux} = \frac{v_{\text{rel}}}{V}. \quad (346)$$

Therefore

$$d\sigma = \frac{dW^{\text{box}}}{v_{\text{rel}}/V}. \quad (347)$$

Using (334) with $a = 2$ gives

$$d\sigma = \frac{1}{4E_1 E_2 v_{\text{rel}}} |\mathcal{M}|^2 d\Phi_n(p_1 + p_2; q_1, \dots, q_n). \quad (348)$$

The relative velocity is most cleanly defined by the invariant relation

$$v_{\text{rel}} = \frac{\sqrt{(p_1 \cdot p_2)^2 - m_1^2 m_2^2}}{E_1 E_2}, \quad F \equiv 4E_1 E_2 v_{\text{rel}} = 4\sqrt{(p_1 \cdot p_2)^2 - m_1^2 m_2^2}. \quad (349)$$

Thus the cross-section formula may be written in the invariant form

$$d\sigma = \frac{|\mathcal{M}|^2}{F} d\Phi_n. \quad (350)$$

Let us unpack the relative velocity. In the center-of-mass frame,

$$\mathbf{p}_1 = -\mathbf{p}_2 \equiv \mathbf{p}_i, \quad (351)$$

so

$$v_{\text{rel}}^{\text{CM}} = |\mathbf{p}_i| \left(\frac{1}{E_1} + \frac{1}{E_2} \right). \quad (352)$$

For equal masses this is $2|\mathbf{p}_i|/E$. In a fixed-target laboratory frame, with particle 2 at rest,

$$p_2 = (m_2, \mathbf{0}), \quad v_{\text{rel}}^{\text{lab}} = \frac{|\mathbf{p}_1|}{E_1}, \quad F = 4m_2 |\mathbf{p}_1|. \quad (353)$$

This is physically transparent: the incident flux is projectile speed times target density.

Two-body phase space and differential cross sections

For a two-body final state, $1 + 2 \rightarrow 3 + 4$, the invariant phase space can be evaluated explicitly in the center-of-mass frame. Writing

$$p_1 + p_2 = (\sqrt{s}, \mathbf{0}), \quad s = (p_1 + p_2)^2, \quad (354)$$

one finds

$$d\Phi_2 = \frac{1}{16\pi^2} \frac{|\mathbf{p}_f|}{\sqrt{s}} d\Omega. \quad (355)$$

Since in the center-of-mass frame

$$F = 4|\mathbf{p}_i|\sqrt{s}, \quad (356)$$

we obtain

$$\frac{d\sigma}{d\Omega_{\text{CM}}} = \frac{1}{64\pi^2 s} \frac{|\mathbf{p}_f|}{|\mathbf{p}_i|} |\mathcal{M}|^2, \quad (357)$$

with an additional factor $1/S_f$ if the final particles are identical. The differential cross section tells us how the final particles are distributed in angle. A detector covering a small solid angle $d\Omega$ around direction (θ, φ) records, in an idealized experiment,

$$dN = \mathcal{L} dt \frac{d\sigma}{d\Omega} d\Omega, \quad (358)$$

where $\mathcal{L}$ is the luminosity. Thus $d\sigma/d\Omega$ is not merely a mathematical density; it is the event yield per unit incident flux per unit solid angle.

Example: $\phi\phi \rightarrow \phi\phi$ in $\lambda\phi^4$. At tree level, $\mathcal{M} = -\lambda$. For identical final scalars in the center-of-mass frame,

$$\frac{d\sigma}{d\Omega_{\text{CM}}} = \frac{1}{2} \frac{\lambda^2}{64\pi^2 s} \frac{|\mathbf{p}_f|}{|\mathbf{p}_i|}. \quad (359)$$

For elastic equal-mass scattering, $|\mathbf{p}_f| = |\mathbf{p}_i|$, hence

$$\frac{d\sigma}{d\Omega_{\text{CM}}} = \frac{\lambda^2}{128\pi^2 s}. \quad (360)$$

Lorentz invariance and physical interpretation

The invariant form (350) shows why the total cross section is Lorentz invariant. First,

$$\frac{d^3\mathbf{p}}{2E_{\mathbf{p}}} = d^4p \theta(p^0) \delta(p^2 - m^2) \quad (361)$$

is Lorentz invariant. Therefore $d\Phi_n$ is invariant. Second, the flux factor F in (349) is built from Lorentz scalars. Third, after summing over final spins and averaging over initial spins or physical polarizations, $|\mathcal{M}|^2$ is a scalar. Hence $d\sigma$, understood as a measure on invariant phase space, is Lorentz invariant.

This statement should not be confused with the claim that every angular distribution is the same in every frame. The variables θ , φ , and $d\Omega$ are frame-dependent. Thus $d\sigma/d\Omega$ in the center-of-mass frame and $d\sigma/d\Omega$ in the laboratory frame are related by a Jacobian. What is invariant is the cross section assigned to the same physical region of final-state phase space.

Physically, a cross section is an effective area. In a fixed-target experiment,

$$\text{event rate} = (\text{incident particle flux}) \times (\text{number of targets}) \times \sigma. \quad (362)$$

In collider language this is written as

$$R = \mathcal{L}\sigma, \quad N_{\text{events}} = \mathcal{L}_{\text{int}}\sigma, \quad \mathcal{L}_{\text{int}} = \int dt \mathcal{L}. \quad (363)$$

A detector surrounding the collision region measures final particles in specified angular and energy bins. The theoretical differential cross section predicts the number of events in those bins after multiplication by luminosity, detector acceptance, and efficiencies.

Cross sections are commonly quoted in barns:

$$1 \text{ barn} = 10^{-24} \text{ cm}^2 = 10^{-28} \text{ m}^2, \quad 1 \text{ mb} = 10^{-3} \text{ barn}, \quad (364)$$

$$1 \text{ pb} = 10^{-12} \text{ barn} = 10^{-36} \text{ cm}^2, \quad 1 \text{ fb} = 10^{-15} \text{ barn} = 10^{-39} \text{ cm}^2. \quad (365)$$

For example, an instantaneous luminosity $\mathcal{L} = 10^{34} \text{ cm}^{-2} \text{ s}^{-1}$ and a cross section $\sigma = 1 \text{ pb}$ give

$$R = 10^{34} \times 10^{-36} \text{ s}^{-1} = 10^{-2} \text{ s}^{-1}, \quad (366)$$

about one event every hundred seconds before efficiencies, cuts, and detector acceptance. With an integrated luminosity of 100 fb^{-1} , the same $1 \text{ pb} = 10^3 \text{ fb}$ process gives roughly 10^5 produced events.

4.4 S-matrix diagrammatics and tree-level examples

We now put the preceding discussion into the form used in practical zero-temperature scattering calculations. The S-matrix element is written as

$$S_{\beta\text{out},\alpha\text{in}} = \langle \beta, \text{out} | \alpha, \text{in} \rangle = \delta_{\beta\alpha} + i(2\pi)^4 \delta^{(4)}(P_\beta - P_\alpha) \mathcal{M}_{\beta\alpha}. \quad (367)$$

The connected part of the S-matrix is therefore encoded in the invariant amplitude $\mathcal{M}_{\beta\alpha}$. The LSZ theorem tells us that this amplitude is obtained from the connected time-ordered Green function by amputating external propagators and putting the external particles on shell. Diagrammatically this becomes the ordinary zero-temperature Feynman prescription.

Diagrammatic rules for the S-matrix. For a tree-level process the rules are:

- (i) Draw all topologically distinct connected diagrams with the required external particles. External legs are not assigned propagators; they are replaced by external wave functions or polarization vectors.
- (ii) Assign a four-momentum to every internal line. At every vertex impose four-momentum conservation. If all momenta are drawn as incoming to a vertex, locality gives the factor

$$(2\pi)^4 \delta^{(4)} \left(\sum_{\text{legs at the vertex}} p_i \right). \quad (368)$$

In a connected tree diagram all but one of these delta functions are used to fix internal momenta. The remaining delta function is the overall conservation law in Eq. (367).

- (iii) Assign the usual Feynman propagator to every internal line. For a scalar of mass m ,

$$\frac{i}{q^2 - m^2 + i\epsilon}; \quad (369)$$

for a Dirac fermion,

$$\frac{i(\not{q} + m)}{q^2 - m^2 + i\epsilon}; \quad (370)$$

and for a photon in Feynman gauge,

$$\frac{-ig_{\mu\nu}}{q^2 + i\epsilon}. \quad (371)$$

- (iv) Assign the vertex factors from the interaction Lagrangian. For $\mathcal{L}_I = -g\phi^3/3!$ the scalar vertex is $-ig$. For Yukawa theory $\mathcal{L}_I = -g\phi\bar{\psi}\psi$ the vertex is $-ig\delta_{\alpha\beta}$, the identity matrix in spinor space. For QED $\mathcal{L}_I = -e\bar{\psi}\gamma^\mu A_\mu\psi$ the vertex is $-ie\gamma^\mu$.
- (v) Attach external wave functions. For scalar external legs LSZ only removes the external propagator. For fermions an incoming fermion gives $u_s(p)$, an outgoing fermion gives $\bar{u}_s(p)$, an incoming antifermion gives $\bar{v}_s(p)$, and an outgoing antifermion gives $v_s(p)$. For external photons use physical polarization vectors: incoming photons contribute $\varepsilon_\mu^{(\lambda)}(k)$ and outgoing photons contribute $\varepsilon_\mu^{(\lambda)*}(k)$.
- (vi) Multiply the factors, include the usual minus sign for each closed fermion loop, and include the relative minus signs required when identical external fermions are exchanged. At tree level there are no loop integrations. At loop level one integrates over every independent loop momentum with measure $\int d^4\ell/(2\pi)^4$.

The following examples show how the rules work in practice. We use the standard Mandelstam variables

$$s = (p_1 + p_2)^2, \quad t = (p_1 - p_3)^2, \quad u = (p_1 - p_4)^2, \quad (372)$$

for a $2 \rightarrow 2$ process $p_1 + p_2 \rightarrow p_3 + p_4$.

Example 1: scalar scattering in $\lambda\phi^3$ theory

Consider a real scalar field with

$$\mathcal{L} = \frac{1}{2}\partial_\mu\phi\partial^\mu\phi - \frac{1}{2}m^2\phi^2 - \frac{g}{3!}\phi^3. \quad (373)$$

At tree level the process

$$\phi(p_1) + \phi(p_2) \rightarrow \phi(p_3) + \phi(p_4) \quad (374)$$

receives three contributions. The intermediate scalar can carry momentum $p_1 + p_2$, $p_1 - p_3$, or $p_1 - p_4$. These are the s , t , and u channels. The connected S-matrix element is

$$S_{fi,c} = i(2\pi)^4 \delta^{(4)}(p_3 + p_4 - p_1 - p_2) \mathcal{M}_{\phi^3}, \quad (375)$$

with

$$\mathcal{M}_{\phi^3} = -g^2 \left[\frac{1}{s - m^2 + i\epsilon} + \frac{1}{t - m^2 + i\epsilon} + \frac{1}{u - m^2 + i\epsilon} \right]. \quad (376)$$

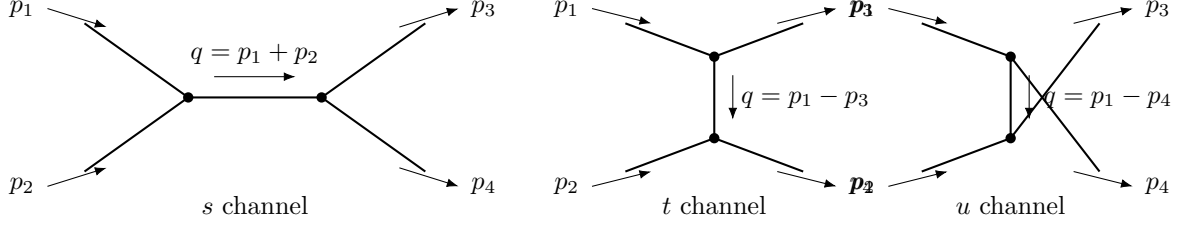


Figure 8: Tree-level diagrams for $\phi(p_1)\phi(p_2) \rightarrow \phi(p_3)\phi(p_4)$ in ϕ^3 theory. Thin arrows show momentum flow. Local momentum conservation fixes the internal momentum to $p_1 + p_2$, $p_1 - p_3$, or $p_1 - p_4$.

This example is useful because students see all three channels in the simplest possible setting. The four-momentum delta function at each vertex fixes the momentum of the internal line; the propagator of that internal line then produces the denominator of the corresponding channel.

Example 2: identical fermion scattering in Yukawa theory

Take

$$\mathcal{L} = \frac{1}{2} \partial_\mu \phi \partial^\mu \phi - \frac{1}{2} m_\phi^2 \phi^2 + \bar{\psi}(i\gamma^\mu \partial_\mu - m)\psi - g\phi\bar{\psi}\psi. \quad (377)$$

Consider

$$\psi(p_1, s_1) + \psi(p_2, s_2) \rightarrow \psi(p_3, s_3) + \psi(p_4, s_4). \quad (378)$$

There are two tree diagrams. In the direct diagram the scalar carries momentum $p_1 - p_3$, so the channel is t . In the exchange diagram it carries momentum $p_1 - p_4$, so the channel is u . Because the external fermions are identical, the exchange contribution has a relative minus sign. The amplitude is

$$\mathcal{M}_{\text{Yukawa}} = -g^2 \left[\frac{(\bar{u}_{s_3}(p_3)u_{s_1}(p_1))(\bar{u}_{s_4}(p_4)u_{s_2}(p_2))}{t - m_\phi^2 + i\epsilon} - \frac{(\bar{u}_{s_4}(p_4)u_{s_1}(p_1))(\bar{u}_{s_3}(p_3)u_{s_2}(p_2))}{u - m_\phi^2 + i\epsilon} \right]. \quad (379)$$



Figure 9: Tree-level identical-fermion scattering in Yukawa theory. Thick arrows are fermion-number arrows on the external and internal fermion lines; the dashed line is the scalar propagator.

This is the relativistic analogue of a force mediated by scalar exchange. The denominator is the propagator of the exchanged scalar, while the spinor bilinears are the external wave functions left after LSZ amputation.

Example 3: Compton scattering in QED

For QED,

$$\mathcal{L} = -\frac{1}{4}F_{\mu\nu}F^{\mu\nu} - \frac{1}{2\xi}(\partial_\mu A^\mu)^2 + \bar{\psi}(i\gamma^\mu\partial_\mu - m)\psi - e\bar{\psi}\gamma^\mu A_\mu\psi. \quad (380)$$

The physical amplitude is independent of the gauge parameter ξ when external photons are transverse and all diagrams of a given order are included. Consider Compton scattering,

$$e^-(p, s) + \gamma(k, \lambda) \rightarrow e^-(p', s') + \gamma(k', \lambda'). \quad (381)$$

There are two tree diagrams: the intermediate electron carries momentum $p + k$ in the s channel, or momentum $p - k'$ in the u channel. With the outgoing photon carrying index μ and the incoming photon carrying index ν , one obtains

$$\begin{aligned} \mathcal{M}_{\text{Comp}} = & -e^2 \bar{u}_{s'}(p') \left[\gamma^\mu \frac{\not{p} + \not{k} + m}{(p + k)^2 - m^2 + i\epsilon} \gamma^\nu \right. \\ & \left. + \gamma^\nu \frac{\not{p} - \not{k}' + m}{(p - k')^2 - m^2 + i\epsilon} \gamma^\mu \right] u_s(p) \varepsilon_\mu^{(\lambda')*}(k') \varepsilon_\nu^{(\lambda)}(k). \end{aligned} \quad (382)$$

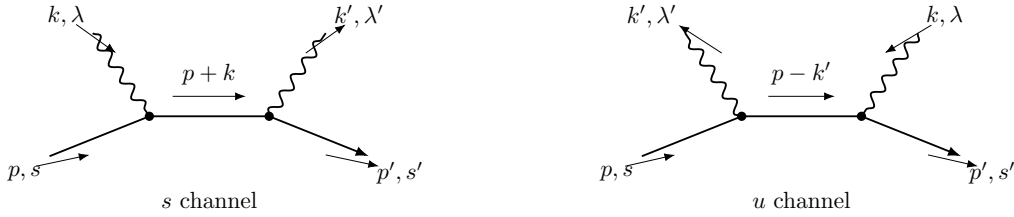


Figure 10: Tree-level Compton scattering. Fermion arrows follow electron number. In the s channel the internal electron carries $p + k$; in the u channel it carries $p - k'$. The photon momentum arrows show which external photon is attached at each vertex.

This example is a good place to emphasize gauge invariance. If we replace the incoming polarization vector by the corresponding photon momentum, the amplitude must vanish:

$$k_\nu \mathcal{M}_{\text{Comp}}^{\mu\nu} = 0, \quad \mathcal{M}_{\text{Comp}} = \varepsilon_\mu^{(\lambda')*}(k') \mathcal{M}_{\text{Comp}}^{\mu\nu} \varepsilon_\nu^{(\lambda)}(k). \quad (383)$$

This is required by gauge invariance because a change of photon polarization by $\varepsilon_\nu(k) \rightarrow \varepsilon_\nu(k) + \alpha k_\nu$ is a gauge transformation of the external photon wave. Physical S-matrix elements cannot depend on this gauge redundancy, so the coefficient of k_ν must vanish. To prove (383), contract (382) with k_ν :

$$k_\nu \mathcal{M}_{\text{Comp}}^{\mu\nu} = -e^2 \bar{u}_{s'}(p') [\gamma^\mu S_F(p + k) \not{k} + \not{k} S_F(p - k') \gamma^\mu] u_s(p), \quad (384)$$

where $S_F(q) = (\not{q} + m)/(q^2 - m^2 + i\epsilon) = 1/(\not{q} - m + i\epsilon)$. Now use the identities

$$\not{k} = S_F^{-1}(p + k) - S_F^{-1}(p), \quad \not{k}' = S_F^{-1}(p') - S_F^{-1}(p - k'), \quad (385)$$

which follow from momentum conservation $p + k = p' + k'$. Then

$$\gamma^\mu S_F(p + k) \not{k} = \gamma^\mu [1 - S_F(p + k) S_F^{-1}(p)], \quad (386)$$

$$\not{k} S_F(p - k') \gamma^\mu = [S_F^{-1}(p') S_F(p - k') - 1] \gamma^\mu. \quad (387)$$

Acting on the external spinors, the extra terms vanish by the Dirac equations $(\not{p} - m)u_s(p) = 0$ and $\bar{u}_{s'}(p')(\not{p}' - m) = 0$. Therefore

$$k_\nu \mathcal{M}_{\text{Comp}}^{\mu\nu} = -e^2 \bar{u}_{s'}(p') [\gamma^\mu - \gamma^\mu] u_s(p) = 0. \quad (388)$$

Exactly the same argument shows $k'_\mu \mathcal{M}_{\text{Comp}}^{\mu\nu} = 0$. Thus the vanishing is not an accident; it is the tree-level manifestation of gauge invariance of the full QED S-matrix.

Example 4: Bhabha scattering in QED

The process

$$e^-(p_1, s_1) + e^+(p_2, s_2) \rightarrow e^-(p_3, s_3) + e^+(p_4, s_4) \quad (389)$$

is called Bhabha scattering. It is pedagogically useful because two different QED channels contribute already at tree level. There is an annihilation diagram in the s channel and a scattering diagram in the t channel. With the standard spinor assignment, the amplitude can be written as

$$\mathcal{M}_{\text{Bhabha}} = -e^2 \left[\frac{(\bar{v}_{s_2}(p_2) \gamma^\mu u_{s_1}(p_1)) (\bar{u}_{s_3}(p_3) \gamma_\mu v_{s_4}(p_4))}{s + i\epsilon} - \frac{(\bar{u}_{s_3}(p_3) \gamma^\mu u_{s_1}(p_1)) (\bar{v}_{s_2}(p_2) \gamma_\mu v_{s_4}(p_4))}{t + i\epsilon} \right]. \quad (390)$$

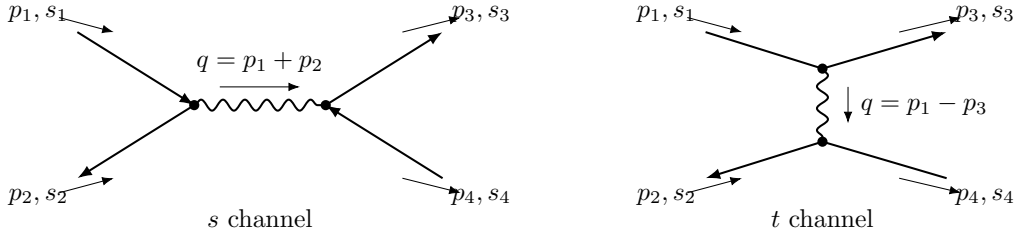


Figure 11: Tree-level Bhabha scattering. Fermion arrows follow electron number; positron momentum flows opposite to the fermion arrow. The exchanged photon carries $p_1 + p_2$ in the annihilation channel and $p_1 - p_3$ in the scattering channel.

The relative sign is fixed by the fermion operator ordering. In practice, Bhabha scattering is also an important luminosity-monitoring process in electron-positron colliders, because it has a clean QED prediction and a large rate in suitable angular regions.

CPT and the S-matrix. The CPT theorem says that every local scalar interaction vertex built from elementary fields is invariant under the combined antiunitary transformation $\Theta = CPT$. We now use this statement where it actually belongs: on the interaction-picture evolution operator. Assume that H_0 contains the physical masses and is also CPT invariant. Since the interaction Hamiltonian is the spatial integral of a local scalar density, CPT gives

$$\Theta H_I^I(t) \Theta^{-1} = H_I^I(-t). \quad (391)$$

The minus sign in the argument is the effect of time reversal. The antiunitarity of Θ also sends $i \rightarrow -i$ and changes time ordering into anti-time ordering. Therefore, at finite cutoff time T ,

$$\begin{aligned}\Theta U_I(T, -T) \Theta^{-1} &= \Theta T \exp \left[-i \int_{-T}^T dt H_I^I(t) \right] \Theta^{-1} \\ &= \bar{T} \exp \left[+i \int_{-T}^T dt H_I^I(-t) \right] \\ &= \bar{T} \exp \left[+i \int_{-T}^T dt H_I^I(t) \right] = U_I(T, -T)^\dagger.\end{aligned}\tag{392}$$

In the second line $\bar{T}$ denotes anti-time ordering. In the third line we changed integration variable $t \mapsto -t$. Taking the strong scattering limit gives

$$\Theta S \Theta^{-1} = S^\dagger, \quad \Theta S^\dagger \Theta^{-1} = S.\tag{393}$$

This is the non-tautological content: CPT invariance of local vertices has become a constraint on the full operator $U_I(+\infty, -\infty)$, hence on S .

For any antiunitary operator one has

$$\langle \beta | S | \alpha \rangle = \langle \Theta \alpha | \Theta S^\dagger \Theta^{-1} | \Theta \beta \rangle.\tag{394}$$

Using (393), this becomes

$$\langle \beta | S | \alpha \rangle = \langle \Theta \alpha | S | \Theta \beta \rangle.\tag{395}$$

Equivalently,

$$\langle \beta, \text{out} | \alpha, \text{in} \rangle = \langle \Theta \alpha, \text{out} | \Theta \beta, \text{in} \rangle,\tag{396}$$

up to conventional one-particle CPT phases. Time reversal is why the role of the initial and final multiparticle labels is exchanged.

Diagrammatically, CPT reverses the direction of time flow, flips particles into antiparticles, and reverses all external momenta and spin labels. Because each local vertex is CPT invariant, the CPT-reversed graph has the same value as the original graph after the external wave functions are transformed appropriately. In the scalar ϕ^3 example the field is self-conjugate, so CPT relates the same s -, t -, and u -channel amplitude with all external momenta reversed. In the Yukawa example it relates fermion scattering to the corresponding antifermion process. In QED it relates Compton scattering of an electron to the CPT-reversed Compton process of a positron with reversed momenta and spins, and likewise relates Bhabha scattering to its reversed positron-electron process.

4.5 Spin-summed amplitudes and cross sections

We now add one practical ingredient which is needed before amplitudes can be compared with scattering experiments. The amplitudes written above still contain external spinors and photon polarization vectors. Experiments usually do not prepare or measure every spin and polarization state. Therefore one sums over final spins and polarizations and averages over initial spins and polarizations. For an unpolarized process,

$$\overline{|\mathcal{M}|^2} = \frac{1}{N_{\text{initial}}} \sum_{\text{spins, pols}} |\mathcal{M}|^2.\tag{397}$$

The bar means “summed over final quantum numbers and averaged over initial quantum numbers.” Once this quantity is known, the two-body center-of-mass formula derived in Section 4.3 gives

$$\frac{d\sigma}{d\Omega} = \frac{1}{64\pi^2 s} \frac{|\mathbf{p}_f|}{|\mathbf{p}_i|} \overline{|\mathcal{M}|^2}.\tag{398}$$

For massless $2 \rightarrow 2$ scattering in the center-of-mass frame, $|\mathbf{p}_f| = |\mathbf{p}_i|$, so only the factor $1/(64\pi^2 s)$ remains.

Spin and polarization sums. The basic spin sums are

$$\sum_s u_s(p) \bar{u}_s(p) = \not{p} + m, \quad \sum_s v_s(p) \bar{v}_s(p) = \not{p} - m. \quad (399)$$

They are the reason spin sums become traces. For example,

$$\sum_{s,s'} (\bar{u}_{s'}(p') \Gamma u_s(p)) (\bar{u}_{s'}(p') \Gamma' u_s(p))^* = \text{Tr} [(\not{p}' + m) \Gamma (\not{p} + m) \bar{\Gamma}'], \quad (400)$$

where

$$\bar{\Gamma}' = \gamma^0 (\Gamma')^\dagger \gamma^0. \quad (401)$$

For external photons, gauge invariance allows us to use

$$\sum_\lambda \varepsilon_\mu^{(\lambda)}(k) \varepsilon_\nu^{(\lambda)*}(k) \longrightarrow -g_{\mu\nu}. \quad (402)$$

This replacement is legitimate only because the Ward identity has already been proved: terms proportional to k_μ or k_ν vanish when contracted into the full gauge-invariant amplitude.

We shall repeatedly use the elementary trace identities (prove these yourself – hint: the tensor structures on the right hand side is fixed by Lorentz covariance up to multiplicative constants)

$$\text{Tr}(\gamma^\mu \gamma^\nu) = 4g^{\mu\nu}, \quad (403)$$

$$\text{Tr}(\gamma^\mu \gamma^\nu \gamma^\rho \gamma^\sigma) = 4(g^{\mu\nu} g^{\rho\sigma} - g^{\mu\rho} g^{\nu\sigma} + g^{\mu\sigma} g^{\nu\rho}), \quad (404)$$

$$\gamma_\mu \gamma^\alpha \gamma^\mu = -2\gamma^\alpha, \quad \gamma_\mu \gamma^\alpha \gamma^\beta \gamma^\mu = 4g^{\alpha\beta}. \quad (405)$$

These are enough for the massless Compton and Bhabha calculations below.

Fierz identities. A Fierz identity is a way of rearranging spinor contractions. It is useful when two amplitudes have different spinor pairings, as happens in Bhabha scattering. The starting point is the completeness of the Dirac matrix basis. With

$$\Gamma^A \in \{1, \gamma^5, \gamma^\mu, \gamma^\mu \gamma^5, \sigma^{\mu\nu}\}, \quad (406)$$

one has the schematic completeness relation (prove this yourself):

$$\delta_{ij} \delta_{kl} = \frac{1}{4} \left[(1)_{il} (1)_{kj} + (\gamma^5)_{il} (\gamma^5)_{kj} + (\gamma^\mu)_{il} (\gamma_\mu)_{kj} - (\gamma^\mu \gamma^5)_{il} (\gamma_\mu \gamma^5)_{kj} + \frac{1}{2} (\sigma^{\mu\nu})_{il} (\sigma_{\mu\nu})_{kj} \right]. \quad (407)$$

Inserting this identity between spinors changes a product of bilinears with pairing (12)(34) into a sum of products with pairing (14)(32). A useful vector example is

$$\begin{aligned} (\bar{\psi}_1 \gamma^\mu \psi_2) (\bar{\psi}_3 \gamma_\mu \psi_4) &= -(\bar{\psi}_1 \psi_4) (\bar{\psi}_3 \psi_2) - (\bar{\psi}_1 \gamma^5 \psi_4) (\bar{\psi}_3 \gamma^5 \psi_2) \\ &\quad + \frac{1}{2} (\bar{\psi}_1 \gamma^\mu \psi_4) (\bar{\psi}_3 \gamma_\mu \psi_2) + \frac{1}{2} (\bar{\psi}_1 \gamma^\mu \gamma^5 \psi_4) (\bar{\psi}_3 \gamma_\mu \gamma^5 \psi_2). \end{aligned} \quad (408)$$

For massless QED it is often simpler to use chiral projectors. Since the vector current preserves chirality, the chiral Fierz identity

$$(\bar{\psi}_1 \gamma^\mu P_L \psi_2) (\bar{\psi}_3 \gamma_\mu P_L \psi_4) = (\bar{\psi}_1 \gamma^\mu P_L \psi_4) (\bar{\psi}_3 \gamma_\mu P_L \psi_2) \quad (409)$$

is especially efficient. The point is not that Fierz identities are new dynamics. They are only algebraic identities that make interference terms look like traces.

Compton scattering

We first compute the unpolarized high-energy Compton cross section in the massless limit. The massive answer is stated afterwards. Let

$$e^-(p) + \gamma(k) \rightarrow e^-(p') + \gamma(k'), \quad (410)$$

and define

$$s = (p + k)^2, \quad t = (k - k')^2 = (p' - p)^2, \quad u = (p - k')^2, \quad s + t + u = 0 \quad (411)$$

for $m = 0$. Removing the external polarization vectors from (382), write

$$\mathcal{M}_{\text{Comp}} = -e^2 \bar{u}(p') A^{\mu\nu} u(p) \varepsilon_\mu^*(k') \varepsilon_\nu(k), \quad (412)$$

with

$$A^{\mu\nu} = \gamma^\mu \frac{\not{p} + \not{k}}{s} \gamma^\nu + \gamma^\nu \frac{\not{p}' - \not{k}'}{u} \gamma^\mu. \quad (413)$$

There are two initial electron spins and two initial photon polarizations, so the average gives a factor 1/4. Using the spin sums and the polarization sum,

$$|\overline{\mathcal{M}_{\text{Comp}}}|^2 = \frac{e^4}{4} (-g_{\mu\mu'}) (-g_{\nu\nu'}) \text{Tr} \left[\not{p}' A^{\mu\nu} \not{p} \bar{A}^{\mu'\nu'} \right], \quad (414)$$

where

$$\bar{A}^{\mu'\nu'} = \gamma^0 (A^{\mu'\nu'})^\dagger \gamma^0. \quad (415)$$

For real external momenta this reverses the order of the gamma matrices inside each chain. The trace splits into s -channel squared, u -channel squared, and interference terms. Using $p^2 = p'^2 = k^2 = k'^2 = 0$ and

$$2p \cdot k = s, \quad 2p \cdot k' = -u, \quad 2p \cdot p' = -t, \quad (416)$$

the gamma algebra gives

$$\sum_{\text{spins, pols}} |\mathcal{M}_{\text{Comp}}|^2 = -8e^4 \left(\frac{s}{u} + \frac{u}{s} \right). \quad (417)$$

Dividing by the four initial states,

$$\boxed{|\overline{\mathcal{M}_{\text{Comp}}}|^2 = -2e^4 \left(\frac{s}{u} + \frac{u}{s} \right)}. \quad (418)$$

This is positive because in the physical Compton region $s > 0$ and $u < 0$. Therefore

$$\boxed{\frac{d\sigma}{d\Omega} = \frac{\alpha^2}{2s} \left[-\frac{s}{u} - \frac{u}{s} \right]} \quad (m = 0). \quad (419)$$

For comparison, if the electron mass is kept and the initial electron is at rest, the same trace calculation gives the Klein–Nishina formula

$$\boxed{\frac{d\sigma}{d\Omega} = \frac{\alpha^2}{2m^2} \left(\frac{\omega'}{\omega} \right)^2 \left(\frac{\omega'}{\omega} + \frac{\omega}{\omega'} - \sin^2 \theta \right), \quad \frac{\omega'}{\omega} = \frac{1}{1 + \frac{\omega}{m}(1 - \cos \theta)}}. \quad (420)$$

In the low-energy limit $\omega \ll m$, this reduces to the Thomson result

$$\frac{d\sigma}{d\Omega} \longrightarrow \frac{\alpha^2}{2m^2} (1 + \cos^2 \theta). \quad (421)$$

Bhabha scattering

Now consider

$$e^-(p_1) + e^+(p_2) \rightarrow e^-(p_3) + e^+(p_4), \quad (422)$$

and take the high-energy limit $m = 0$. The Mandelstam variables are

$$s = (p_1 + p_2)^2, \quad t = (p_1 - p_3)^2, \quad u = (p_1 - p_4)^2, \quad s + t + u = 0. \quad (423)$$

The amplitude in (390) is the sum of the annihilation channel and the scattering channel,

$$\mathcal{M}_s = -e^2 \frac{(\bar{v}_2 \gamma^\mu u_1)(\bar{u}_3 \gamma_\mu v_4)}{s}, \quad \mathcal{M}_t = +e^2 \frac{(\bar{u}_3 \gamma^\mu u_1)(\bar{v}_2 \gamma_\mu v_4)}{t}, \quad (424)$$

where the relative sign is the one already discussed in Section 4.4.

The squared s -channel term is a direct trace:

$$\begin{aligned} \sum |\mathcal{M}_s|^2 &= \frac{e^4}{s^2} \text{Tr}(\not{p}_2 \gamma^\mu \not{p}_1 \gamma^\nu) \text{Tr}(\not{p}_3 \gamma_\mu \not{p}_4 \gamma_\nu) \\ &= 8e^4 \frac{u^2 + t^2}{s^2}. \end{aligned} \quad (425)$$

Similarly,

$$\sum |\mathcal{M}_t|^2 = 8e^4 \frac{u^2 + s^2}{t^2}. \quad (426)$$

The interference term is the only place where the spinor contractions are not immediately in the same order. One may either evaluate it by writing all spinor indices explicitly, or use a Fierz rearrangement to put the two products into the same pairing. The result is (check it yourself)

$$2 \sum \text{Re}(\mathcal{M}_s \mathcal{M}_t^*) = 16e^4 \frac{u^2}{st}. \quad (427)$$

Since $t < 0$ in the physical region, the sign of this term is fixed automatically by $st < 0$. Adding the three pieces and averaging over the two initial electron and two initial positron spins gives

$$\overline{|\mathcal{M}_{\text{Bhabha}}|^2} = 2e^4 \left[\frac{u^2 + t^2}{s^2} + \frac{u^2 + s^2}{t^2} + \frac{2u^2}{st} \right]. \quad (428)$$

Therefore

$$\frac{d\sigma}{d\Omega} = \frac{\alpha^2}{2s} \left[\frac{u^2 + t^2}{s^2} + \frac{u^2 + s^2}{t^2} + \frac{2u^2}{st} \right] \quad (m = 0). \quad (429)$$

In the center-of-mass frame,

$$t = -\frac{s}{2}(1 - \cos \theta), \quad u = -\frac{s}{2}(1 + \cos \theta). \quad (430)$$

The strong forward enhancement comes from the t -channel photon propagator. This is why Bhabha scattering has a large rate at small angles and is so useful for luminosity measurements in electron-positron experiments.

What should be remembered. The practical procedure is always the same:

- (i) compute the invariant amplitude with external wave functions;
- (ii) use Ward identities to replace physical photon polarization sums by $-g_{\mu\nu}$;
- (iii) use spinor completeness relations to turn spin sums into traces;
- (iv) use Fierz identities when interference terms contain different spinor pairings;
- (v) insert $\overline{|\mathcal{M}|^2}$ into the invariant phase-space formula.

This closes the circle: LSZ gives amplitudes, Feynman rules compute them, and phase space and/or spin sums turn them into measurable (un)polarized cross sections.

5 Quasiparticles of a relativistic thermal medium and kinetic theory

The previous sections developed the real-time Schwinger–Keldysh formalism and showed how ordinary zero-temperature scattering theory is recovered when all external operators are placed on the forward branch. We now turn to the genuinely thermal use of the formalism. In a medium there is no empty vacuum through which a disturbance propagates freely. The thermal state is already populated, and any excitation is dressed by its interactions with the surrounding particles. It may acquire a temperature-dependent dispersion relation, it may be damped, and on sufficiently long time scales its distribution may relax toward equilibrium. The natural observables are therefore not exclusive S-matrix elements, but retarded response functions, spectral functions, thermal self-energies, damping rates, and kinetic equations.

The first idea developed in this section is the finite-temperature notion of a quasiparticle. A thermal quasiparticle is not an elementary particle in the vacuum sense. It is a pole, or at least a narrow peak, in a retarded propagator. The real part of the retarded self-energy shifts the location of the peak, while the imaginary part gives its lifetime. This is the real-time replacement of the vacuum mass-shell condition. It is also the simplest setting in which one sees why finite-temperature perturbation theory must often be reorganized: if the thermal self-energy is of the same order as the inverse soft propagator, it must be included already in the leading propagator.

The second idea is that kinetic theory is not a separate classical model added by hand. It emerges from the same Schwinger–Keldysh theory after two controlled approximations: a gradient expansion in slowly varying space-time dependence, and a quasiparticle approximation in which the spectral function is concentrated near its thermal mass shell. In this way the Boltzmann equation appears as the long-wavelength, late-time limit of the Schwinger–Dyson, or Kadanoff–Baym, equations.

The section is organized around one guiding question: what becomes of the particle concept in a hot, interacting relativistic medium? We first answer this question kinematically, by defining quasiparticles from retarded propagators. We then answer it dynamically, by computing one-loop thermal self-energies in a simple Yukawa theory. These examples show how the real part of a retarded self-energy produces a thermal mass, how the imaginary part produces a thermal width, and why soft modes sometimes require resummation. Finally we explain how the same real-time formalism becomes kinetic theory after a gradient expansion and a quasiparticle approximation. The result is the conceptual bridge needed for the following sections: hard thermal loops describe collisionless mean-field response, while transport coefficients require the slower collisional dynamics of kinetic theory.

5.1 Quasiparticles from the retarded propagator

A quasiparticle is a long-lived excitation of the thermal medium. Operationally, it is seen as a sharp peak in a spectral function, or equivalently as a pole of a retarded propagator close to the real frequency axis. For a scalar operator the exact retarded propagator has the form

$$D_R(k) = \frac{i}{k^2 - m^2 - \Pi_R(k)}, \quad k^2 = (k^0)^2 - \mathbf{k}^2, \quad (431)$$

where Π_R is the retarded scalar self-energy. The spectral function is

$$\rho_D(k) = i(D_R(k) - D_A(k)). \quad (432)$$

If Π_R varies slowly near the free energy $E_{\mathbf{k}} = \sqrt{\mathbf{k}^2 + m^2}$, then the pole is approximately

$$k^0 = E_{\mathbf{k}} + \delta E_{\mathbf{k}} - \frac{i}{2}\Gamma_{\mathbf{k}}, \quad (433)$$

with

$$\delta E_{\mathbf{k}} = \frac{\text{Re } \Pi_R(E_{\mathbf{k}}, \mathbf{k})}{2E_{\mathbf{k}}}, \quad \Gamma_{\mathbf{k}} = -\frac{\text{Im } \Pi_R(E_{\mathbf{k}}, \mathbf{k})}{E_{\mathbf{k}}}. \quad (434)$$

Thus the real part of the self-energy shifts the dispersion relation, while the imaginary part gives the inverse lifetime. The sign convention in (434) is chosen so that a stable or damped excitation has $\Gamma_{\mathbf{k}} > 0$, hence $\text{Im } \Pi_R(E_{\mathbf{k}}, \mathbf{k}) < 0$.

For a fermion the exact retarded propagator is

$$S_R(p) = \frac{i}{\not{p} - m - \Sigma_R(p)}. \quad (435)$$

For an excitation with free spinor $u_s(\mathbf{p})$, normalized so that $\bar{u}_s(p)\gamma^0 u_s(p) = 2E_{\mathbf{p}}$, the leading pole shift is

$$\delta E_{\mathbf{p}} - \frac{i}{2}\Gamma_{\mathbf{p}} = \frac{1}{2E_{\mathbf{p}}} \bar{u}_s(p)\Sigma_R(E_{\mathbf{p}}, \mathbf{p})u_s(p). \quad (436)$$

Again the real part changes the dispersion relation and the imaginary part gives damping. At finite temperature these statements are physical: a probe inserted into the medium can propagate, scatter off thermal particles, be absorbed, or stimulate emission. All of these effects are measured by the same retarded self-energy.

This gives the organizing principle for the rest of the section. We will not compute thermal loops merely as formal diagrams. Each loop will be interpreted through the retarded propagator: its real part answers where the quasiparticle peak is located, and its imaginary part answers how long that excitation survives in the medium.

5.2 One-loop self-energies in Yukawa theory from Schwinger–Keldysh rules

We now use Yukawa theory as a small laboratory for thermal self-energies. The point is not that this model is special, but that it contains both a bosonic and a fermionic quasiparticle and therefore displays the two basic structures that recur throughout thermal field theory. The calculation also shows how the branch labels of the Schwinger–Keldysh contour enter ordinary-looking one-loop diagrams. Take

$$\mathcal{L} = \frac{1}{2}\partial_\mu\phi\partial^\mu\phi - \frac{1}{2}m_\phi^2\phi^2 + \bar{\psi}(i\gamma^\mu\partial_\mu - m_\psi)\psi - g\phi\bar{\psi}\psi. \quad (437)$$

The branch vertex is a spinor matrix,

$$(V_a)_{\alpha\beta} = -i\eta_a g \delta_{\alpha\beta}, \quad \eta_+ = +1, \quad \eta_- = -1. \quad (438)$$

In most loop formulae below the identity matrix $\delta_{\alpha\beta}$ is already contracted with the neighbouring fermion propagator. It is nevertheless important to remember that a Yukawa vertex has two spinor indices, unlike a purely scalar vertex.

The useful practical route is first to compute the $++$ and $--$ diagrams, exactly as in ordinary Feynman perturbation theory but with thermal $++$ and $--$ propagators. The lesser and greater components then follow from the same graph with the vertices placed on opposite branches. From these components one reconstructs the retarded self-energy. This is often simpler than trying to guess the retarded diagram directly.

The reader should keep one distinction in mind throughout this subsection. The diagrams below give the self-energies. Their physical interpretation—thermal masses, damping rates, and the need for resummation—will be extracted in the next subsection. Separating these two steps keeps the diagrammatics and the physics from being confused.

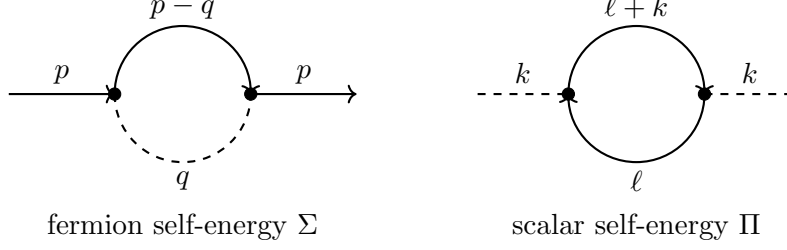


Figure 12: One-loop Yukawa self-energies. Left: a fermion propagates while emitting and reabsorbing a scalar from the thermal bath. Right: a scalar fluctuates into a fermion-antifermion pair. In Schwinger–Keldysh perturbation theory each vertex carries a branch label, and each internal line is one of the thermal components computed in Section 2.

Fermion self-energy. With the convention that the inverse retarded propagator is $S_R^{-1} = S_{0R}^{-1} - \Sigma_R$, the branch self-energy is determined by

$$-i\Sigma_{\alpha\beta}^{ab}(p) = (-i\eta_a g \delta_{\alpha\gamma})(-i\eta_b g \delta_{\delta\beta}) \int \frac{d^4 q}{(2\pi)^4} D_0^{ab}(q) (S_0^{ab}(p-q))_{\gamma\delta}. \quad (439)$$

Thus the identity matrices at the two Yukawa vertices simply transmit the spinor indices through the internal fermion propagator. Suppressing the spinor indices again, this becomes

$$\Sigma^{ab}(p) = ig^2 \eta_a \eta_b \int \frac{d^4 q}{(2\pi)^4} D_0^{ab}(q) S_0^{ab}(p-q). \quad (440)$$

For example,

$$\Sigma^{++}(p) = ig^2 \int \frac{d^4 q}{(2\pi)^4} D_0^{++}(q) S_0^{++}(p-q), \quad (441)$$

$$\Sigma^{--}(p) = ig^2 \int \frac{d^4 q}{(2\pi)^4} D_0^{--}(q) S_0^{--}(p-q). \quad (442)$$

These are the thermal analogues of the ordinary Feynman and anti-Feynman loop integrals. The genuinely thermal information is most transparently displayed by the cut components

$$\Sigma^>(p) = g^2 \int \frac{d^4 q}{(2\pi)^4} D_0^>(q) S_0^>(p-q), \quad \Sigma^<(p) = g^2 \int \frac{d^4 q}{(2\pi)^4} D_0^<(q) S_0^<(p-q), \quad (443)$$

where the overall sign is fixed by the convention used in (439). Writing the free Wightman functions in spectral form,

$$D_0^>(q) = -i(1 + n_B(q^0))\rho_B(q), \quad D_0^<(q) = -in_B(q^0)\rho_B(q), \quad (444)$$

$$S_0^>(r) = -i(1 - n_F(r^0))\rho_F(r), \quad S_0^<(r) = +in_F(r^0)\rho_F(r), \quad (445)$$

we obtain the cut representation

$$\Sigma^>(p) = -g^2 \int \frac{d^4 q}{(2\pi)^4} (1 + n_B(q^0))(1 - n_F(p^0 - q^0))\rho_B(q)\rho_F(p-q), \quad (446)$$

$$\Sigma^<(p) = +g^2 \int \frac{d^4 q}{(2\pi)^4} n_B(q^0)n_F(p^0 - q^0)\rho_B(q)\rho_F(p-q). \quad (447)$$

The factors in these equations have a direct physical interpretation. The factor $1 + n_B$ is Bose enhancement: emission of a scalar into an already occupied mode is stimulated. The factor

n_B is absorption of a scalar already present in the bath. The factors $1 - n_F$ and n_F are Pauli blocking and occupation of thermal fermions.

The retarded self-energy is obtained from its spectral discontinuity. Define

$$\rho_\Sigma(p) = \Sigma^>(p) - \Sigma^<(p). \quad (448)$$

Then

$$\Sigma_R(p) = \Sigma_{\text{ct}}(p) + \int \frac{d\omega}{2\pi} \frac{\rho_\Sigma(\omega, \mathbf{p})}{p^0 - \omega + i\epsilon}, \quad (449)$$

where Σ_{ct} contains counterterms and possible real local terms. Hence

$$\text{Re } \Sigma_R(p) = \Sigma_{\text{ct}}(p) + \mathcal{P} \int \frac{d\omega}{2\pi} \frac{\rho_\Sigma(\omega, \mathbf{p})}{p^0 - \omega}, \quad (450)$$

$$\text{Im } \Sigma_R(p) = -\frac{1}{2} \rho_\Sigma(p). \quad (451)$$

This is the clean separation between the dispersive part and the damping part.

Scalar self-energy. The scalar self-energy is the fermion bubble. The closed fermion loop gives the usual minus sign. With $D_R^{-1} = D_{0R}^{-1} - \Pi_R$,

$$i\Pi^{ab}(k) = -(-i\eta_a g)(-i\eta_b g) \int \frac{d^4\ell}{(2\pi)^4} \text{tr} \left[S_0^{ab}(\ell + k) S_0^{ba}(\ell) \right]. \quad (452)$$

The two identity matrices $\delta_{\alpha\beta}$ from the Yukawa vertices are implicit in the trace; they do not change the trace, but they fix the spinor-index contraction.

Thus the $++$ component is the ordinary thermal Feynman fermion bubble,

$$\Pi^{++}(k) = ig^2 \int \frac{d^4\ell}{(2\pi)^4} \text{tr} \left[S_0^{++}(\ell + k) S_0^{++}(\ell) \right], \quad (453)$$

while Π^{--} is obtained by replacing both propagators by $--$ propagators. The cut components are

$$\Pi^>(k) = -g^2 \int \frac{d^4\ell}{(2\pi)^4} \text{tr} \left[S_0^>(\ell + k) S_0^<(\ell) \right], \quad (454)$$

$$\Pi^<(k) = -g^2 \int \frac{d^4\ell}{(2\pi)^4} \text{tr} \left[S_0^<(\ell + k) S_0^>(\ell) \right]. \quad (455)$$

Substituting (445) gives

$$\Pi^>(k) = -g^2 \int \frac{d^4\ell}{(2\pi)^4} (1 - n_F(\ell^0 + k^0)) n_F(\ell^0) \text{tr} \left[\rho_F(\ell + k) \rho_F(\ell) \right], \quad (456)$$

$$\Pi^<(k) = -g^2 \int \frac{d^4\ell}{(2\pi)^4} n_F(\ell^0 + k^0) (1 - n_F(\ell^0)) \text{tr} \left[\rho_F(\ell + k) \rho_F(\ell) \right]. \quad (457)$$

These equations show Pauli blocking explicitly. The scalar can decay into a fermion-antifermion pair only if the corresponding final states are available; conversely it can be produced by annihilating occupied fermionic modes in the medium.

Define

$$\rho_\Pi(k) = \Pi^>(k) - \Pi^<(k). \quad (458)$$

The retarded scalar self-energy is

$$\Pi_R(k) = \Pi_{\text{ct}}(k) + \int \frac{d\omega}{2\pi} \frac{\rho_\Pi(\omega, \mathbf{k})}{k^0 - \omega + i\epsilon}, \quad (459)$$

so that

$$\text{Re } \Pi_R(k) = \Pi_{\text{ct}}(k) + \mathcal{P} \int \frac{d\omega}{2\pi} \frac{\rho_\Pi(\omega, \mathbf{k})}{k^0 - \omega}, \quad \text{Im } \Pi_R(k) = -\frac{1}{2} \rho_\Pi(k). \quad (460)$$

Fluctuation-dissipation checks. The Wightman functions obey the KMS relations

$$D_0^>(q) = e^{\beta q^0} D_0^<(q), \quad S_0^>(q) = -e^{\beta q^0} S_0^<(q). \quad (461)$$

For the fermion self-energy, the loop energy satisfies $p^0 = q^0 + (p^0 - q^0)$. Therefore

$$\Sigma^>(p) = -e^{\beta p^0} \Sigma^<(p), \quad (462)$$

where the minus sign is the fermionic KMS sign. Equivalently,

$$\Sigma_K(p) = (1 - 2n_F(p^0))(\Sigma_R(p) - \Sigma_A(p)). \quad (463)$$

For the scalar self-energy, the two fermion signs multiply to give a bosonic KMS relation:

$$\Pi^>(k) = e^{\beta k^0} \Pi^<(k), \quad (464)$$

so

$$\Pi_K(k) = (1 + 2n_B(k^0))(\Pi_R(k) - \Pi_A(k)). \quad (465)$$

These are the fluctuation-dissipation relations for self-energies. They say that the same microscopic processes which damp a quasiparticle also determine the noise or fluctuation part of the effective thermal dynamics.

5.3 Explicit thermal masses, widths, and why resummation is necessary

The previous subsection gave the one-loop self-energies directly in Schwinger–Keldysh language. We now extract physical quantities from those formulae. The logic is simple, but it is easy to lose sight of it in the algebra: the real part of the retarded self-energy gives a mass shift, the imaginary part gives a width, and the size of the correction tells us whether the propagator must be resummed.

This subsection is deliberately explicit because two technical points often confuse students the first time they see finite-temperature loops.

First, a finite-temperature loop is not renormalized by inventing new temperature-dependent counterterms. The same ultraviolet divergences are present at zero temperature. The reason is physical and simple: ultraviolet divergences come from very short distances, or equivalently very large loop momenta. At such short distances the system cannot tell whether the state is the vacuum or a thermal ensemble. In a renormalizable local QFT, the divergent part is therefore a local polynomial in the external momenta and masses, and it is the same local divergence already removed at zero temperature. Thus the clean object to compute is the thermal correction

$$\Pi_R^{\text{th}}(K) := \Pi_R(K; T) - \Pi_R(K; 0), \quad \Sigma_R^{\text{th}}(P) := \Sigma_R(P; T) - \Sigma_R(P; 0). \quad (466)$$

After this subtraction the integrals are ultraviolet finite. In real time this finiteness is also visible from the propagators themselves. For example,

$$G_T^{++}(Q) = G_{T=0}^{++}(Q) + 2\pi n_B(|q^0|)\delta(Q^2), \quad (467)$$

and similarly for fermions with a Fermi–Dirac factor. Every term in the difference between the finite- T loop and the zero-temperature loop has at least one factor n_B or n_F . These factors behave as $e^{-E/T}$ at large energy, so the thermal part is ultraviolet convergent.

Second, we must be clear about the relation between the real-time Feynman component computed in Section 5.2, the retarded self-energy which controls quasiparticles, and the Euclidean Matsubara expression which is easiest for static thermal masses. The result in Section 5.2 was a $++$ self-energy. This is the thermal Feynman, or time-ordered, self-energy:

$$G_F \equiv G^{++}, \quad \Pi_F \equiv \Pi^{++}, \quad \Sigma_F \equiv \Sigma^{++}. \quad (468)$$

The retarded object is not identical to the Feynman object. In the $+/-$ basis one obtains it by the causal combination

$$G_R = G^{++} - G^{+-}, \quad \Pi_R = \Pi^{++} + \Pi^{+-}, \quad (469)$$

with the corresponding sign for fermionic self-energies fixed by the convention for the inverse propagator. The most useful statement in equilibrium is spectral. For a bosonic two-point function,

$$G_R(p^0, \mathbf{p}) = \int \frac{d\omega}{2\pi} \frac{\rho(\omega, \mathbf{p})}{p^0 - \omega + i0^+}, \quad (470)$$

whereas the Feynman component is another boundary value of the same spectral data. Equivalently,

$$\text{Re } G_F(p) = \text{Re } G_R(p), \quad \text{Im } G_F(p) = \coth \frac{\beta p^0}{2} \text{Im } G_R(p) \quad (471)$$

for bosons, with $\coth(\beta p^0/2)$ replaced by $\tanh(\beta p^0/2)$ for fermionic two-point functions. The same logic applies to self-energies:

$$\text{Re } \Pi_F(p) = \text{Re } \Pi_R(p), \quad \text{Re } \Sigma_F(p) = \text{Re } \Sigma_R(p), \quad (472)$$

away from singular thresholds, with the imaginary parts related by the corresponding equilibrium thermal factors. This is the precise reason why the real thermal mass may be extracted from the Feynman, or $++$, self-energy computed in the previous subsection. For the thermal mass we only need the real static part of the self-energy. Therefore one may either compute the Euclidean Matsubara self-energy and analytically continue it to the retarded function, or equivalently compute the real part of the thermal Feynman self-energy and use (472). These two routes give the same answer for the mass shift. The greater and lesser self-energies are essential when we compute imaginary parts and widths, because damping is controlled by the discontinuity across real-time cuts.

Now we explain why Euclidean finite-temperature perturbation theory contains Matsubara sums. Analytic continuation alone does not turn an integral into a sum. Instead, the thermal trace $Z = \text{Tr } e^{-\beta H}$ is represented by evolution in imaginary time around a circle of circumference β :

$$\tau \sim \tau + \beta. \quad (473)$$

Bosonic fields are periodic around this Euclidean thermal circle, while fermionic fields are antiperiodic:

$$\phi(\tau + \beta, \mathbf{x}) = \phi(\tau, \mathbf{x}), \quad \psi(\tau + \beta, \mathbf{x}) = -\psi(\tau, \mathbf{x}). \quad (474)$$

Therefore the Fourier modes in the Euclidean time direction are discrete,

$$\omega_n^B = 2\pi n T, \quad \omega_n^F = (2n + 1)\pi T. \quad (475)$$

This is the actual reason for the finite-temperature replacement

$$\int \frac{dq_E^0}{2\pi} \longrightarrow T \sum_n. \quad (476)$$

The analytic-continuation statement has a different role: it tells us how the Euclidean answer is related to the real-time retarded answer. In equilibrium the KMS condition implies analyticity in the strip of complex time. Equivalently, in frequency space the Euclidean correlator and the retarded correlator are boundary values of one analytic function. For a bosonic self-energy one first computes the Euclidean function at discrete external frequencies $K_E = (\omega_m^B, \mathbf{k})$ and then obtains the retarded function by analytic continuation

$$\Pi_R(k^0, \mathbf{k}) = -\Pi_E(\omega_m^B \rightarrow -i(k^0 + i0^+), \mathbf{k}) \quad (477)$$

with the sign convention used in the inverse scalar propagator in these notes. For the static mass we set $K_E = 0$. No dissipative cut is crossed in this static limit, so this Euclidean calculation gives the real static retarded self-energy after applying the above sign convention.

Once the Euclidean loop has been written as a Matsubara sum, the contour C_M is introduced only as a method for evaluating that sum. It is not the original physical object. For a generic loop-energy integrand $f(z)$ define

$$I_B[f] := T \sum_{n \in \mathbb{Z}} f(i\omega_n^B), \quad I_F[f] := T \sum_{n \in \mathbb{Z}} f(i\omega_n^F), \quad (478)$$

where

$$\omega_n^B = 2\pi nT, \quad \omega_n^F = (2n+1)\pi T. \quad (479)$$

The corresponding zero-temperature object is the continuous Euclidean energy integral

$$I_0[f] := \int_{-\infty}^{\infty} \frac{dq_E^0}{2\pi} f(iq_E^0), \quad (480)$$

because at zero temperature Euclidean time is non-compact and q_E^0 is continuous. The thermal correction is the difference $I_{B,F}[f] - I_0[f]$. The goal of the contour trick is simply to compute the discrete sum in (478) efficiently and then subtract the $T = 0$ piece.

To convert the sum into residues, introduce the auxiliary functions

$$h_B(z) = \frac{1}{2} \coth \frac{\beta z}{2}, \quad h_F(z) = \frac{1}{2} \tanh \frac{\beta z}{2}. \quad (481)$$

The important point is this: the Matsubara poles are poles of h_B or h_F , not poles of the original loop integrand $f(z)$. We assume that $f(z)$ is analytic at all Matsubara points and falls fast enough at infinity. In actual loop integrals $f(z)$ is meromorphic and has its own physical poles, such as propagator poles at $z = \pm E$. The original contour C_M in the left panel of Fig. 13 encloses the Matsubara poles on the imaginary axis but is chosen not to enclose the physical poles of f . Since every Matsubara pole has residue T , residue theory gives

$$\oint_{C_M} \frac{dz}{2\pi i} h_B(z) f(z) = T \sum_n f(i\omega_n^B) = I_B[f], \quad \oint_{C_M} \frac{dz}{2\pi i} h_F(z) f(z) = T \sum_n f(i\omega_n^F) = I_F[f]. \quad (482)$$

This equation is the connection between the original Euclidean sum and the contour integral.

Now deform the contour. Because $h_{B,F}(z)f(z)$ falls sufficiently fast at infinity, the integral over a large contour enclosing all singularities vanishes. Hence the contour around the Matsubara poles can be moved to small contours around the poles of f , as shown in the right panel of Fig. 13. Each small contour has the orientation opposite to the original C_M contour; the small contours around $+E$ and $-E$ have the same orientation as each other. Therefore

$$I_B[f] = - \sum_{z_p \in \text{poles}(f)} \text{Res}_{z=z_p} [h_B(z) f(z)], \quad (483)$$

$$I_F[f] = - \sum_{z_p \in \text{poles}(f)} \text{Res}_{z=z_p} [h_F(z) f(z)]. \quad (484)$$

If f has no poles inside the deformation region and the arc at infinity gives no contribution, then the right-hand side is empty and the sum vanishes. In the self-energy integrals below f does have poles: these are precisely the Euclidean propagator poles. Thus the logic is causal: first identify the Matsubara sum (478); then represent it by the C_M contour in (482); finally move the contour as in Fig. 13 to obtain (483) and (484).

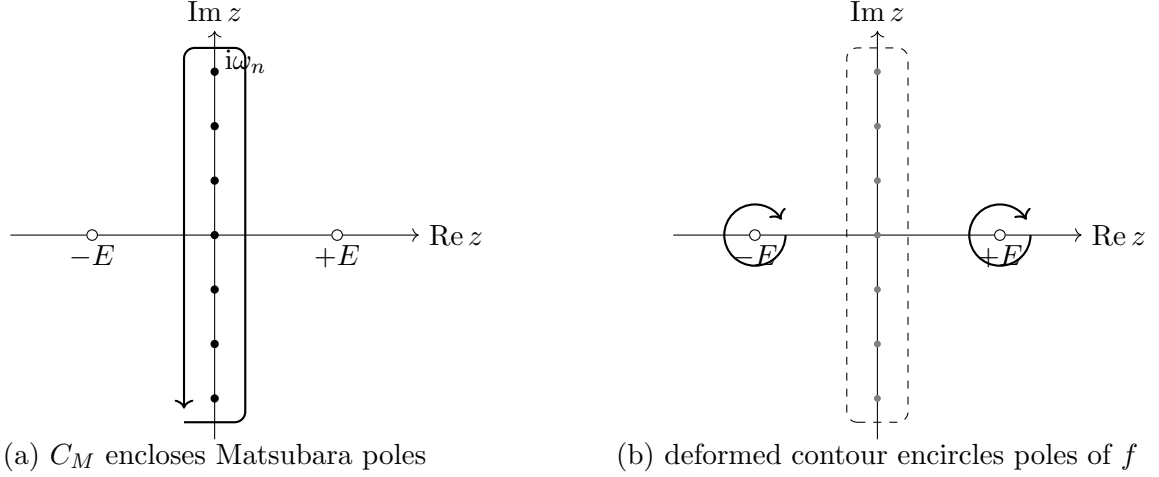


Figure 13: Contour derivation of Matsubara sums. In (a), C_M encloses the poles of $h_B(z) = \frac{1}{2} \coth(\beta z/2)$ or $h_F(z) = \frac{1}{2} \tanh(\beta z/2)$ on the imaginary axis. These poles produce the Matsubara sum. In (b), the contour has been moved away from the imaginary axis. The new small circles encircle the physical poles of the loop integrand $f(z)$, here drawn at $z = \pm E$. Both small circles have the same orientation; it is opposite to the original C_M orientation.

Let us now apply this to the elementary denominator which will appear in the scalar thermal mass. Choose

$$f_E(z) = \frac{1}{E^2 - z^2}, \quad f_E(i\omega_n^{B,F}) = \frac{1}{(\omega_n^{B,F})^2 + E^2}. \quad (485)$$

This makes the connection between (478) and the following sums explicit: the sums below are simply $I_B[f_E]$ and $I_F[f_E]$. The poles of f_E are at $z = +E$ and $z = -E$, so the deformed contours in Fig. 13(b) encircle precisely these two poles. Using (483) and (484), one obtains

$$T \sum_n \frac{1}{(\omega_n^B)^2 + E^2} = \frac{1}{2E} \coth \frac{E}{2T} = \frac{1 + 2n_B(E)}{2E}, \quad (486)$$

$$T \sum_n \frac{1}{(\omega_n^F)^2 + E^2} = \frac{1}{2E} \tanh \frac{E}{2T} = \frac{1 - 2n_F(E)}{2E}. \quad (487)$$

These two formulae are the precise way in which the continuous zero-temperature energy integral is replaced by a thermal sum. The vacuum piece is the $1/(2E)$ term; the remaining term is the thermal correction. For fermions, for instance,

$$T \sum_n \frac{1}{(\omega_n^F)^2 + E^2} - \int \frac{dq_E^0}{2\pi} \frac{1}{(q_E^0)^2 + E^2} = -\frac{n_F(E)}{E}. \quad (488)$$

For bosons the analogous difference is $n_B(E)/E$.

We are now ready to apply these general remarks to the simplest physical quantity: the thermal mass. In this part of the calculation we keep only the real static part of the self-energy. Widths and cuts will be discussed afterward.

Scalar thermal mass from the fermion bubble. The scalar thermal mass is the real part of the scalar self-energy in the static long-wavelength limit,

$$m_{\phi,T}^2 = \Pi_R^{\text{th}}(0, \mathbf{0}). \quad (489)$$

Let us derive this slowly from the one-loop expression of Section 5.2. The real-time ++ component was

$$\Pi^{++}(k) = ig^2 \int \frac{d^4\ell}{(2\pi)^4} \text{tr}[S_0^{++}(\ell+k)S_0^{++}(\ell)]. \quad (5.2 \text{ result})$$

This equation is the thermal Feynman self-energy. For the static mass we use the real part, and therefore we may compute the corresponding Euclidean self-energy and analytically continue it as explained above. The external scalar is bosonic, so the Euclidean external energy is a bosonic Matsubara frequency ω_m^B . For the thermal mass we take the static long-wavelength limit $K_E = (0, \mathbf{0})$. The internal loop lines are fermions; therefore their Euclidean energies are fermionic Matsubara frequencies,

$$\int \frac{dq_E^0}{2\pi} \longrightarrow T \sum_{n \in \mathbb{Z}}, \quad q_E^0 = \omega_n^F = (2n+1)\pi T. \quad (490)$$

Thus the Euclidean continuation of (5.2 result) at zero external momentum is

$$\Pi_E(0) = -g^2 T \sum_n \int \frac{d^3\mathbf{q}}{(2\pi)^3} \text{tr} \left[\frac{-i\gamma_E \cdot Q}{Q^2} \frac{-i\gamma_E \cdot Q}{Q^2} \right] + \text{counterterms}, \quad Q^2 = (\omega_n^F)^2 + q^2. \quad (491)$$

The overall minus sign is the closed fermion loop. Since

$$\text{tr} [(-i\gamma_E \cdot Q)(-i\gamma_E \cdot Q)] = -4Q^2, \quad (492)$$

we obtain

$$\Pi_E(0) = 4g^2 T \sum_n \int \frac{d^3\mathbf{q}}{(2\pi)^3} \frac{1}{(\omega_n^F)^2 + q^2} + \text{counterterms}. \quad (493)$$

The sign convention in these notes is such that the real-time retarded self-energy in the inverse propagator is obtained from the Euclidean result by the usual continuation. For the static mass, the thermal part is therefore read from the thermal part of (493).

Now use the contour identity derived above. Take

$$f(z) = \frac{1}{q^2 - z^2}, \quad f(i\omega_n^F) = \frac{1}{q^2 + (\omega_n^F)^2}. \quad (494)$$

The only poles of f are at $z = \pm q$. Equation (484) gives

$$\begin{aligned} T \sum_n \frac{1}{(\omega_n^F)^2 + q^2} &= - \sum_{z_p = \pm q} \text{Res}_{z=z_p} \left[\frac{1}{2} \tanh \frac{\beta z}{2} \frac{1}{q^2 - z^2} \right] \\ &= - \left[\frac{1}{2} \tanh \frac{\beta q}{2} \left(-\frac{1}{2q} \right) + \frac{1}{2} \tanh \left(-\frac{\beta q}{2} \right) \left(\frac{1}{2q} \right) \right] \\ &= \frac{1}{2q} \tanh \frac{\beta q}{2} = \frac{1 - 2n_F(q)}{2q}. \end{aligned} \quad (495)$$

The zero-temperature Euclidean integral is the same expression with a continuous energy:

$$\int \frac{dq_E^0}{2\pi} \frac{1}{(q_E^0)^2 + q^2} = \frac{1}{2q}. \quad (496)$$

Subtracting the vacuum piece therefore leaves only the part proportional to the occupation number:

$$\begin{aligned} \Pi_E^{\text{th}}(0) &= 4g^2 \int \frac{d^3\mathbf{q}}{(2\pi)^3} \left[T \sum_n \frac{1}{(\omega_n^F)^2 + q^2} - \int \frac{dq_E^0}{2\pi} \frac{1}{(q_E^0)^2 + q^2} \right] \\ &= 4g^2 \int \frac{d^3\mathbf{q}}{(2\pi)^3} \left[\frac{1 - 2n_F(q)}{2q} - \frac{1}{2q} \right] = -4g^2 \int \frac{d^3\mathbf{q}}{(2\pi)^3} \frac{n_F(q)}{q}. \end{aligned} \quad (497)$$

With the retarded inverse propagator convention used in real time, the mass correction is $m_{\phi,T}^2 = -\Pi_E^{\text{th}}(0)$. Hence

$$\begin{aligned} m_{\phi,T}^2 &= 4g^2 \int \frac{d^3\mathbf{q}}{(2\pi)^3} \frac{n_F(q)}{q} = 4g^2 \frac{1}{2\pi^2} \int_0^\infty dq q \frac{1}{e^{q/T} + 1} \\ &= \frac{2g^2 T^2}{\pi^2} \int_0^\infty dx \frac{x}{e^x + 1} = \frac{2g^2 T^2}{\pi^2} \frac{\pi^2}{12} = \frac{g^2 T^2}{6}. \end{aligned} \quad (498)$$

Thus

$$m_{\phi,T}^2 = \frac{g^2 T^2}{6}. \quad (499)$$

The important lesson is not the numerical coefficient alone. The important lesson is the method: the finite answer is obtained by subtracting the zero-temperature integral from the finite-temperature sum, and the contour identity converts the difference into a simple integral over a thermal occupation number.

Fermion thermal mass and the hard-thermal-loop approximation. The fermion thermal mass is extracted from the retarded fermion self-energy of Section 5.2. We now spell out how the hard-thermal-loop expression follows from the one-loop Schwinger–Keldysh formula. The branch expression was

$$\Sigma^{ab}(P) = ig^2 \eta_a \eta_b \int \frac{d^4 Q}{(2\pi)^4} D_0^{ab}(Q) S_0^{ab}(P - Q). \quad (500)$$

The retarded thermal part may be obtained by taking one line to be a thermal on-shell line and the other line to be the retarded or advanced vacuum line. This is just the real-time version of subtracting the zero-temperature part. Schematically,

$$\Sigma_R^{\text{th}}(P) = g^2 \int \frac{d^4 Q}{(2\pi)^4} [D_{\text{th}}(Q) S_R(P - Q) + D_A(Q) S_{\text{th}}(P - Q)], \quad (501)$$

where

$$D_{\text{th}}(Q) = 2\pi n_B(|q^0|) \delta(Q^2), \quad (502)$$

$$S_{\text{th}}(R) = -2\pi n_F(|r^0|) \delta(R^2). \quad (503)$$

The delta-functions are the statement that the thermal particles in the bath are real on-shell particles. This is the first simplification.

The hard-thermal-loop approximation is the controlled limit

$$P = (p^0, \mathbf{p}) \sim gT, \quad Q \sim T. \quad (504)$$

The loop momentum is hard because the distribution functions $n_B(q)$ and $n_F(q)$ weight momenta of order T . The external disturbance is soft because we are interested in collective long-wavelength excitations. Hence

$$P^2 \sim g^2 T^2, \quad P \cdot Q \sim gT^2, \quad P^2 \ll P \cdot Q. \quad (505)$$

Consider first the term in which the scalar line is thermal. The scalar delta-function gives

$$\int dq^0 \delta((q^0)^2 - q^2) F(q^0) = \frac{F(q) + F(-q)}{2q}. \quad (506)$$

For the positive-energy pole $Q = q(1, \hat{\mathbf{q}})$,

$$\begin{aligned} S_R(P - Q) &= \frac{P - Q}{(P - Q)^2 + i\epsilon(p^0 - q^0)} \simeq \frac{-q(\gamma^0 - \hat{\mathbf{q}} \cdot \boldsymbol{\gamma})}{-2q(p^0 - \hat{\mathbf{q}} \cdot \mathbf{p}) + i\epsilon} \\ &= \frac{\gamma^0 - \hat{\mathbf{q}} \cdot \boldsymbol{\gamma}}{2(p^0 - \hat{\mathbf{q}} \cdot \mathbf{p}) + i\epsilon}. \end{aligned} \quad (507)$$

The negative-energy pole gives the same expression after the change of integration variable $\hat{\mathbf{q}} \rightarrow -\hat{\mathbf{q}}$. The two poles therefore remove the factor of $1/2$ from the on-shell measure and give the contribution

$$\Sigma_{R,B}^{\text{th}}(P) = \frac{g^2}{2} \int \frac{d^3\mathbf{q}}{(2\pi)^3} \frac{n_B(q)}{q} \frac{\gamma^0 - \hat{\mathbf{q}} \cdot \boldsymbol{\gamma}}{p^0 - \hat{\mathbf{q}} \cdot \mathbf{p} + i\epsilon}. \quad (508)$$

Now consider the term in which the fermion line is thermal. Write the on-shell fermion momentum as $R = r(1, \hat{\mathbf{r}})$. The scalar denominator is

$$(P - R)^2 \simeq -2r(p^0 - \hat{\mathbf{r}} \cdot \mathbf{p}), \quad (509)$$

while the numerator of the thermal fermion spectral function is

$$\mathcal{R} = r(\gamma^0 - \hat{\mathbf{r}} \cdot \boldsymbol{\gamma}). \quad (510)$$

after the same relabeling of the negative-energy pole. Thus

$$\Sigma_{R,F}^{\text{th}}(P) = \frac{g^2}{2} \int \frac{d^3\mathbf{q}}{(2\pi)^3} \frac{n_F(q)}{q} \frac{\gamma^0 - \hat{\mathbf{q}} \cdot \boldsymbol{\gamma}}{p^0 - \hat{\mathbf{q}} \cdot \mathbf{p} + i\epsilon}. \quad (511)$$

This explains the previously compressed statement. The scalar thermal pole and the fermion thermal pole produce the same leading denominator because both hard particles are nearly lightlike and see the same soft Doppler-shifted frequency $p^0 - \hat{\mathbf{q}} \cdot \mathbf{p}$. Their occupation numbers add because the soft fermion can scatter off either a thermal scalar or a thermal fermion in the bath. Adding (508) and (511) gives

$$\Sigma_R^{\text{th}}(P) = \frac{g^2}{2} \int \frac{d^3\mathbf{q}}{(2\pi)^3} \frac{n_B(q) + n_F(q)}{q} \frac{\gamma^0 - \hat{\mathbf{q}} \cdot \boldsymbol{\gamma}}{p^0 - \hat{\mathbf{q}} \cdot \mathbf{p} + i\epsilon} + O(g^2 T P/T). \quad (512)$$

The radial and angular integrations now factorize. Define

$$m_{\psi,T}^2 := \frac{g^2}{2} \int \frac{d^3\mathbf{q}}{(2\pi)^3} \frac{n_B(q) + n_F(q)}{q}. \quad (513)$$

Using

$$\begin{aligned} \int \frac{d^3\mathbf{q}}{(2\pi)^3} \frac{n_B(q) + n_F(q)}{q} &= \frac{T^2}{2\pi^2} \int_0^\infty dx x \left(\frac{1}{e^x - 1} + \frac{1}{e^x + 1} \right) \\ &= \frac{T^2}{2\pi^2} \left(\frac{\pi^2}{6} + \frac{\pi^2}{12} \right) = \frac{T^2}{8}, \end{aligned} \quad (514)$$

we find

$$m_{\psi,T}^2 = \frac{g^2 T^2}{16}. \quad (515)$$

Therefore the leading soft self-energy is

$$\Sigma_R^{\text{HTL}}(P) = m_{\psi,T}^2 \int \frac{d\Omega_{\hat{\mathbf{q}}}}{4\pi} \frac{\gamma^0 - \hat{\mathbf{q}} \cdot \boldsymbol{\gamma}}{p^0 - \hat{\mathbf{q}} \cdot \mathbf{p} + i\epsilon}. \quad (516)$$

Equation (516) is not a new assumption. It is exactly the leading P/T term of the one-loop Schwinger–Keldysh self-energy of Section 5.2 after the zero-temperature piece has been removed.

The angular integral is elementary. Choose the polar axis along $\mathbf{p}$ and write $p = |\mathbf{p}|$, $z = \hat{\mathbf{q}} \cdot \hat{\mathbf{p}}$. Then

$$I_0(p^0, p) = \frac{1}{2} \int_{-1}^1 \frac{dz}{p^0 - pz + i\epsilon} = \frac{1}{2p} \log \frac{p^0 + p + i\epsilon}{p^0 - p + i\epsilon}, \quad (517)$$

$$I_1(p^0, p) = \frac{1}{2} \int_{-1}^1 \frac{z dz}{p^0 - pz + i\epsilon} = \frac{p^0}{p} I_0(p^0, p) - \frac{1}{p}. \quad (518)$$

Thus

$$\Sigma_R^{\text{HTL}}(p^0, \mathbf{p}) = m_{\psi,T}^2 [\gamma^0 I_0(p^0, p) - \hat{\mathbf{p}} \cdot \boldsymbol{\gamma} I_1(p^0, p)]. \quad (519)$$

For $|p^0| < p$, the logarithm crosses the negative real axis. Hence

$$\text{Im } I_0(p^0, p) = -\frac{\pi}{2p}, \quad \text{Im } I_1(p^0, p) = -\frac{\pi p^0}{2p^2}, \quad (520)$$

and

$$\text{Im } \Sigma_R^{\text{HTL}}(p^0, \mathbf{p}) = -\frac{\pi m_{\psi,T}^2}{2p} \left[\gamma^0 - \frac{p^0}{p} \hat{\mathbf{p}} \cdot \boldsymbol{\gamma} \right] \theta(p^2 - (p^0)^2). \quad (521)$$

This is the imaginary part of the same one-loop retarded self-energy. It is called Landau damping: a space-like soft disturbance can be absorbed by hard particles already present in the bath. For a fermion at zero spatial momentum the HTL self-energy is real,

$$\Sigma_R^{\text{HTL}}(p^0, \mathbf{0}) = \frac{m_{\psi,T}^2}{p^0 + i\epsilon} \gamma^0, \quad (522)$$

so the pole equation gives $(p^0)^2 = m_{\psi,T}^2$.

Thermal masses came from the real part of the retarded self-energy. The next question is what the imaginary part means. Its physical meaning is a rate: the excitation can decay, scatter, or be absorbed by particles already present in the medium. The cleanest way to see this is to put the external particle at rest and rewrite the cut of the one-loop self-energy as an on-shell phase-space integral.

Exact one-loop thermal widths at rest. The imaginary parts of the one-loop self-energies are ultraviolet finite because they are cuts. A cut puts the internal lines on shell and turns the loop into a physical phase-space integral. Let us show this explicitly, rather than merely state it.

For a scalar at rest, write the external momentum as

$$K = (M, \mathbf{0}), \quad (523)$$

where M is the energy of the scalar excitation whose width we want to compute. The relation between the width and the retarded self-energy is

$$\Gamma_\phi(M) = -\frac{1}{M} \text{Im } \Pi_R(M, \mathbf{0}). \quad (524)$$

The cut equations of Section 5.2 give

$$\text{Im } \Pi_R(K) = -\frac{1}{2} [\Pi^>(K) - \Pi^<(K)]. \quad (525)$$

Substitute the fermion spectral functions in the scalar bubble,

$$\rho_F(Q) = 2\pi \mathcal{Q} \text{sgn}(q^0) \delta(Q^2) = 2\pi \mathcal{Q} [\delta(q^0 - q) - \delta(q^0 + q)] \frac{1}{2q}. \quad (526)$$

The two delta-functions in the product of spectral functions enforce

$$Q^2 = 0, \quad R^2 = 0, \quad K = Q + R. \quad (527)$$

This is what “the internal lines go on shell” means. For $M > 0$ and at zero spatial momentum, the decay cut has

$$Q = (q, \mathbf{q}), \quad R = (r, -\mathbf{q}), \quad M = q + r. \quad (528)$$

Since both particles are massless, $q = r = |\mathbf{q}|$, and the energy delta-function becomes

$$\delta(M - q - r) = \delta(M - 2q). \quad (529)$$

The cut therefore gives

$$\begin{aligned} -\text{Im } \Pi_R(M, \mathbf{0}) &= \frac{1}{2} \int \frac{d^3 \mathbf{q}}{(2\pi)^3 2q} \frac{d^3 \mathbf{r}}{(2\pi)^3 2r} (2\pi)^4 \delta^{(4)}(K - Q - R) \\ &\quad \times \sum_{\text{spins}} |\mathcal{M}(\phi \rightarrow \psi \bar{\psi})|^2 [1 - n_F(q) - n_F(r)]. \end{aligned} \quad (530)$$

The thermal factor is also easy to understand. The “1” is spontaneous decay, while the two negative terms are Pauli blocking of the two final fermionic states. At rest, $q = r = M/2$, so the factor becomes $1 - 2n_F(M/2)$.

Now do the phase-space integral. The spatial delta-function sets $\mathbf{r} = -\mathbf{q}$:

$$\begin{aligned} \int d\Phi_2 &:= \int \frac{d^3 \mathbf{q}}{(2\pi)^3 2q} \frac{d^3 \mathbf{r}}{(2\pi)^3 2r} (2\pi)^4 \delta^{(4)}(K - Q - R) \\ &= \int \frac{d^3 \mathbf{q}}{(2\pi)^3 4q^2} (2\pi) \delta(M - 2q) = \frac{4\pi}{(2\pi)^3} \frac{2\pi}{4} \int_0^\infty dq \delta(M - 2q) = \frac{1}{8\pi}. \end{aligned} \quad (531)$$

The spin sum is

$$\sum_{\text{spins}} |\mathcal{M}|^2 = g^2 \text{tr}(\not{Q} \not{R}) = 4g^2 Q \cdot R = 2g^2 M^2, \quad (532)$$

because $Q \cdot R = M^2/2$. Combining (530), (531), and (532), we obtain

$$-\text{Im } \Pi_R(M, \mathbf{0}) = \frac{g^2 M^2}{8\pi} \left[1 - 2n_F\left(\frac{M}{2}\right) \right]. \quad (533)$$

Therefore

$$\Gamma_\phi(M) = \frac{g^2 M}{8\pi} \left[1 - 2n_F\left(\frac{M}{2}\right) \right] = \frac{g^2 M}{8\pi} \tanh \frac{M}{4T}. \quad (534)$$

This proves explicitly that the width is the imaginary part of the same retarded self-energy whose real part shifted the quasiparticle mass.

For a fermionic quasiparticle at rest, write

$$P = (M, \mathbf{0}). \quad (535)$$

Here M denotes the energy of the quasiparticle pole. The matrix self-energy must be projected onto the positive-energy spinor:

$$\Gamma_\psi(M) = -\frac{1}{2M} \bar{u}_s(P) \text{Im } \Sigma_R(P) u_s(P). \quad (536)$$

The cut of the fermion self-energy contains one internal scalar and one internal fermion. Their spectral delta-functions impose

$$Q^2 = 0, \quad R^2 = 0, \quad P = Q + R. \quad (537)$$

For $P = (M, \mathbf{0})$ this again implies $Q^0 = R^0 = M/2$. The cut phase-space formula is

$$\begin{aligned} \Gamma_\psi(M) &= \frac{1}{2M} \int \frac{d^3 \mathbf{q}}{(2\pi)^3 2q} \frac{d^3 \mathbf{r}}{(2\pi)^3 2r} (2\pi)^4 \delta^{(4)}(P - Q - R) \\ &\quad \times \sum_{\text{final spin}} |\mathcal{M}(\psi \rightarrow \psi \phi)|^2 [1 + n_B(r) - n_F(q)]. \end{aligned} \quad (538)$$

The thermal factor has the expected signs: emission of a boson is Bose-enhanced by $1 + n_B$, while occupation of the final fermion state blocks the process by $-n_F$. The two-body phase space is again $1/(8\pi)$. For a fixed initial spin,

$$\sum_{\text{final spin}} |\mathcal{M}|^2 = g^2 \bar{u}_s(P) \not{Q} u_s(P). \quad (539)$$

At rest, $Q^0 = M/2$ and $\bar{u}_s(P) \gamma^0 u_s(P) = 2M$, while the spatial part averages to zero against the rest spinor. Hence

$$\sum_{\text{final spin}} |\mathcal{M}|^2 = g^2 M^2. \quad (540)$$

Thus

$$\begin{aligned} \Gamma_\psi(M) &= \frac{1}{2M} \frac{1}{8\pi} g^2 M^2 \left[1 + n_B \left(\frac{M}{2} \right) - n_F \left(\frac{M}{2} \right) \right] \\ &= \frac{g^2 M}{16\pi} \coth \frac{M}{2T}. \end{aligned} \quad (541)$$

In the last step we used

$$1 + n_B(x) - n_F(x) = 1 + \frac{1}{e^{x/T} - 1} - \frac{1}{e^{x/T} + 1} = \frac{e^{2x/T} + 1}{e^{2x/T} - 1} = \coth \frac{x}{T}. \quad (542)$$

With $x = M/2$ this gives the factor $\coth(M/2T)$. Again the width is UV finite because the cut is a physical on-shell phase-space integral, not a short-distance local divergence.

The mass shifts and widths just computed are perturbative corrections to a propagator. Usually this is enough. However, if the external momentum is soft, the inverse free propagator can be as small as the thermal correction itself. Then the correction is no longer a small insertion; it must be included in the leading propagator.

Why thermal mass resummation is necessary. Thermal mass resummation is necessary for the basic reason that near a soft quasiparticle pole, the loop correction is the same size as the free inverse propagator. For a soft scalar momentum $K \sim gT$,

$$K^2 \sim g^2 T^2, \quad \Pi_R^{\text{th}}(K) \sim g^2 T^2. \quad (543)$$

If we expand the propagator in powers of Π_R ,

$$D_R = D_{0R} + D_{0R} \Pi_R D_{0R} + D_{0R} \Pi_R D_{0R} \Pi_R D_{0R} + \cdots, \quad (544)$$

then $D_{0R} \sim 1/(g^2 T^2)$, so each insertion satisfies

$$\Pi_R D_{0R} \sim \frac{g^2 T^2}{g^2 T^2} \sim 1. \quad (545)$$

The second term is not smaller than the first, and the third is not smaller than the second. This is what it means to say that expanding in powers of Π_R is not a controlled approximation for soft modes. One must sum the geometric series:

$$D_R^{\text{resum}}(K) = \frac{i}{K^2 - m_\phi^2 - \Pi_R(K)}. \quad (546)$$

In practice this is implemented by adding and subtracting a thermal mass term,

$$\mathcal{L}_0 \rightarrow \mathcal{L}_0 - \frac{1}{2} m_{\phi,T}^2 \phi^2, \quad \mathcal{L}_I \rightarrow \mathcal{L}_I + \frac{1}{2} m_{\phi,T}^2 \phi^2. \quad (547)$$

The first copy changes the free propagator, so the zeroth-order quasiparticle already has the correct thermal mass. The second copy is treated as a counterterm insertion, so no contribution is counted twice. This is a reorganization of perturbation theory around the actual excitations of the thermal medium.

Figure 14: Dyson resummation of self-energy insertions. For soft thermal momenta, each insertion is as important as the free inverse propagator, so the whole geometric series must be summed.

5.4 From Schwinger–Dyson equations to the Boltzmann equation

So far we have used Schwinger–Keldysh perturbation theory to characterize quasiparticles in equilibrium. The same formalism also explains why kinetic theory is the correct language for their slow evolution. This is the payoff of using a real-time contour: it does not only compute equilibrium spectral data, but also gives evolution equations for correlation functions when the state is allowed to vary slowly in space and time.

We sketch the standard route, following the spectral/statistical organization used by Berges in his introduction to nonequilibrium quantum field theory.¹ For a real scalar field define

$$F(x, y) = \frac{1}{2} \langle \{ \phi(x), \phi(y) \} \rangle, \quad \rho(x, y) = i \langle [\phi(x), \phi(y)] \rangle. \quad (548)$$

The contour propagator can be written as

$$G(x, y) = F(x, y) - \frac{i}{2} \rho(x, y) \operatorname{sgn}_C(x^0 - y^0). \quad (549)$$

The self-energy is decomposed in the same way,

$$\Sigma(x, y) = \Sigma_F(x, y) - \frac{i}{2} \Sigma_\rho(x, y) \operatorname{sgn}_C(x^0 - y^0). \quad (550)$$

The Schwinger–Dyson equation on the closed time path is

$$\int_C d^4 z G^{-1}(x, z) G(z, y) = i \delta_C^{(4)}(x - y), \quad G^{-1} = G_0^{-1} - \Sigma. \quad (551)$$

Resolving the contour equation into its spectral and statistical parts gives the Kadanoff–Baym equations,

$$[\square_x + M^2(x)] F(x, y) = - \int_{t_0}^{x^0} d^4 z \Sigma_\rho(x, z) F(z, y) + \int_{t_0}^{y^0} d^4 z \Sigma_F(x, z) \rho(z, y), \quad (552)$$

$$[\square_x + M^2(x)] \rho(x, y) = - \int_{y^0}^{x^0} d^4 z \Sigma_\rho(x, z) \rho(z, y). \quad (553)$$

Here $M^2(x)$ contains the local mass term and local tadpole corrections. Equation (552) evolves the occupation information. Equation (553) evolves the spectrum. This is the quantum generalization of the statement that kinetic theory needs both “how many excitations there are” and “what their dispersion relation is.”

To see the Boltzmann equation, introduce center and relative coordinates,

$$X = \frac{x + y}{2}, \quad s = x - y, \quad (554)$$

and Wigner transform, for example

$$F(X, p) = \int d^4 s e^{ip \cdot s} F\left(X + \frac{s}{2}, X - \frac{s}{2}\right). \quad (555)$$

¹J. Berges, *Introduction to Nonequilibrium Quantum Field Theory*, arXiv:hep-ph/0409233.

If the state varies slowly on microscopic time scales, products of convolutions may be expanded in gradients. To leading order one obtains the kinetic equation

$$2p^\mu \partial_{X^\mu} F(X, p) = \Sigma_\rho(X, p) F(X, p) - \Sigma_F(X, p) \rho(X, p). \quad (556)$$

This equation is still fully quantum in the sense that it is written for the statistical and spectral functions, not for classical particles.

The quasiparticle limit is the final approximation. For weak damping the spectral function is sharply peaked near the mass shell,

$$\rho(X, p) \simeq 2\pi \operatorname{sgn}(p^0) \delta(p^2 - m_T^2(X)), \quad (557)$$

and the statistical function is parameterized by a phase-space distribution,

$$F(X, p) = \left(\frac{1}{2} + f(X, \mathbf{p}) \right) \rho(X, p) \quad (p^0 > 0). \quad (558)$$

Putting this ansatz into (556), integrating over positive energy, and using $\mathbf{v}_\mathbf{p} = \nabla_\mathbf{p} E_\mathbf{p}$, one obtains the Boltzmann equation

$$(\partial_t + \mathbf{v}_\mathbf{p} \cdot \nabla_\mathbf{x}) f(t, \mathbf{x}, \mathbf{p}) = C[f](t, \mathbf{x}, \mathbf{p}). \quad (559)$$

The collision term is the on-shell version of the right-hand side of (556). In greater/lesser notation it has the schematic form

$$C[f] = \frac{1}{2E_\mathbf{p}} \left[i\Sigma^<(X, p) iG^>(X, p) - i\Sigma^>(X, p) iG^<(X, p) \right]_{p^0=E_\mathbf{p}}. \quad (560)$$

The first term is gain: interactions produce a particle in the mode $\mathbf{p}$. The second term is loss: interactions remove a particle from that mode. In a $2 \leftrightarrow 2$ process these terms become the usual products

$$(1 + f_1)(1 + f_2)f_3f_4 - f_1f_2(1 + f_3)(1 + f_4) \quad (561)$$

for bosons, with $(1 - f)$ replacing $(1 + f)$ for fermionic final-state blocking.

Thermal equilibrium is a fixed point already in the full quantum equations. In equilibrium the KMS relation implies

$$G_{\text{eq}}^>(p) = e^{\beta p^0} G_{\text{eq}}^<(p), \quad \Sigma_{\text{eq}}^>(p) = e^{\beta p^0} \Sigma_{\text{eq}}^<(p) \quad (562)$$

for bosonic correlators, with the corresponding fermionic minus sign for fermionic two-point functions. Equivalently,

$$F_{\text{eq}}(p) = \left(\frac{1}{2} + n_B(p^0) \right) \rho_{\text{eq}}(p), \quad \Sigma_{F, \text{eq}}(p) = \left(\frac{1}{2} + n_B(p^0) \right) \Sigma_{\rho, \text{eq}}(p). \quad (563)$$

Substituting (563) into the right-hand side of (556) gives zero identically. Thus equilibrium is not merely a fixed point of the approximate Boltzmann equation. It is a fixed point of the underlying Schwinger–Dyson /Kadanoff–Baym evolution itself. The Boltzmann equation inherits this property after the gradient and quasiparticle approximations.

The main lesson of this section is therefore not a particular mass formula or decay rate. It is the hierarchy of real-time descriptions. Retarded propagators define thermal quasiparticles. Their self-energies determine mass shifts and widths. When the relevant fields vary slowly, the same theory becomes kinetic theory. The next section applies this hierarchy to gauge fields in a relativistic plasma, where soft electromagnetic disturbances are governed by the hard thermal loop effective theory.

6 Hard thermal loop effective theory of the QED plasma

The preceding section described thermal self-energies and kinetic theory in general terms. Gauge theory plasmas bring in a new feature: long-wavelength electromagnetic fields are themselves collective degrees of freedom of the medium. In a systematic approximation, these long-wavelength (soft) electromagnetic fields are self-consistent mean fields generated by the particle-like (hard) degrees of freedom. In a weakly coupled QED plasma, the typical momentum of a charged particle in the heat bath is hard, of order T , whereas the collective gauge fields of interest may carry soft momenta of order eT . This separation of scales makes certain one-loop diagrams as important as the tree-level soft inverse propagator. The resulting contributions are the hard thermal loops.

The importance of hard thermal loops is not merely that they give another finite-temperature correction. They are the leading gauge-invariant effective theory for soft electromagnetic fields interacting with hard charged particles. They determine Debye screening, plasma oscillations, Landau damping, and the soft fermion quasiparticle branches. Equivalently, they describe the collisionless response of an almost free gas of hard particles to slowly varying gauge fields. This is why the same formulas can be derived in two apparently different ways: from one-loop Schwinger–Keldysh perturbation theory and from the Vlasov equation.

This section first derives the retarded photon self-energy in the hard-loop limit and uses it to discuss screening, plasmon dispersion relations, and Landau damping. It then formulates the corresponding HTL effective theory. In QED this effective theory is especially transparent: the photon HTL action generates the polarization tensor, while the fermion HTL action generates the soft fermion self-energy and the associated gauge-invariant photon–fermion vertices. The same add-and-subtract logic will later reappear in transport, where the relevant resummation is not the HTL resummation of soft mean fields but the ladder resummation associated with collisions.

6.1 Self-energy, Debye screening, plasma oscillations and Landau damping

We begin with the simplest and most important HTL observable: the retarded photon self-energy. It is the kernel which relates a soft electromagnetic field to the current induced in the plasma. Once this kernel is known, one can read off electric screening, collective plasma oscillations, and the absorptive response responsible for Landau damping.

Consider a weakly coupled QED plasma in equilibrium. The hard particles in the loop have momenta of order the temperature, while the external electromagnetic disturbance is soft:

$$Q^\mu \sim T, \quad K^\mu \sim eT. \quad (564)$$

The loop momentum Q is therefore carried by ordinary thermal electrons and positrons in the bath. The external momentum K belongs to a slowly varying collective field. The expression “hard thermal loop” refers precisely to this kinematics: the loop momentum is hard, the external legs are soft, and the leading answer survives because the hard thermal population is large.

This scale separation has a simple physical meaning. A hard particle with momentum $\mathbf{q}$ moves with almost fixed velocity

$$v^\mu = (1, \hat{\mathbf{q}}), \quad \hat{\mathbf{q}} = \frac{\mathbf{q}}{q}, \quad q = |\mathbf{q}|. \quad (565)$$

A soft gauge field varies slowly on the microscopic scale $1/T$, so the hard particle sees it as a slowly varying background. Algebraically this is why denominators simplify:

$$(Q + K)^2 = Q^2 + 2Q \cdot K + K^2 \simeq 2qv \cdot K, \quad v \cdot K = k^0 - \hat{\mathbf{q}} \cdot \mathbf{k}. \quad (566)$$

The factor $q \sim T$ is hard, while $v \cdot K \sim gT$ is soft. This is the origin of all HTL denominators.

Perturbative Schwinger–Keldysh derivation of the longitudinal polarization. Consider QED with one massless Dirac fermion. The one-loop retarded photon self-energy is the fermion bubble shown in Fig. 15. We derive it directly from Schwinger–Keldysh perturbation theory, because this makes clear that the answer is a real-time response function.

The electromagnetic current is

$$j^\mu(x) = e\bar{\psi}(x)\gamma^\mu\psi(x), \quad (567)$$

and the retarded polarization is the current-current response

$$\Pi_R^{\mu\nu}(x-y) = -i\theta(x^0-y^0) \langle [j^\mu(x), j^\nu(y)] \rangle. \quad (568)$$

Equivalently, in terms of greater and lesser current correlators,

$$\rho_\Pi^{\mu\nu}(K) = \Pi^{>\mu\nu}(K) - \Pi^{<\mu\nu}(K), \quad \Pi_R^{\mu\nu}(K) = \int \frac{d\omega}{2\pi} \frac{\rho_\Pi^{\mu\nu}(\omega, \mathbf{k})}{k^0 - \omega + i0^+}. \quad (569)$$

The one-loop SK rules give

$$\Pi^{>\mu\nu}(K) = -e^2 \int \frac{d^4Q}{(2\pi)^4} \text{tr}[\gamma^\mu S^>(Q+K) \gamma^\nu S^<(Q)], \quad (570)$$

$$\Pi^{<\mu\nu}(K) = -e^2 \int \frac{d^4Q}{(2\pi)^4} \text{tr}[\gamma^\mu S^<(Q+K) \gamma^\nu S^>(Q)]. \quad (571)$$

These equations say that the absorptive part is obtained by cutting the two fermion lines in the loop. The full retarded function is then reconstructed with the dispersion relation (569). Insert the free thermal Wightman functions

$$S^>(Q) = -i(1 - n_F(q^0))\rho_F(Q), \quad S^<(Q) = +in_F(q^0)\rho_F(Q), \quad (572)$$

with

$$\rho_F(Q) = 2\pi \text{sgn}(q^0) \mathcal{Q} \delta(Q^2). \quad (573)$$

The two spectral delta-functions put the two internal fermion lines on shell:

$$q^0 = \pm q, \quad q^0 + k^0 = \pm |\mathbf{q} + \mathbf{k}|. \quad (574)$$

There are four sign choices. The particle-hole channel has both lines on the positive-energy branch and gives the denominator

$$k^0 + q - |\mathbf{q} + \mathbf{k}| + i0^+. \quad (575)$$

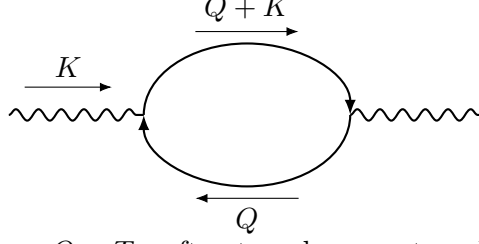
The antiparticle-hole channel gives the same leading contribution after relabelling the loop momentum. The pair-creation and pair-annihilation channels have denominators of order $q + |\mathbf{q} + \mathbf{k}| \sim T$ and are subleading in the HTL limit. Keeping the particle-hole and antiparticle-hole contributions gives

$$\Pi_R^{00}(k^0, \mathbf{k}) = 4e^2 \int \frac{d^3\mathbf{q}}{(2\pi)^3} \frac{n_F(q) - n_F(|\mathbf{q} + \mathbf{k}|)}{k^0 + q - |\mathbf{q} + \mathbf{k}| + i0^+} + \text{subleading HTL terms}. \quad (576)$$

The factor 4 is a factor 2 from spin times a factor 2 from particle-hole plus antiparticle-hole channels. Thus the formula is not an assumed kinetic expression; it is the leading soft-external-momentum limit of the real-time SK one-loop polarization.

For $q \sim T$ and $k^0, k \sim eT \ll q$,

$$|\mathbf{q} + \mathbf{k}| = q + \hat{\mathbf{q}} \cdot \mathbf{k} + O(k^2/q), \quad n_F(q) - n_F(|\mathbf{q} + \mathbf{k}|) = -\frac{dn_F}{dq} \hat{\mathbf{q}} \cdot \mathbf{k} + \dots \quad (577)$$



hard loop $Q \sim T$, soft external momentum $K \sim eT$

Figure 15: The one-loop photon self-energy. In the HTL approximation the loop momentum is hard and the external photon momentum is soft. Expanding the hard fermion propagators in K/Q produces the universal HTL denominator $v \cdot K + i0^+$.

Therefore

$$\Pi_R^{00}(K) = 4e^2 \int \frac{d^3\mathbf{q}}{(2\pi)^3} \left(-\frac{dn_F}{dq} \right) \frac{\hat{\mathbf{q}} \cdot \mathbf{k}}{k^0 - \hat{\mathbf{q}} \cdot \mathbf{k} + i0^+}. \quad (578)$$

Using

$$\frac{\hat{\mathbf{q}} \cdot \mathbf{k}}{k^0 - \hat{\mathbf{q}} \cdot \mathbf{k} + i0^+} = -1 + \frac{k^0}{k^0 - \hat{\mathbf{q}} \cdot \mathbf{k} + i0^+}, \quad (579)$$

and defining $\Pi_L^R = -\Pi_R^{00}$, we obtain

$$\Pi_L^R(K) = m_D^2 \left[1 - k^0 \int \frac{d\Omega_{\hat{\mathbf{q}}}}{4\pi} \frac{1}{k^0 - \hat{\mathbf{q}} \cdot \mathbf{k} + i0^+} \right], \quad (580)$$

where

$$m_D^2 = -4e^2 \int \frac{d^3\mathbf{q}}{(2\pi)^3} \frac{dn_F(q)}{dq} = \frac{e^2 T^2}{3}. \quad (581)$$

The angular integral is elementary. Taking $\mathbf{k}$ along the z -axis,

$$\int \frac{d\Omega}{4\pi} \frac{1}{k^0 - k \cos \theta + i0^+} = \frac{1}{2k} \log \frac{k^0 + k + i0^+}{k^0 - k + i0^+}. \quad (582)$$

Thus

$$\Pi_L^R(k^0, k) = m_D^2 \left[1 - \frac{k^0}{2k} \log \frac{k^0 + k + i0^+}{k^0 - k + i0^+} \right]. \quad (583)$$

Kinetic derivation. The same formula follows from the collisionless Boltzmann equation. A soft electric field perturbs the hard fermion distribution $f = f_0 + \delta f$. Linearizing,

$$(\partial_t + \hat{\mathbf{q}} \cdot \nabla_{\mathbf{x}}) \delta f + e \mathbf{E} \cdot \hat{\mathbf{q}} \frac{df_0(q)}{dq} = 0. \quad (584)$$

For a plane wave $e^{-ik^0 t + i\mathbf{k} \cdot \mathbf{x}}$,

$$\delta f(K, \mathbf{q}) = \frac{e \mathbf{E}(K) \cdot \hat{\mathbf{q}}}{k^0 - \hat{\mathbf{q}} \cdot \mathbf{k} + i0^+} \frac{df_0(q)}{dq}. \quad (585)$$

The induced charge density is

$$\rho_{\text{ind}}(K) = 2e \int \frac{d^3\mathbf{q}}{(2\pi)^3} \delta f(K, \mathbf{q}), \quad (586)$$

where the factor 2 is spin. Including electrons and positrons gives the same factor 4 as in the loop calculation. The radial integral gives m_D^2 and the angular integral gives precisely (583). The kinetic derivation is therefore the classical-looking limit of the same quantum loop: hard particles move almost freely and respond to a slowly varying soft field.

Debye screening. The static limit of (583) is

$$\Pi_L^R(0, k) = m_D^2. \quad (587)$$

Therefore the static longitudinal propagator is screened:

$$D_L^R(0, k) = \frac{1}{k^2 + m_D^2}. \quad (588)$$

A test charge no longer produces a Coulomb potential but a Yukawa potential,

$$V(r) = \frac{e^2}{4\pi r} e^{-m_D r}. \quad (589)$$

The screening length is m_D^{-1} . This is Debye screening.

Our discussion shows that the thermal mass should not be thought of as a mysterious new bare mass generated from nothing. It is the leading local imprint of the medium on long-wavelength propagation. In gauge theory this imprint has two complementary descriptions. In finite-temperature perturbation theory it is the hard thermal loop self-energy. In kinetic theory it is the linear response of hard plasma particles to a soft field. The agreement is not accidental: HTL theory is precisely the field-theoretic completion of collisionless kinetic theory, with the correct quantum statistics, spin, and gauge structure.

Longitudinal and transverse plasmon poles. Plasmons are poles of resummed soft gauge-field propagators. For the longitudinal electric field the pole condition is

$$k^2 + \Pi_L^R(\omega_L, k) = 0. \quad (590)$$

For a timelike mode $\omega > k$ one may expand

$$\log \frac{\omega + k}{\omega - k} = 2 \left(\frac{k}{\omega} + \frac{k^3}{3\omega^3} + \frac{k^5}{5\omega^5} + \cdots \right). \quad (591)$$

Substituting this into (583) gives

$$\Pi_L^R(\omega, k) = -\frac{m_D^2}{3} \frac{k^2}{\omega^2} + O(k^4/\omega^4). \quad (592)$$

Then (590) gives

$$\omega_L^2(0) = \omega_{\text{pl}}^2, \quad \omega_{\text{pl}}^2 = \frac{m_D^2}{3}, \quad \omega_L^2(k) = \omega_{\text{pl}}^2 + \frac{3}{5}k^2 + O(k^4/m_D^2). \quad (593)$$

The longitudinal plasmon is an oscillation of the electric field and the induced charge density of the plasma.

There is also a transverse polarization describing transverse electromagnetic waves. It is obtained by projecting the spatial part of the same one-loop polarization:

$$P_T^{ij} = \delta^{ij} - \hat{k}^i \hat{k}^j, \quad \Pi_T^R = \frac{1}{2} P_T^{ij} \Pi_R^{ij}. \quad (594)$$

The derivation is completely parallel to the longitudinal calculation and is a useful exercise. The result is

$$\Pi_T^R(k^0, k) = \frac{m_D^2}{2} \frac{(k^0)^2}{k^2} \left[1 - \frac{(k^0)^2 - k^2}{2k^0 k} \log \frac{k^0 + k + i0^+}{k^0 - k + i0^+} \right]. \quad (595)$$

The transverse propagator has inverse denominator

$$(D_T^R)^{-1}(k^0, k) = (k^0)^2 - k^2 - \Pi_T^R(k^0, k), \quad (596)$$

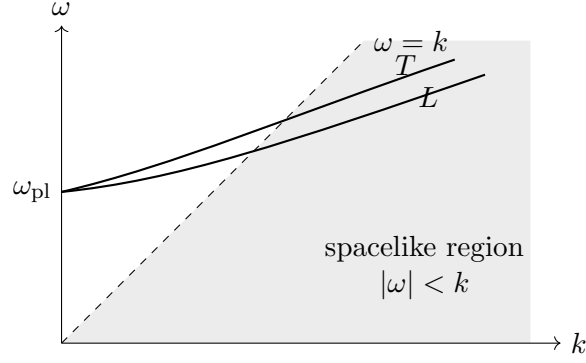


Figure 16: Schematic HTL collective modes. The longitudinal and transverse plasmon poles are timelike at small momentum, $\omega_{L,T}(k) > k$, and therefore lie outside the spacelike Landau-damping region.

so the pole equation is

$$\omega_T^2 - k^2 - \Pi_T^R(\omega_T, k) = 0. \quad (597)$$

Using (591),

$$\omega_T^2(0) = \omega_{\text{pl}}^2, \quad \omega_T^2(k) = \omega_{\text{pl}}^2 + \frac{6}{5}k^2 + O(k^4/m_D^2). \quad (598)$$

Thus both longitudinal and transverse soft electromagnetic fields have plasma modes. At $k = 0$ they share the same plasma frequency; at nonzero k they are different collective branches.

Landau damping: what is being damped. The phrase “Landau damping of a spacelike disturbance” means damping of a *spacelike Fourier component of the soft electromagnetic field*. For a Fourier component $K^\mu = (k^0, \mathbf{k})$, spacelike means

$$K^2 = (k^0)^2 - k^2 < 0, \quad \text{equivalently} \quad |k^0| < k. \quad (599)$$

The damping comes from the common HTL denominator

$$\frac{1}{k^0 - \mathbf{k} \cdot \hat{\mathbf{q}} + i0^+} = \mathcal{P} \frac{1}{k^0 - \mathbf{k} \cdot \hat{\mathbf{q}}} - i\pi \delta(k^0 - \mathbf{k} \cdot \hat{\mathbf{q}}). \quad (600)$$

The delta-function can be satisfied only if

$$\hat{\mathbf{q}} \cdot \hat{\mathbf{k}} = \frac{k^0}{k}, \quad (601)$$

which requires $|k^0| < k$. Physically, the soft field has phase velocity k^0/k , and hard particles with that velocity component can exchange energy with the field while remaining on shell. The coherent field amplitude is then damped.

For the longitudinal polarization, the imaginary part follows from the logarithm in (583). For $|k^0| < k$ and $k^0 > 0$,

$$\text{Im } \Pi_L^R(k^0, k) = \frac{\pi m_D^2}{2} \frac{k^0}{k} \theta(k^2 - (k^0)^2), \quad (602)$$

with the sign convention used in (583). This is damping of a longitudinal electric field, or equivalently of a soft charge-density disturbance. The transverse channel has the same cut but a different angular weight:

$$\text{Im } \Pi_T^R(k^0, k) = -\frac{\pi m_D^2}{4} \frac{k^0}{k} \left(1 - \frac{(k^0)^2}{k^2}\right) \theta(k^2 - (k^0)^2) \quad (k^0 > 0). \quad (603)$$

This is damping of a transverse electromagnetic field component. The long-wavelength plasmon poles above are timelike, $\omega_{L,T}(k) > k$, so they are not Landau damped at leading collisionless HTL order. Their widths arise from higher-order collisional processes, not from the spacelike HTL cut.

6.2 Hard thermal loop effective action

Why resummation is needed. For a soft gauge field, the bare inverse propagator is of order

$$K^2 \sim g^2 T^2. \quad (604)$$

The one-loop HTL polarization is of the same order,

$$\Pi_{\text{HTL}} \sim g^2 T^2. \quad (605)$$

Therefore the expansion

$$D_0 + D_0 \Pi D_0 + D_0 \Pi D_0 \Pi D_0 + \dots \quad (606)$$

is not ordered term by term for $K \sim gT$, because

$$D_0 \Pi \sim \frac{\Pi}{K^2} \sim 1. \quad (607)$$

The leading soft propagator is therefore the resummed one,

$$D_{\text{HTL}}(K) = \frac{1}{D_0^{-1}(K) - \Pi_{\text{HTL}}(K)}. \quad (608)$$

This is the gauge-theory version of thermal-mass resummation. In gauge theory, however, propagators and vertices are related by Ward identities. A systematic resummation must therefore also include the HTL vertices.

The QED photon HTL effective action. Let us first construct the abelian gauge-field part. The kinetic derivation above gives an induced current linear in the soft electromagnetic field. In Fourier space it can be written as

$$j_{\text{ind}}^\mu(K) = -\Pi_{\text{HTL}}^{\mu\nu}(K) A_\nu(K), \quad (609)$$

where the HTL tensor is

$$\Pi_{\text{HTL}}^{\mu\nu}(K) = m_D^2 \left[-\delta_0^\mu \delta_0^\nu + k^0 \int \frac{d\Omega_{\hat{\mathbf{v}}}}{4\pi} \frac{v^\mu v^\nu}{k^0 - \mathbf{k} \cdot \hat{\mathbf{v}} + i0^+} \right], \quad v^\mu = (1, \hat{\mathbf{v}}). \quad (610)$$

This tensor is transverse: $K_\mu \Pi_{\text{HTL}}^{\mu\nu} = 0$. The corresponding quadratic effective action is

$$\Gamma_{\gamma, \text{HTL}}^{\text{QED}}[A] = \frac{1}{2} \int \frac{d^4 K}{(2\pi)^4} A_\mu(-K) \Pi_{\text{HTL}}^{\mu\nu}(K) A_\nu(K). \quad (611)$$

Equivalently, in a compact coordinate-space notation,

$$\Gamma_{\gamma, \text{HTL}}^{\text{QED}}[A] = -\frac{m_D^2}{2} \int d^4 x \int \frac{d\Omega_{\hat{\mathbf{v}}}}{4\pi} F_{\mu\alpha}(x) v^\alpha \frac{1}{(v \cdot \partial)^2} F^{\mu\beta}(x) v_\beta. \quad (612)$$

The inverse derivative means propagation along the straight lightlike trajectory of a hard particle:

$$\frac{1}{v \cdot \partial + 0^+} f(x) = \int_0^\infty ds f(x - vs), \quad (613)$$

up to the convention-dependent factor of i in Fourier space. Varying (612) twice with respect to A_μ gives the two-point HTL vertex $i\Pi_{\text{HTL}}^{\mu\nu}(K)$. Because QED is abelian, $F_{\mu\nu}$ is linear in A_μ and $v \cdot \partial$ contains no gauge field. Therefore the pure-photon QED HTL action is quadratic: it produces the HTL photon self-energy but no HTL three-photon or four-photon vertices. This is different from nonabelian gauge theory.

The QED fermion HTL effective action and its vertices. QED nevertheless has nontrivial HTL vertices involving soft fermions. The fermionic HTL effective action is

$$\Gamma_{\psi,\text{HTL}}^{\text{QED}} = m_f^2 \int d^4x \int \frac{d\Omega_{\hat{\mathbf{v}}}}{4\pi} \bar{\psi}(x) \frac{\not{v}}{iv \cdot D} \psi(x), \quad m_f^2 = \frac{e^2 T^2}{8}, \quad (614)$$

where $D_\mu = \partial_\mu + ieA_\mu$. The inverse covariant derivative is the gauge-covariant version of propagation along the hard particle trajectory. In abelian theory it is a Wilson-line operator along the straight line in the v^μ direction. For perturbative calculations we expand it in powers of the soft field:

$$\frac{1}{iv \cdot D} = \frac{1}{iv \cdot \partial - ev \cdot A} = \frac{1}{iv \cdot \partial} + \frac{1}{iv \cdot \partial} ev \cdot A \frac{1}{iv \cdot \partial} + \frac{1}{iv \cdot \partial} ev \cdot A \frac{1}{iv \cdot \partial} ev \cdot A \frac{1}{iv \cdot \partial} + \dots \quad (615)$$

The zeroth term gives the HTL fermion self-energy:

$$\Sigma_{\text{HTL}}(P) = m_f^2 \int \frac{d\Omega_{\hat{\mathbf{v}}}}{4\pi} \frac{\not{v}}{p^0 - \hat{\mathbf{v}} \cdot \mathbf{p} + i0^+}. \quad (616)$$

This is the same structure as the Yukawa fermion HTL self-energy derived earlier, with the appropriate QED thermal mass.

The term linear in A_μ gives the HTL correction to the photon-fermion vertex. If the incoming fermion has momentum P and the outgoing fermion has momentum $P + K$, the HTL vertex correction is

$$\delta\Gamma_{\text{HTL}}^\mu(P + K, P) = em_f^2 \int \frac{d\Omega_{\hat{\mathbf{v}}}}{4\pi} \frac{\not{v} v^\mu}{(v \cdot (P + K) + i0^+)(v \cdot P + i0^+)}. \quad (617)$$

This formula is not guessed. It is obtained directly from the second term in (615): each inverse $v \cdot \partial$ becomes a denominator $v \cdot P$, and the inserted gauge field contributes ev^μ . The vertex satisfies the Ward identity

$$\begin{aligned} K_\mu \delta\Gamma_{\text{HTL}}^\mu(P + K, P) &= em_f^2 \int \frac{d\Omega_{\hat{\mathbf{v}}}}{4\pi} \not{v} \frac{v \cdot K}{(v \cdot (P + K))(v \cdot P)} \\ &= e [\Sigma_{\text{HTL}}(P) - \Sigma_{\text{HTL}}(P + K)]. \end{aligned} \quad (618)$$

Together with the bare vertex this is exactly what is required by gauge invariance of the resummed fermion propagator.

The next term in (615) gives a two-photon/two-fermion HTL vertex. For photons with momenta K_1, K_2 inserted along the fermion line, one contribution is

$$\delta\Gamma_{\text{HTL}}^{\mu\nu}(P + K_1 + K_2, P) = e^2 m_f^2 \int \frac{d\Omega_{\hat{\mathbf{v}}}}{4\pi} \frac{\not{v} v^\mu v^\nu}{v \cdot (P + K_1 + K_2) v \cdot (P + K_1) v \cdot P} + (K_1, \mu) \leftrightarrow (K_2, \nu), \quad (619)$$

with the same retarded $+i0^+$ prescription understood in each denominator. Higher vertices are obtained in the same way: every extra soft photon insertion contributes a factor ev^μ and one extra eikonal denominator. This is the concrete meaning of “expanding the HTL effective action.”

What changes in nonabelian gauge theory. In nonabelian gauge theory the gauge-field HTL action is genuinely nonlinear:

$$\Gamma_{\text{HTL}}^{\text{nonabelian}}[A] = \frac{m_D^2}{2} \int \frac{d\Omega_{\hat{\mathbf{v}}}}{4\pi} \int d^4x \text{tr} \left[F_{\mu\alpha}(x) \frac{v^\alpha v^\beta}{(v \cdot D)^2} F^\mu{}_\beta(x) \right]. \quad (620)$$

Now $F_{\mu\nu}$ contains A^2 , and D_μ contains A_μ . Expanding (620) therefore generates HTL three-gluon, four-gluon, and higher gluon vertices, in addition to the gluon self-energy. These vertices are not optional; they are required by nonabelian Ward identities.

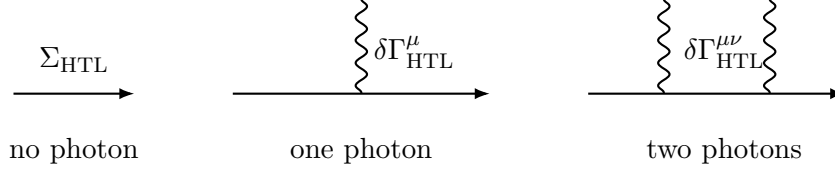


Figure 17: HTL vertices from the QED fermion effective action. Expanding the inverse covariant derivative in (615) generates a fermion self-energy and an infinite sequence of vertices with two external fermions and any number of soft photons.

Systematic HTL perturbation theory. The practical prescription is an “add and subtract” reorganization of perturbation theory. The point is not to change the theory. The point is to choose a better zeroth-order problem for soft external momenta. For a soft photon or a soft fermion, the bare inverse propagator and the HTL self-energy are of the same order. For example,

$$P \sim eT, \quad S_0^{-1}(P) \sim eT, \quad \Sigma_{\text{HTL}}(P) \sim eT. \quad (621)$$

Therefore expanding repeatedly in $S_0 \Sigma_{\text{HTL}}$ is not controlled: the nominal correction is of order one. The cure is to include the HTL term already in the propagator used for soft lines.

The prescription is:

- (i) **Separate hard and soft scales.** Hard loop momenta are of order T . Soft external gauge or fermion momenta are of order eT . The hard particles have already been integrated out into $\mathcal{L}_{\text{HTL}}$.
- (ii) **Use HTL-resummed soft propagators.** For a soft photon, use

$$D_*^{-1}(K) = D_0^{-1}(K) - \Pi_{\text{HTL}}(K). \quad (622)$$

For a soft fermion, use

$$S_*^{-1}(P) = S_0^{-1}(P) - \Sigma_{\text{HTL}}(P). \quad (623)$$

These formulae mean that screening, plasma poles, Landau damping, and thermal fermion quasiparticles are included already at leading order for soft propagation.

- (iii) **Use HTL vertices when they are part of the same soft approximation.** In QED the pure photon HTL effective action is quadratic in A_μ , so it corrects the photon two-point function but does not generate pure three-photon or four-photon HTL vertices. The fermionic HTL effective action, however, generates soft photon-fermion vertices such as the one-photon vertex $\Gamma_{\text{HTL}}^\mu(P+K, P)$ and the two-photon/two-fermion vertex. These vertices are needed so that the resummed amplitudes obey the Ward identities. In nonabelian gauge theory there are also pure soft gluon HTL vertices, because $F_{\mu\nu}$ and D_μ are nonlinear in the gauge field.
- (iv) **Avoid double counting by adding and subtracting the same HTL term.** Schematically,

$$\mathcal{L} = \underbrace{(\mathcal{L}_0 + \mathcal{L}_{\text{HTL}})}_{\text{new zeroth-order soft theory}} + \underbrace{(\mathcal{L}_{\text{int}} - \mathcal{L}_{\text{HTL}})}_{\text{interactions and HTL counterterms}}. \quad (624)$$

The term $+\mathcal{L}_{\text{HTL}}$ tells us to use HTL-resummed propagators and HTL vertices for soft lines. The term $-\mathcal{L}_{\text{HTL}}$ is a counterterm insertion. It removes the HTL contribution whenever the same contribution would otherwise be counted again in a higher-order diagram.

It is useful to see the double-counting subtraction in one explicit example. Suppose a soft photon line is computed using the resummed propagator

$$D_*(K) = \frac{1}{D_0^{-1}(K) - \Pi_{\text{HTL}}(K)}. \quad (625)$$

If we expanded this propagator formally, we would find

$$D_*(K) = D_0(K) + D_0(K)\Pi_{\text{HTL}}(K)D_0(K) + \cdots. \quad (626)$$

Thus the HTL self-energy has already been included inside the zeroth-order soft photon line. At the next order, the interaction part contains a counterterm vertex with Feynman rule $-\Pi_{\text{HTL}}^{\mu\nu}(K)$. A diagram with one such counterterm insertion subtracts precisely the piece that would be double counted. The same logic applies to soft fermion lines, where the counterterm is $-\Sigma_{\text{HTL}}(P)$, and to HTL fermion-photon vertices, where the counterterm is the negative of the corresponding HTL vertex. This is the finite-temperature analogue of adding and subtracting a thermal mass, but here the added object is the full gauge-invariant HTL effective action rather than a single local mass term.

What HTLs can describe. Hard thermal loop resummation addresses a large class of soft collective phenomena:

- Debye screening of static electric fields;
- longitudinal and transverse plasma oscillations;
- Landau damping of spacelike Fourier components of soft gauge fields;
- thermal fermion quasiparticles, including the normal branch and the plasmino branch;
- gauge-invariant soft scattering amplitudes in a hot plasma;
- the collisionless limit of kinetic theory for hard particles interacting with soft fields.

It does not by itself describe every transport coefficient. Quantities such as shear viscosity, electrical conductivity, and color conductivity require collisions and ladder resummations in addition to HTLs. Conceptually, the division of labor is clear: HTLs describe coherent mean-field response of hard thermal particles to soft fields; the Boltzmann equation adds the slower collisional evolution of the distribution function.

7 Electrical conductivity of a relativistic plasma from Schwinger–Keldysh theory

Electrical conductivity is the simplest transport coefficient in which the full real-time logic of finite-temperature field theory becomes visible. The observable itself is defined by a linear response experiment: apply a weak external electric field to an equilibrium plasma and measure the induced current. This makes the Kubo formula look like an ordinary retarded two-point function. However, the dc limit of this two-point function is qualitatively different from the quasiparticle pole calculations of the previous sections. At small external frequency, products of retarded and advanced propagators become nearly pinched between opposite sides of the real axis. This signals a long-lived disturbance: the total electric current can relax only through collisions.

The main lesson of this section is that transport coefficients cannot be obtained from a naive finite-order perturbative expansion. A thermal width must first regulate the pinch singularity, but even this is not enough: the width appearing in conductivity is a transport relaxation rate, not simply the single-particle damping rate. Diagrammatically, the same nearly pinching denominators enhance an infinite sequence of ladder diagrams. The Schwinger–Keldysh Bethe–Salpeter equation for the effective current vertex is the field-theoretic origin of the linearized Boltzmann equation.

We develop this logic in a self-contained way. First we derive the Kubo formula from the Schwinger–Keldysh generating functional. We then evaluate the one-loop current bubble and identify the pinch problem. Next we show how the Drude form follows both from the regulated real-time correlator and, remarkably, exactly from kinetic theory in the relaxation-time approximation. The leading-log estimate of the QED transport rate then gives $\Gamma_{\text{tr}} \sim e^4 T \log(1/e)$ and hence $\sigma_{\text{dc}} \sim T/[e^2 \log(1/e)]$. Finally we discuss the optical conductivity, where the same physics appears as the broadening of the free-theory zero-frequency delta function into a Drude peak.

7.1 The Kubo formula for conductivity from the Schwinger–Keldysh generating functional

The physical question. Apply a weak, slowly varying external electromagnetic field to a plasma in thermal equilibrium. The induced current defines the conductivity:

$$\delta \langle J^i(\omega, \mathbf{k}) \rangle = \sigma^{ij}(\omega, \mathbf{k}) E_j(\omega, \mathbf{k}). \quad (627)$$

For the dc conductivity we take the homogeneous limit first and then the small-frequency limit,

$$\sigma_{\text{dc}} = \lim_{\omega \rightarrow 0} \sigma(\omega, \mathbf{0}). \quad (628)$$

For the optical conductivity we keep the frequency finite at zero spatial momentum:

$$\sigma(\omega) \equiv \sigma(\omega, \mathbf{0}). \quad (629)$$

The central object is therefore a real-time retarded current-current correlator.

In QED the electric current is

$$J^\mu(x) = e \bar{\psi}(x) \gamma^\mu \psi(x). \quad (630)$$

We treat the applied electromagnetic field as a classical external source. To avoid confusion with the dynamical photon field, denote the applied field by A_μ^{ext} . Its coupling to the current is

$$S_{\text{ext}} = \int d^4x J^\mu(x) A_\mu^{\text{ext}}(x), \quad (631)$$

or, equivalently, $H_{\text{ext}} = - \int d^3x J^\mu A_\mu^{\text{ext}}$ in the Hamiltonian language.

Why Schwinger–Keldysh is the natural formalism. Conductivity is an expectation value at a finite time in the presence of a source. It is not an in-out matrix element. Therefore the correct generating functional is the Schwinger–Keldysh one:

$$Z[A_+, A_-] = \text{Tr} \left[U_{A_+}(+\infty, -\infty) \rho_0 U_{A_-}^\dagger(+\infty, -\infty) \right]. \quad (632)$$

Here A_+ couples to the current on the forward branch and A_- couples to the current on the backward branch. The physical source is obtained by setting $A_+ = A_- = A$ at the end. It is useful to introduce the r/a combinations

$$A_r = \frac{1}{2}(A_+ + A_-), \quad A_a = A_+ - A_-, \quad (633)$$

and similarly for the current,

$$J_r = \frac{1}{2}(J_+ + J_-), \quad J_a = J_+ - J_-. \quad (634)$$

The source coupling on the closed time contour has the difference form

$$\int d^4x (J_+^\mu A_{+\mu} - J_-^\mu A_{-\mu}) = \int d^4x (J_a^\mu A_{r\mu} + J_r^\mu A_{a\mu}). \quad (635)$$

This equation is the simplest way to remember the physical meaning of the r/a basis: A_r is the ordinary physical source, while A_a is the source that differentiates the generating functional to produce a physical expectation value.

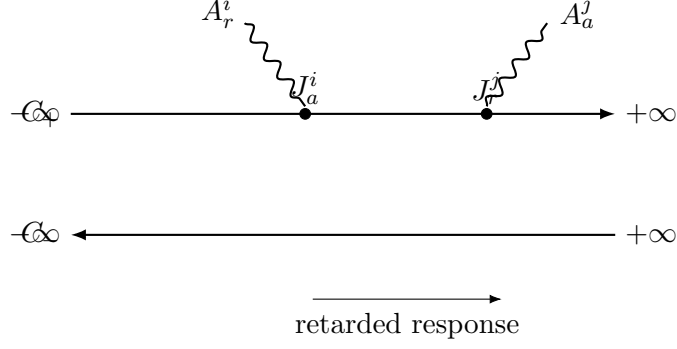


Figure 18: Schwinger–Keldysh source logic for conductivity. The physical external field is the r -source A_r . Differentiating with respect to the a -source extracts the expectation value of the physical current. The mixed r/a correlator is the retarded current-current correlator.

The retarded correlator from $Z[A_+, A_-]$. Let

$$W[A_+, A_-] = -i \ln Z[A_+, A_-]. \quad (636)$$

The expectation value of the physical current in the presence of A_r is

$$\langle J^i(x) \rangle_A = \left. \frac{\delta W}{\delta A_{ai}(x)} \right|_{A_a=0}. \quad (637)$$

Expanding to first order in A_r gives

$$\delta \langle J^i(x) \rangle = - \int d^4 y G_R^{ij}(x-y) A_j^{\text{ext}}(y), \quad (638)$$

with

$$G_R^{ij}(x-y) = -i \theta(x^0 - y^0) \langle [J^i(x), J^j(y)] \rangle. \quad (639)$$

The minus sign in (638) follows from the Hamiltonian perturbation $H_{\text{ext}} = - \int J^i A_i^{\text{ext}}$. Some authors absorb this sign into the definition of the response kernel. The final spectral formula for the real conductivity is independent of this convention.

For a spatially homogeneous external vector potential, take temporal gauge $A_0^{\text{ext}} = 0$ and $\mathbf{k} = 0$. With Fourier convention $e^{-i\omega t}$,

$$E_j(\omega) = i\omega A_j^{\text{ext}}(\omega). \quad (640)$$

Combining (627), (638), and (640), we obtain

$$\sigma^{ij}(\omega) = -\frac{1}{i\omega} G_R^{ij}(\omega, \mathbf{0}). \quad (641)$$

For an isotropic plasma,

$$G_R^{ij}(\omega, \mathbf{0}) = \delta^{ij} G_R(\omega), \quad \sigma(\omega) = -\frac{1}{3i\omega} G_R^{ii}(\omega, \mathbf{0}). \quad (642)$$

This is the Kubo formula in the form we will use below.

Spectral form and positivity. Define the current spectral function by

$$\rho_J^{ij}(\omega, \mathbf{0}) = \int d^4 x e^{i\omega t} \langle [J^i(x), J^j(0)] \rangle. \quad (643)$$

With the convention (639),

$$\rho_J^{ij}(\omega, \mathbf{0}) = -2 \operatorname{Im} G_R^{ij}(\omega, \mathbf{0}). \quad (644)$$

Therefore the real part of the optical conductivity is

$$\operatorname{Re} \sigma(\omega) = \frac{1}{6\omega} \rho_J^{ii}(\omega, \mathbf{0}) = -\frac{1}{3\omega} \operatorname{Im} G_R^{ii}(\omega, \mathbf{0}), \quad \omega > 0. \quad (645)$$

This expression makes positivity transparent: for positive frequency, the spectral weight of a physical current is positive, so $\operatorname{Re} \sigma(\omega) \geq 0$.

The dc conductivity is obtained from the small-frequency slope of the spectral function:

$$\sigma_{\text{dc}} = \lim_{\omega \rightarrow 0^+} \operatorname{Re} \sigma(\omega) = \frac{1}{6} \lim_{\omega \rightarrow 0^+} \frac{\rho_J^{ii}(\omega, \mathbf{0})}{\omega}. \quad (646)$$

This formula is the starting point for the perturbative calculation.

Why this is already a Schwinger–Keldysh statement. Equation (646) looks like an equilibrium formula, but its derivation used the closed-time-path structure. The conductivity is a retarded response: the current at time x^0 responds only to sources at earlier times $y^0 < x^0$. This causal ordering is exactly what the mixed r/a correlator in Schwinger–Keldysh perturbation theory computes. In the next subsection we will evaluate G_R^{ij} at one loop. The calculation will look innocent, but its small-frequency limit will reveal the central difficulty of transport: products of nearly pinching retarded and advanced propagators.

7.2 The one-loop current bubble and the origin of the pinch problem

We now compute the first diagram that contributes to the current-current correlator. The goal of this subsection is not yet to obtain the final dc conductivity. The goal is more important pedagogically: to see why the dc conductivity cannot be obtained by a naive finite-order perturbative expansion. The one-loop bubble already knows about the quasiparticles that carry current, but it also exposes the singular small frequency structure which will force us to resum widths and ladder diagrams.

For QED the electric current is

$$J^i(x) = e \bar{\psi}(x) \gamma^i \psi(x). \quad (647)$$

The one-loop contribution to the retarded correlator is the fermion bubble with one external current insertion on the response leg and one on the source leg. In the r/a basis it is convenient to write it as

$$G_{R,1\text{-loop}}^{ij}(K) = -ie^2 \int \frac{d^4 P}{(2\pi)^4} \operatorname{tr} [\gamma^i S_R(P+K) \gamma^j S_K(P) + \gamma^i S_K(P+K) \gamma^j S_A(P)]. \quad (648)$$

Here $K = (\omega, \mathbf{k})$ is the external momentum. This is the same structure we have seen several times in Schwinger–Keldysh perturbation theory: one internal line is spectral/statistical, and the other is retarded or advanced. The trace is over Dirac indices. The two terms are required by causality. They are also the real-time version of the statement that the retarded correlator is a commutator, not a time-ordered correlator.

Relation to the greater and lesser form. It is useful to see explicitly that (648) computes a retarded commutator. The current spectral function is

$$\rho_J^{ij}(K) = G_J^{>ij}(K) - G_J^{<ij}(K), \quad (649)$$

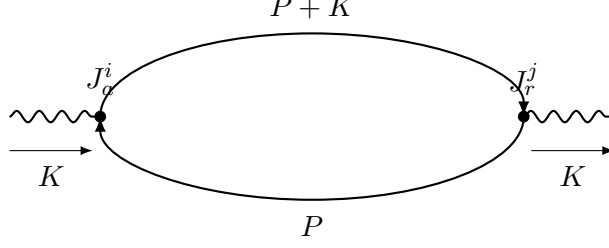


Figure 19: The one-loop current bubble for the retarded conductivity correlator. The external legs are not dynamical photons here; they are insertions of the current coupled to a weak probe field. In the r/a basis the retarded correlator is built from one retarded or advanced fermion line and one Keldysh line, as in (648).

where, at one loop,

$$G_J^{>ij}(K) = -e^2 \int \frac{d^4 P}{(2\pi)^4} \text{tr}[\gamma^i S^>(P+K) \gamma^j S^<(P)], \quad (650)$$

$$G_J^{<ij}(K) = -e^2 \int \frac{d^4 P}{(2\pi)^4} \text{tr}[\gamma^i S^<(P+K) \gamma^j S^>(P)]. \quad (651)$$

In equilibrium,

$$S^>(P) = [1 - n_F(p^0)] \rho_D(P), \quad S^<(P) = -n_F(p^0) \rho_D(P), \quad (652)$$

with the sign convention of Section 2. Substituting these expressions shows that the cut version of the bubble has a simple interpretation: it counts the difference between thermally allowed absorption and emission processes. This is the form that is best for spectral functions. The r/a form (648) is best for seeing causal response and pinch singularities.

Homogeneous electric field and the transport kinematics. For conductivity we take

$$K = (\omega, \mathbf{0}). \quad (653)$$

This is a special limit. It does not create a finite-momentum particle-hole pair. Instead, it asks how an almost homogeneous electric field changes the occupation of already existing charged quasiparticles. The part of the bubble relevant for transport therefore comes from putting the two fermion lines on nearly the same quasiparticle pole. This is the origin of the product

$$S_R(P+K) S_A(P). \quad (654)$$

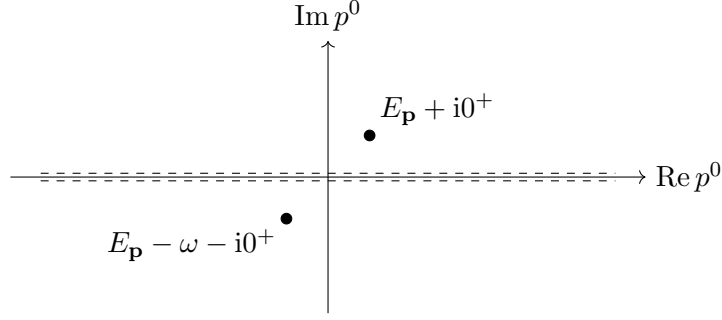
One should distinguish this transport contribution from the high-frequency pair-creation contribution. Pair creation controls optical absorption at frequencies of order the hard scale. The dc limit is instead controlled by quasiparticles close to their mass shell whose momenta are changed only very slowly by the external field.

To display the singularity, approximate the retarded and advanced fermion propagators near a positive-energy quasiparticle pole by scalar denominators multiplying smooth spin projectors:

$$S_R(P) \simeq \frac{Z_{\mathbf{p}} \Lambda_+(\mathbf{p}) \gamma^0}{p^0 - E_{\mathbf{p}} + i0^+}, \quad S_A(P) \simeq \frac{Z_{\mathbf{p}} \Lambda_+(\mathbf{p}) \gamma^0}{p^0 - E_{\mathbf{p}} - i0^+}. \quad (655)$$

For $K = (\omega, \mathbf{0})$, the two denominators become

$$p^0 + \omega - E_{\mathbf{p}} + i0^+, \quad p^0 - E_{\mathbf{p}} - i0^+. \quad (656)$$



opposite $i0^+$ prescriptions pinch the real contour as $\omega \rightarrow 0$

Figure 20: Pinch singularity. A retarded pole and an advanced pole approach the real integration contour from opposite sides. In the dc limit this produces the factor $1/(\omega + i0^+)$ in (657).

The poles lie on opposite sides of the real p^0 axis. As $\omega \rightarrow 0$, they squeeze the integration contour. This is the *pinch singularity*. More explicitly,

$$\int \frac{dp^0}{2\pi} \frac{1}{p^0 + \omega - E_{\mathbf{p}} + i0^+} \frac{1}{p^0 - E_{\mathbf{p}} - i0^+} = \frac{i}{\omega + i0^+}. \quad (657)$$

The factor $1/\omega$ is the diagrammatic origin of the infinite dc conductivity of a gas of perfectly stable charged particles. In the Kubo formula there is already a factor $1/\omega$; therefore the small-frequency limit is not a harmless Taylor expansion.

Including a thermal width. Interactions give a quasiparticle a finite lifetime. At the level of propagators this means replacing

$$0^+ \longrightarrow \Gamma_{\mathbf{p}}/2. \quad (658)$$

The same integral then becomes

$$\int \frac{dp^0}{2\pi} \frac{1}{p^0 + \omega - E_{\mathbf{p}} + i\Gamma_{\mathbf{p}}/2} \frac{1}{p^0 - E_{\mathbf{p}} - i\Gamma_{\mathbf{p}}/2} = \frac{i}{\omega + i\Gamma_{\mathbf{p}}}. \quad (659)$$

This is the simplest way to see the emergence of a Drude denominator. The width regulates the pinch. However, this is not yet the final answer for the conductivity, because the relevant width for current relaxation is the *transport* width, not necessarily the single-particle damping rate.

The reason is physical. A small-angle Coulomb scattering can strongly affect the phase of a quasiparticle and therefore contribute to its damping rate, but it barely changes the direction of its velocity and hence barely relaxes the electric current. Conductivity is controlled by processes that degrade current. The field-theoretic expression of this statement is that vertex corrections are as important as self-energy insertions.

Quasiparticle reduction of the one-loop bubble. Keeping only the pinch part, the one-loop result reduces to the familiar kinetic-theory structure

$$G_{R,\text{pinch}}^{ij}(\omega, \mathbf{0}) \simeq -e^2 \sum_{s=\pm} \int \frac{d^3p}{(2\pi)^3} \nu_s v_{\mathbf{p}}^i v_{\mathbf{p}}^j \left(-\frac{\partial n_F(E_{\mathbf{p}})}{\partial E_{\mathbf{p}}} \right) \frac{\omega}{\omega + i\Gamma_{\mathbf{p}}}. \quad (660)$$

Here $s = +$ and $s = -$ label particles and antiparticles, ν_s denotes the spin degeneracy for each branch, and

$$\mathbf{v}_{\mathbf{p}} = \frac{\partial E_{\mathbf{p}}}{\partial \mathbf{p}} \quad (661)$$

is the quasiparticle velocity. This formula should be read as a bridge, not as the final leading-order QED answer. It says that the current-current correlator is controlled by charge carriers on shell, weighted by how easily their occupation changes. The factor $-\partial n_F/\partial E$ localizes the response to the thermally occupied hard momenta $p \sim T$.

Substituting (660) into the Kubo formula gives the Drude form

$$\sigma^{ij}(\omega) \simeq e^2 \sum_{s=\pm} \int \frac{d^3p}{(2\pi)^3} \nu_s v_{\mathbf{p}}^i v_{\mathbf{p}}^j \left(-\frac{\partial n_F(E_{\mathbf{p}})}{\partial E_{\mathbf{p}}} \right) \frac{1}{\Gamma_{\mathbf{p}} - i\omega}. \quad (662)$$

Remarkably, this is exactly what is obtained from kinetic theory in the relaxation-time approximation. Let us show this explicitly, because it makes the physical meaning of (662) transparent.

Consider a spatially homogeneous electric field

$$E^j(t) = E^j(\omega) e^{-i\omega t}. \quad (663)$$

Let q_s denote the charge of branch s : for a QED plasma $q_+ = e$ for particles and $q_- = -e$ for antiparticles. Write the distribution function as

$$f_s(t, \mathbf{p}) = n_F(E_{\mathbf{p}}) + \delta f_s(t, \mathbf{p}). \quad (664)$$

The homogeneous linearized Boltzmann equation in the relaxation-time approximation is

$$\frac{\partial \delta f_s}{\partial t} + q_s E^j(t) \frac{\partial n_F(E_{\mathbf{p}})}{\partial p^j} = -\Gamma_{\mathbf{p}} \delta f_s. \quad (665)$$

Here $\Gamma_{\mathbf{p}}$ should be read as the relaxation rate of the current-carrying distortion of the distribution. In a full treatment this is the transport rate obtained from the linearized collision operator. Using

$$\frac{\partial n_F(E_{\mathbf{p}})}{\partial p^j} = v_{\mathbf{p}}^j \frac{\partial n_F(E_{\mathbf{p}})}{\partial E_{\mathbf{p}}}, \quad (666)$$

and substituting $\delta f_s(t, \mathbf{p}) = \delta f_s(\omega, \mathbf{p}) e^{-i\omega t}$, we obtain

$$(\Gamma_{\mathbf{p}} - i\omega) \delta f_s(\omega, \mathbf{p}) = q_s E^j(\omega) v_{\mathbf{p}}^j \left(-\frac{\partial n_F(E_{\mathbf{p}})}{\partial E_{\mathbf{p}}} \right). \quad (667)$$

Therefore

$$\delta f_s(\omega, \mathbf{p}) = \frac{q_s E^j(\omega) v_{\mathbf{p}}^j}{\Gamma_{\mathbf{p}} - i\omega} \left(-\frac{\partial n_F(E_{\mathbf{p}})}{\partial E_{\mathbf{p}}} \right). \quad (668)$$

The induced current is

$$\delta J^i(\omega) = \sum_{s=\pm} \nu_s q_s \int \frac{d^3p}{(2\pi)^3} v_{\mathbf{p}}^i \delta f_s(\omega, \mathbf{p}). \quad (669)$$

Substituting (668) gives

$$\delta J^i(\omega) = \sum_{s=\pm} \nu_s q_s^2 \int \frac{d^3p}{(2\pi)^3} v_{\mathbf{p}}^i v_{\mathbf{p}}^j \left(-\frac{\partial n_F(E_{\mathbf{p}})}{\partial E_{\mathbf{p}}} \right) \frac{1}{\Gamma_{\mathbf{p}} - i\omega} E^j(\omega) \quad (670)$$

$$= e^2 \sum_{s=\pm} \int \frac{d^3p}{(2\pi)^3} \nu_s v_{\mathbf{p}}^i v_{\mathbf{p}}^j \left(-\frac{\partial n_F(E_{\mathbf{p}})}{\partial E_{\mathbf{p}}} \right) \frac{1}{\Gamma_{\mathbf{p}} - i\omega} E^j(\omega), \quad (671)$$

because $q_s^2 = e^2$ for both particles and antiparticles. Reading off $\delta J^i = \sigma^{ij} E^j$ reproduces exactly (662). The important lesson is that Schwinger–Keldysh perturbation theory has produced the same structure as kinetic theory, and has also told us why a naive one-loop calculation is incomplete: the denominator in (662) comes from a regulated pinch singularity, so the rate in that denominator must be computed with the same accuracy as the current vertex.

Why the next subsection must resum ladders. In weakly coupled QED,

$$\Gamma_{\mathbf{p}} \sim e^4 T \log(1/e) \quad (672)$$

for transport relaxation. Since $\Gamma_{\mathbf{p}}$ is parametrically small, the nearly pinching product $S_R S_A$ is parametrically large. A ladder rung which would normally look like a higher-order correction is multiplied by another nearly pinching pair of propagators. The result is that an infinite set of ladder diagrams contributes at the same order to the dc conductivity. The next subsection first extracts the parametric transport rate from the same soft scattering kernel. The following subsection will then turn the ladder sum into a Bethe–Salpeter equation for an effective current vertex, and show that this equation is the Schwinger–Keldysh form of the linearized Boltzmann equation.

7.3 The leading-log transport rate in a weakly coupled QED plasma

The previous subsection showed that the dc conductivity is controlled by nearly on-shell charged quasiparticles and that the small denominator is regulated by a transport relaxation rate. We now estimate this rate in weakly coupled QED. The goal is not yet to solve the full integral equation for the conductivity. The goal is to understand why the relevant scale is

$$\Gamma_{\text{tr}} \sim e^4 T \log(1/e). \quad (673)$$

This estimate is one of the central lessons of plasma transport: the rate is small because scattering is weak, but it is enhanced by a logarithm because many momentum transfers between eT and T contribute.

What quantity is being estimated? A single-particle damping rate counts how often a particle scatters. A transport rate counts how quickly the current relaxes. These are not the same. If a hard electron is deflected by a very small angle, it has scattered, but its velocity and hence its contribution to the electric current have barely changed. Therefore the transport rate contains an angular weight,

$$\Gamma_{\text{tr}}(p) \simeq \sum_b n_b \int d\sigma_{ab} v_{\text{rel}} (1 - \cos \theta), \quad (674)$$

where a is the hard charge whose current we follow, b runs over thermal scatterers, $n_b \sim T^3$ is their number density, and θ is the deflection angle of the hard charge. For relativistic particles $v_{\text{rel}} = O(1)$.

In Schwinger–Keldysh perturbation theory this same quantity appears inside the ladder kernel. A rung of the ladder is a cut scattering process: the two nearly pinching propagators put the charge carriers on shell, and the rung supplies the squared scattering amplitude together with thermal occupation factors. The angular weight $1 - \cos \theta$ appears only after one asks for current relaxation rather than merely particle damping. This is the diagrammatic origin of the word “transport” in Γ_{tr} .

Screened Coulomb scattering. The dominant weak-coupling scattering is small-angle t -channel photon exchange between hard charged particles. The exchanged momentum is soft compared with the hard particle momentum,

$$Q = (q^0, \mathbf{q}), \quad q_{\perp} \ll p \sim T. \quad (675)$$

For a parametric leading-log estimate, it is enough to keep the screened Coulomb denominator. Debye screening replaces

$$\frac{1}{q_{\perp}^2} \longrightarrow \frac{1}{q_{\perp}^2 + m_D^2}, \quad m_D \sim eT. \quad (676)$$

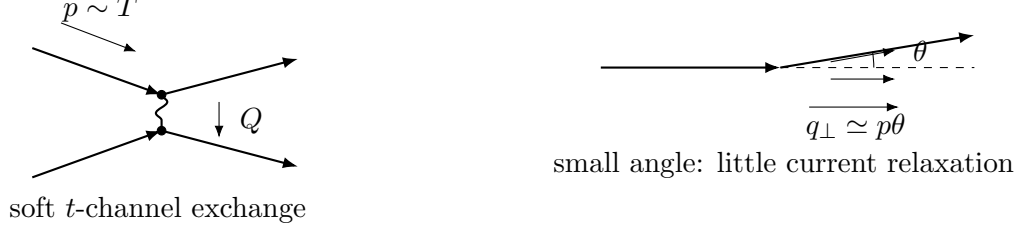


Figure 21: The leading-log transport rate comes from many soft, small-angle Coulomb scatterings. The left panel shows the soft exchanged photon in the scattering amplitude. The right panel shows why the transport rate is weighted by $1 - \cos \theta \simeq q_\perp^2 / (2p^2)$: very small angle scattering barely changes the current.

The small-angle differential cross section then has the form

$$\frac{d\sigma}{dq_\perp^2} \simeq \frac{C e^4}{(q_\perp^2 + m_D^2)^2}. \quad (677)$$

Here C is a numerical factor of order one depending on which charged species scatter and on spin/statistics. For the leading-log argument the precise value of C is not important; the important structure is the Coulomb factor $(q_\perp^2 + m_D^2)^{-2}$.

The transport weight for a hard particle of momentum $p \sim T$ is

$$1 - \cos \theta \simeq \frac{\theta^2}{2} \simeq \frac{q_\perp^2}{2p^2}. \quad (678)$$

Therefore

$$\begin{aligned} \sigma_{\text{tr}}(p) &\equiv \int d\sigma (1 - \cos \theta) \\ &\simeq \frac{C e^4}{2p^2} \int_0^{q_{\text{max}}^2} dq_\perp^2 \frac{q_\perp^2}{(q_\perp^2 + m_D^2)^2}. \end{aligned} \quad (679)$$

The upper limit is of order the hard scale, $q_{\text{max}} \sim T$. This is because the small-angle soft approximation is certainly valid below T , while the logarithm is insensitive to the exact order-one choice of the upper cutoff. The integral is elementary:

$$\begin{aligned} \int_0^{q_{\text{max}}^2} dx \frac{x}{(x + m_D^2)^2} &= \left[\log(x + m_D^2) + \frac{m_D^2}{x + m_D^2} \right]_0^{q_{\text{max}}^2} \\ &= \log \frac{q_{\text{max}}^2 + m_D^2}{m_D^2} - 1 + \frac{m_D^2}{q_{\text{max}}^2 + m_D^2}. \end{aligned} \quad (680)$$

For $q_{\text{max}} \sim T$ and $m_D \sim eT$, this gives

$$\int_0^{q_{\text{max}}^2} dq_\perp^2 \frac{q_\perp^2}{(q_\perp^2 + m_D^2)^2} = 2 \log(1/e) + O(1). \quad (681)$$

Thus

$$\sigma_{\text{tr}}(p \sim T) \sim \frac{e^4}{T^2} \log(1/e). \quad (682)$$

Multiplying by the density of scatterers, $n \sim T^3$, gives

$$\Gamma_{\text{tr}} \sim n \sigma_{\text{tr}} \sim e^4 T \log(1/e). \quad (683)$$

This is the promised result.

Where the logarithm comes from. It is useful to isolate the physics of the logarithm. The integrand in (679) behaves as

$$\frac{q_\perp^2}{(q_\perp^2 + m_D^2)^2} \simeq \frac{1}{q_\perp^2} \quad \text{for } m_D^2 \ll q_\perp^2 \ll T^2. \quad (684)$$

Therefore each decade of transverse momentum transfer contributes equally. The lower end is cut off by Debye screening, $q_\perp \sim m_D \sim eT$. The upper end is the hard scale $q_\perp \sim T$. Hence

$$\int_{m_D^2}^{T^2} \frac{dq_\perp^2}{q_\perp^2} \sim \log \frac{T^2}{m_D^2} \sim 2 \log(1/e). \quad (685)$$

The logarithm is not an ultraviolet divergence of the quantum field theory. It is a physical hierarchy of scales in the plasma: there is a wide interval of momentum transfers between the screening scale and the hard thermal scale.

How this appears in SK ladder perturbation theory. The same estimate can be read directly from real-time perturbation theory. The conductivity bubble contains nearly pinching pairs of fermion propagators. A soft photon rung connecting the two sides of the ladder represents one scattering event. Cutting the rung puts the scattering particles on shell and produces the Boltzmann kernel

$$\mathcal{C}[f] \sim \int d\Pi_b d\Pi_c d\Pi_d (2\pi)^4 \delta^{(4)}(p + b - c - d) |\mathcal{M}_{pb \rightarrow cd}|^2 \times \text{thermal occupation factors}. \quad (686)$$

For soft t -channel exchange,

$$|\mathcal{M}|^2 \propto \frac{e^4}{(q_\perp^2 + m_D^2)^2}. \quad (687)$$

The current vertex is odd under momentum reversal and measures the direction of the charge carrier's velocity. Therefore the collision operator acting on this vertex automatically inserts the same transport weight $1 - \cos \theta$. This is the SK diagrammatic version of the elementary kinetic estimate above.

A single rung is not enough. Since the nearly pinching pair gives a factor of order $1/\Gamma_{\text{tr}}$, and one soft scattering rung gives a factor of order Γ_{tr} , every additional ladder rung contributes at the same order. This is why transport coefficients require an infinite ladder resummation even at leading order. The leading-log calculation above identifies the scale that will appear in the resummed answer:

$$\tau_{\text{tr}} \sim \frac{1}{e^4 T \log(1/e)}. \quad (688)$$

Consequence for the size of the conductivity. In a relaxation-time estimate, the dc conductivity scales as

$$\sigma_{\text{dc}} \sim e^2 \int \frac{d^3 p}{(2\pi)^3} \left(-\frac{\partial n_F}{\partial p} \right) \frac{1}{\Gamma_{\text{tr}}}. \quad (689)$$

The integral is of order T^2 . Therefore

$$\sigma_{\text{dc}} \sim \frac{T}{e^2 \log(1/e)}. \quad (690)$$

The precise numerical coefficient requires solving the full linearized collision equation with the correct spin, particle-antiparticle, and exchange channels. The leading-log derivation above is nevertheless the essential physics: screened Coulomb scattering gives the current relaxation rate $e^4 T \log(1/e)$, and SK perturbation theory explains why the corresponding collision kernel must be ladder resummed.

7.4 Optical conductivity and the Drude peak

The dc conductivity is the zero-frequency limit of a more general object: the optical conductivity $\sigma(\omega)$. In the Schwinger–Keldysh language it is defined by the same retarded current-current correlator introduced in the Kubo formula, but now we keep the external frequency finite:

$$\sigma^{ij}(\omega) = \frac{1}{i\omega} G_R^{ij}(\omega, \mathbf{0}), \quad G_R^{ij}(t, \mathbf{x}) = -i\theta(t) \langle [J^i(t, \mathbf{x}), J^j(0)] \rangle. \quad (691)$$

For an isotropic plasma,

$$\sigma(\omega) = \frac{1}{3i\omega} G_R^{ii}(\omega, \mathbf{0}). \quad (692)$$

The word “optical” simply means that the electric field oscillates in time. We perturb the plasma by a spatially homogeneous field

$$\mathbf{E}(t) = \mathbf{E}_\omega e^{-i\omega t}, \quad (693)$$

and ask for the induced current

$$\mathbf{J}(t) = \sigma(\omega) \mathbf{E}_\omega e^{-i\omega t}. \quad (694)$$

Thus $\sigma_{\text{dc}} = \lim_{\omega \rightarrow 0} \sigma(\omega)$, while finite ω measures how quickly the plasma current can follow an oscillating electric field.

The relaxation-time derivation. The simplest derivation is the finite-frequency version of the kinetic calculation in the previous subsection. Consider particles and antiparticles of charge $q_s e$, with $q_s = \pm 1$. In a homogeneous electric field the linearized Boltzmann equation in the relaxation-time approximation is

$$\frac{\partial \delta f_s}{\partial t} + q_s e \mathbf{E}(t) \cdot \nabla_{\mathbf{p}} n_F(p) = -\frac{\delta f_s}{\tau_{\text{tr}}}. \quad (695)$$

The equilibrium distribution is isotropic, so

$$\nabla_{\mathbf{p}} n_F(p) = \hat{\mathbf{p}} \frac{dn_F}{dp}. \quad (696)$$

Using $\delta f_s(t, \mathbf{p}) = \delta f_s(\omega, \mathbf{p}) e^{-i\omega t}$, Eq. (695) gives

$$\delta f_s(\omega, \mathbf{p}) = \frac{q_s e \tau_{\text{tr}}}{1 - i\omega \tau_{\text{tr}}} \left(-\frac{dn_F}{dp} \right) \mathbf{E}_\omega \cdot \hat{\mathbf{p}}. \quad (697)$$

The current is

$$J^i(\omega) = e \sum_s q_s \int \frac{d^3 p}{(2\pi)^3} v^i \delta f_s(\omega, \mathbf{p}), \quad v^i = \hat{p}^i \quad (698)$$

for ultrarelativistic charge carriers. Since $q_s^2 = 1$, particles and antiparticles add. The angular integral is

$$\int \frac{d\Omega_{\hat{\mathbf{p}}}}{4\pi} \hat{p}^i \hat{p}^j = \frac{\delta^{ij}}{3}. \quad (699)$$

Therefore

$$J^i(\omega) = \frac{\sigma_{\text{dc}}}{1 - i\omega \tau_{\text{tr}}} E_\omega^i, \quad (700)$$

where

$$\sigma_{\text{dc}} = \frac{e^2 \tau_{\text{tr}}}{3} \sum_s \int \frac{d^3 p}{(2\pi)^3} \left(-\frac{dn_F}{dp} \right). \quad (701)$$

Thus the optical conductivity in the relaxation approximation is

$$\boxed{\sigma(\omega) = \frac{\sigma_{\text{dc}}}{1 - i\omega \tau_{\text{tr}}}} \quad (702)$$

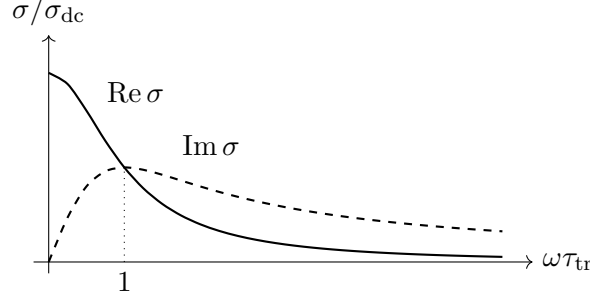


Figure 22: Drude optical conductivity. The dissipative part is a peak of width $1/\tau_{\text{tr}} = \Gamma_{\text{tr}}$ around $\omega = 0$. The imaginary part describes the phase lag between current and applied electric field.

This is the Drude form. It is not merely a guess: it is the direct solution of the linearized kinetic equation when the complicated collision operator is approximated by a single transport relaxation time.

Separating real and imaginary parts gives

$$\text{Re } \sigma(\omega) = \frac{\sigma_{\text{dc}}}{1 + \omega^2 \tau_{\text{tr}}^2}, \quad \text{Im } \sigma(\omega) = \frac{\sigma_{\text{dc}} \omega \tau_{\text{tr}}}{1 + \omega^2 \tau_{\text{tr}}^2}. \quad (703)$$

The real part is the dissipative part: it measures the rate at which the oscillating electric field does work on the plasma. The imaginary part is reactive: it measures the part of the current that is out of phase with the electric field.

The same structure from SK perturbation theory. In the previous subsections we saw that the one-loop current bubble contains nearly pinching pairs of propagators. Keeping a finite external frequency and a finite transport width, the important factor has the form

$$S_R(P + K)S_A(P) \longrightarrow \frac{2\pi Z_{\mathbf{p}} \delta(p^0 - E_{\mathbf{p}})}{-i\omega + \Gamma_{\text{tr}}}. \quad (704)$$

This is the diagrammatic origin of the denominator in (702). The full leading-order SK calculation does not simply insert Γ_{tr} by hand. Instead one solves the frequency-dependent ladder equation for the effective current vertex,

$$\Gamma^i(P; \omega) = v^i + \int_K \mathcal{K}(P, K) S_R(K + \omega) \Gamma^i(K; \omega) S_A(K), \quad (705)$$

where $\mathcal{K}$ is the cut scattering kernel. In the quasiparticle limit this Bethe–Salpeter equation is the linearized Boltzmann equation at frequency ω . The relaxation-time approximation replaces the full collision operator by one number, $\Gamma_{\text{tr}} = 1/\tau_{\text{tr}}$, and therefore reduces the full ladder equation to the Drude answer.

This explains the relation between the three descriptions without producing an overlong formula:

$$\begin{aligned} \text{SK ladder resummation} &\Longleftrightarrow \text{linearized Boltzmann equation,} \\ \text{linearized Boltzmann equation} &\Longrightarrow \text{Drude form in the relaxation approximation.} \end{aligned} \quad (706)$$

The first equivalence is a controlled statement in weakly coupled kinetic theory. The final arrow is an additional simplification: it assumes that the angular and momentum dependence of the collision operator can be represented by a single relaxation time.

Frequency regimes. The optical conductivity is useful because it separates several physical regimes.

- (i) **Hydrodynamic or dc regime:** $\omega\tau_{\text{tr}} \ll 1$. The current has enough time to relax during each oscillation of the electric field. The conductivity is approximately real and equal to σ_{dc} .
- (ii) **Drude crossover:** $\omega\tau_{\text{tr}} \sim 1$. The current cannot fully follow the electric field. The real part decreases, while the imaginary part becomes important.
- (iii) **Collisionless transport regime:** $\Gamma_{\text{tr}} \ll \omega \ll T$. Collisions are too slow to relax the current during one cycle. The Drude form gives

$$\text{Re } \sigma(\omega) \simeq \frac{\sigma_{\text{dc}}}{\omega^2 \tau_{\text{tr}}^2}, \quad \text{Im } \sigma(\omega) \simeq \frac{\sigma_{\text{dc}}}{\omega \tau_{\text{tr}}}. \quad (707)$$

In this window the response is mostly reactive.

- (iv) **Hard-frequency regime:** $\omega \sim T$. The simple kinetic description is no longer enough. The current correlator is sensitive to pair creation, annihilation, and the full thermal spectral function. One must return to the microscopic SK correlator rather than the low-frequency kinetic approximation.

Free theory and the role of collisions. The Drude peak has a useful limiting interpretation. If we formally take $\tau_{\text{tr}} \rightarrow \infty$, then

$$\text{Re } \sigma(\omega) = \frac{\sigma_{\text{dc}}}{1 + \omega^2 \tau_{\text{tr}}^2} \quad (708)$$

becomes infinitely narrow and infinitely tall, with fixed spectral weight. In distributional language it approaches a delta function at $\omega = 0$. This is what one expects for a perfectly clean gas of charged particles: without collisions, a constant electric field accelerates the charges indefinitely, so the dc conductivity is infinite. Interactions broaden this delta function into a Drude peak of width Γ_{tr} . In weakly coupled QED,

$$\Gamma_{\text{tr}} \sim e^4 T \log(1/e), \quad \sigma_{\text{dc}} \sim \frac{T}{e^2 \log(1/e)}. \quad (709)$$

Thus the width of the optical peak and the size of the dc conductivity are two sides of the same physical process: the relaxation of electric current by screened Coulomb scattering.